

Diffusion Models: Theory Note

Quick Guide

Luo Yi

May 5, 2026

1	Variational Autoencoder	4
1.1	Latent-variable generative models	4
1.2	KL divergence.	5
1.3	Evidence lower bound	6
1.4	Reparameterization and Gaussian VAE	7
1.5	Hierarchical VAE and the bridge to DDPM.	9
2	DDPM and DDIM	10
2.1	Gaussian tools for diffusion.	10
2.1.1	Tweedie's identity	10
2.2	Forward diffusion chain	11
2.2.1	Noise schedules	12
2.3	Denoising posterior	12
2.4	Reverse model and ELBO	13
2.5	Parameterizations, SNR, and v -prediction	14
2.5.1	Three equivalent prediction targets.	14
2.5.2	SNR equivalence of the denoising ELBO	14
2.5.3	v -prediction and the rotation structure	14
2.6	DDIM: generalized non-Markovian samplers	15
2.6.1	Underdetermined system and the generalized update	15
2.6.2	Sub-sequence sampling and the probability flow ODE	16
3	Continuous-Time Score-Based Models	17
3.1	Itô calculus prerequisites.	17
3.2	Score function and Langevin dynamics	18
3.3	Score matching	19
3.4	Noise Conditional Score Networks	20
3.4.1	Three failure modes of vanilla score matching.	20
3.4.2	NCSN training objective.	20
3.4.3	Annealed Langevin dynamics	21
3.5	The SDE framework.	21
3.5.1	From discrete chains to continuous SDEs	22
3.5.2	Transition kernels and boundary conditions	22
3.6	Reverse-time SDE and predictor–corrector samplers	22
3.6.1	Anderson's reverse-time SDE	22
3.6.2	Continuous-time score matching loss	23
3.6.3	Predictor–corrector samplers	23
4	Probability Flow ODEs and Flow Matching	25
4.1	Flow models and the continuity equation.	25
4.2	The probability flow ODE	26
4.2.1	An equivalence class of forward processes	26
4.3	Instantaneous change of variables.	27
4.4	Flow matching	28
4.4.1	The flow matching loss.	28
4.4.2	Conditional probability paths	28

4.4.3	The marginalization trick	28
4.4.4	Conditional flow matching and gradient equivalence	29
4.5	Affine conditional flows and connections to diffusion	30
4.5.1	Affine paths and their velocities	30
4.5.2	The OT–Gaussian (linear) path	30
4.5.3	Connection to diffusion models: recovering the score	30
5	Unification	32
5.1	The discrete-to-continuous limit: DDPM $\rightarrow$ VP-SDE	32
5.2	The SDE equivalence class and the probability flow ODE	33
5.3	Objective equivalence	34
5.4	Flow matching and score-based models	35
5.5	Sampling as numerical integration	36
5.6	The unified picture	36
5.7	Stochastic Interpolants	36
5.7.1	Motivation	36
5.7.2	Construction	38
5.7.3	Velocity and Score as Quadratic Minimisers	38
5.7.4	Special Cases	40
6	Extensions	41
6.1	Conditional Generation	41
6.1.1	Classifier guidance	41
6.1.2	Classifier-free guidance	42
6.1.3	Energy-based and loss-guided conditioning	42
6.2	Latent Diffusion	42
6.2.1	Setup	42
6.2.2	Score in latent vs. data space	42
6.3	Sampling Acceleration	43
6.3.1	Consistency models	43
6.3.2	Progressive distillation	43
6.4	Discrete Diffusion	44
6.4.1	Forward process: continuous-time Markov chain	44
6.4.2	Transition matrices and marginals	44
6.4.3	Closed-form reverse kernel	45
6.4.4	Training objective: discrete ELBO	45
7	Diffusion for Inverse Problems	46
7.1	Problem Setup	46
7.2	Approximation Taxonomy	47
7.3	Linear and Spectral Methods	47
7.3.1	DDRM: Denoising diffusion restoration models	47
7.3.2	DDNM: Denoising diffusion null-space model	48
7.3.3	PIGdM: Pseudoinverse-guided diffusion	48
7.4	Gradient-Based Methods	49
7.4.1	DPS: Complete derivation	49

7.4.2	Loss-guided diffusion and extensions	50
7.5	Plug-and-Play Diffusion Priors	50
7.5.1	Half-quadratic splitting (HQS)	50
7.5.2	Denoiser–score connection	51
7.5.3	DiffPIR and DAPS	51
7.6	Variational Inference Perspective	51
7.6.1	Variational ELBO with diffusion score	52
7.6.2	Comparison with trajectory-based methods	52
7.7	Sequential Monte Carlo	52
7.7.1	State-space formulation	52
7.7.2	Twisted proposals and MCGDiff	52
7.7.3	Filtering perspective	53
7.8	Error Analysis	53
8	Conclusion	55
8.1	Mathematical Synthesis	55
8.2	Open Problems	56
8.3	Research Outlook: Diffusion for Inverse Problems	57
A	Omitted Proofs	59
A.1	Lemma: Gaussian tools	59
A.2	Chapter 3: Itô Calculus	61
A.3	Theorem 3.7: Langevin Stationary Distribution	63
A.4	Theorem 3.10: DSM = ESM up to a constant	67
A.5	Theorem 3.17: Reverse-time SDE	69
A.6	Mass Conservation (Theorem 4.2)	71
A.7	Marginalization Trick (Theorem 4.12)	71
A.8	Proposition 5.1: DDPM $\rightarrow$ VP-SDE weak convergence	72

1. Variational Autoencoder

1.1 Latent-variable generative models

The goal of generative modeling is to draw new samples from the unknown data distribution $p_{\text{data}}(x)$. A *latent-variable model* does this by introducing an unobserved variable $z \in \mathbb{R}^k$ and factoring the joint distribution as

$$p_{\theta}(x, z) = p_{\theta}(x|z)p(z), \quad (1.1)$$

where $p(z) = \mathcal{N}(0, I_k)$ is a designer-chosen prior and $p_{\theta}(x|z)$ is a decoder parameterized by θ (typically a neural network). The model's marginal $p_{\theta}(x) = \int p_{\theta}(x|z)p(z) dz$ is the distribution over observations, but this integral is intractable for nonlinear decoders.

Four key distributions. Every VAE analysis revolves around four objects [6]:

- $p(x)$: the true, unknown data distribution.
- $p(z) = \mathcal{N}(0, I_k)$: the prior over latents, chosen by the designer.
- $p(z|x)$: the true posterior, intractable for nonlinear decoders.
- $p_{\theta}(x|z)$: the decoder likelihood, approximated by a neural network.

Since $p(z|x) \propto p_{\theta}(x|z)p(z)$ requires normalizing an intractable integral, we introduce an *encoder* $q_{\phi}(z|x)$ —a neural network outputting a Gaussian over z —as a tractable proxy for the true posterior. Figure 1.1 shows the VAE structure.

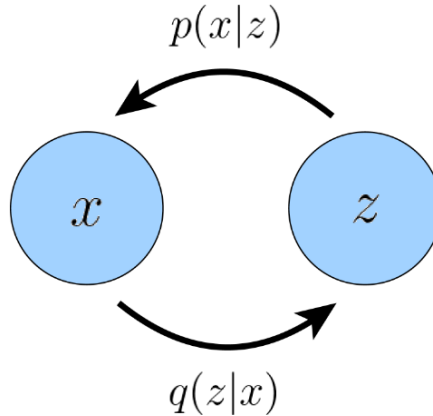


Figure 1.1: Schematic structure of VAE

Key Idea

VAE trains simultaneously: (1) a decoder $p_\theta(x|z)$ that generates realistic x from z , and (2) an encoder $q_\phi(z|x)$ that maps x to a latent code close to the prior $p(z)$. Both objectives are unified in the ELBO.

1.2 KL divergence

Definition 1.1 (Kullback–Leibler divergence). For densities p, q with $p \ll q$:

$$\text{KL}(p||q) = \int p(x) \log \frac{p(x)}{q(x)} dx. \quad (1.2)$$

where $p \ll q$ means $q(x) = 0 \Rightarrow p(x) = 0$ a.e., ensuring the ratio is well-defined.

Proposition 1.2 (Nonnegativity). $\text{KL}(p||q) \geq 0$, with equality iff $p = q$ a.e.

Proof. The map $\varphi(u) = u \log u$ is strictly convex on $(0, \infty)$ with $\varphi(1) = 0$. By Jensen's inequality applied to φ under the measure $q(x) dx$,

$$\text{KL}(p||q) = \int q(x) \varphi\left(\frac{p(x)}{q(x)}\right) dx \geq \varphi\left(\int q(x) \frac{p(x)}{q(x)} dx\right) = \varphi(1) = 0 \quad (1.3)$$

Strict convexity forces equality if and only if $p(x)/q(x)$ is q a.s. constant, which combined with $\int p dx = \int q dx = 1$ gives $p = q$ a.e. $\square$

Proposition 1.3 (Chain rule). For joint densities $p(x, y), q(x, y)$:

$$\text{KL}(p(x, y)||q(x, y)) = \text{KL}(p(x)||q(x)) + \mathbb{E}_{p(x)}[\text{KL}(p(y|x)||q(y|x))]. \quad (1.4)$$

Proof. Write $p(x, y) = p(x)p(y|x)$ and $q(x, y) = q(x)q(y|x)$. Then

$$\begin{aligned} \text{KL}(p(x, y)||q(x, y)) &= \int p(x, y) \log \frac{p(x)p(y|x)}{q(x)q(y|x)} dx dy \\ &= \int p(x) \log \frac{p(x)}{q(x)} dx + \int p(x) \left[\int p(y|x) \log \frac{p(y|x)}{q(y|x)} dy \right] dx \end{aligned} \quad (1.5)$$

which is the right-hand side of Equation (1.4). $\square$

Proposition 1.4 (KL between Gaussians). For $p = \mathcal{N}(\mu_1, \Sigma_1), q = \mathcal{N}(\mu_2, \Sigma_2)$ on $\mathbb{R}^d$:

$$\text{KL}(p||q) = \frac{1}{2} \left[\text{tr}(\Sigma_2^{-1} \Sigma_1) + (\mu_1 - \mu_2)^\top \Sigma_2^{-1} (\mu_1 - \mu_2) - d + \log \frac{\det \Sigma_2}{\det \Sigma_1} \right]. \quad (1.6)$$

Proof. Expand $\log(p/q)$ using the Gaussian log-density and take expectation under $p = \mathcal{N}(\mu_1, \Sigma_1)$. The two quadratic-form expectations are

$$\begin{aligned} \mathbb{E}_p[(X - \mu_1)^\top \Sigma_1^{-1} (X - \mu_1)] &= \text{tr}(\Sigma_1^{-1} \Sigma_1) = d, \\ \mathbb{E}_p[(X - \mu_2)^\top \Sigma_2^{-1} (X - \mu_2)] &= \text{tr}(\Sigma_2^{-1} \Sigma_1) + (\mu_1 - \mu_2)^\top \Sigma_2^{-1} (\mu_1 - \mu_2); \end{aligned}$$

collecting terms and simplifying the log-determinant ratio gives 1.6. $\square$

Corollary 1.5 (KL between isotropic Gaussians). For $p = \mathcal{N}(\mu_1, \sigma_1^2 I)$ and $q = \mathcal{N}(\mu_2, \sigma_2^2 I)$ on $\mathbb{R}^d$,

$$\text{KL}(p||q) = \frac{d}{2} \left(\frac{\sigma_1^2}{\sigma_2^2} - 1 - \log \frac{\sigma_1^2}{\sigma_2^2} \right) + \frac{\|\mu_1 - \mu_2\|^2}{2\sigma_2^2}. \quad (1.7)$$

In the equipoise case $\sigma_1 = \sigma_2 = \sigma$ the variance terms cancel and

$$\text{KL}(p||q) = \frac{\|\mu_1 - \mu_2\|^2}{2\sigma^2}. \quad (1.8)$$

Proof. Apply [Proposition 1.4](#) with $\Sigma_i = \sigma_i^2 I$: $\Sigma_2^{-1} \Sigma_1 = (\sigma_1^2 / \sigma_2^2) I$, $\text{tr}(\Sigma_2^{-1} \Sigma_1) = d\sigma_1^2 / \sigma_2^2$, $\det \Sigma_i = \sigma_i^{2d}$, $(\mu_2 - \mu_1)^\top \Sigma_2^{-1} (\mu_2 - \mu_1) = \|\mu_1 - \mu_2\|^2 / \sigma_2^2$. Substituting yields [Equation \(1.7\)](#). $\square$

1.3 Evidence lower bound

Definition 1.6 (Evidence Lower Bound). The Evidence Lower Bound is defined as

$$\text{ELBO}(\mathbf{x}) := \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \left[\log \frac{p(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right] \quad (1.9)$$

Proof. ELBO is a lower bound for the logarithm of the prior distribution $\log p(\mathbf{x})$:

$$\begin{aligned} \log p(\mathbf{x}) &= \log \int p(\mathbf{x}, \mathbf{z}) d\mathbf{z} \\ &= \log \int \frac{p(\mathbf{x}, \mathbf{z}) q_\phi(\mathbf{z}|\mathbf{x})}{q_\phi(\mathbf{z}|\mathbf{x})} d\mathbf{z} \\ &= \log \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \left[\frac{p(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right] \\ &\geq \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \left[\log \frac{p(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right] \end{aligned} \quad (1.10)$$

where the inequality follows from the Jensen's inequality. $\square$

Theorem 1.7 (ELBO decompositions). Define the ELBO as $\mathcal{L}(\phi, \theta; \mathbf{x}) := \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [\log p_\theta(\mathbf{x}, \mathbf{z}) - \log q_\phi(\mathbf{z}|\mathbf{x})]$. Then:

$$\log p_\theta(\mathbf{x}) = \mathcal{L}(\phi, \theta; \mathbf{x}) + \text{KL}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z}|\mathbf{x})), \quad (1.11)$$

$$\mathcal{L}(\phi, \theta; \mathbf{x}) = \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [\log p_\theta(\mathbf{x}|\mathbf{z})]}_{\text{reconstruction}} - \underbrace{\text{KL}(q_\phi(\mathbf{z}|\mathbf{x})||p(\mathbf{z}))}_{\text{prior matching}}. \quad (1.12)$$

Proof. The joint probability density in ELBO has two decompositions according to the chain rule:

$$p(\mathbf{x}, \mathbf{z}) = p(\mathbf{z})p(\mathbf{x}|\mathbf{z}) = p(\mathbf{x})p(\mathbf{z}|\mathbf{x}) \quad (1.13)$$

1. *Gap form* [Equation \(1.11\)](#). Write $\log p_\theta(\mathbf{x}) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [\log p_\theta(\mathbf{x})]$ (multiplying by $\int q_\phi(\mathbf{z}|\mathbf{x}) d\mathbf{z} = 1$). Apply Bayes:

$$\log p_\theta(\mathbf{x}) = \log p_\theta(\mathbf{x}, \mathbf{z}) - \log p_\theta(\mathbf{z}|\mathbf{x}) \quad (1.14)$$

so

$$\begin{aligned}\mathbb{E}_{q_\phi}[\log p_\theta(x)] &= \mathbb{E}_{q_\phi}[\log p_\theta(x, z) - \log q_\phi(z|x)] + \mathbb{E}_{q_\phi}[\log q_\phi(z|x) - \log p_\theta(z|x)] \\ &= \mathcal{L} + \text{KL}(q_\phi \| p_\theta(z|x)).\end{aligned}$$

2. *Reconstruction+prior form Equation (1.12)*. Substitute $\log p_\theta(x, z) = \log p_\theta(x|z) + \log p(z)$ into $\mathcal{L}$:

$$\begin{aligned}\mathcal{L}(\phi, \theta; x) &= \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x, z) - \log q_\phi(z|x)] \\ &= \mathbb{E}_{q_\phi(z|x)}[\log p(z) + \log p_\theta(x|z) - \log q_\phi(z|x)] \\ &= \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] + \mathbb{E}_{q_\phi(z|x)}[\log p(z) - \log q_\phi(z|x)] \\ &= \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \text{KL}(q_\phi(z|x) \| p(z))\end{aligned}\tag{1.15}$$

□

Interpretation.

1. *Gap form Equation (1.11)*: It shows $\mathcal{L} \leq \log p_\theta(x)$, with equality iff $q_\phi(z|x) = p_\theta(z|x)$ a.e. This also means maximizing ELBO tightens the variational lower bound.
2. *Reconstruction+prior form Equation (1.12)*: The two terms balance two competing objectives. The *reconstruction* term pushes the decoder to assign high probability to the true x given the encoded z . The *prior-matching* term forces the encoder's distribution $q_\phi(z|x)$ to stay close to $\mathcal{N}(0, I_k)$, so the latent space is regularly populated and new samples generated by $z \sim p(z)$ land in regions where the decoder behaves sensibly.

Common Pitfall

Without the KL term, training minimizes only reconstruction error. The encoder then collapses $\sigma_{\phi,j} \rightarrow 0$, turning $q_\phi(z|x)$ into a point mass—the randomness vanishes and generation from $z \sim \mathcal{N}(0, I_k)$ produces arbitrary outputs. The KL term in the ELBO is the mechanism that prevents this variance collapse [22].

1.4 Reparameterization and Gaussian VAE

VAE is a model that aims to approximate the true distribution $p(x)$ so that we can draw samples. A VAE is parameterized by (ϕ, θ) . Therefore, training a VAE is equivalent to solving an optimization problem whose solution recovers $p(x)$. However, since $p(x)$ is not accessible, instead, we maximize the ELBO, a tractable lower bound on $\log p(x)$.

Definition 1.8. The optimization objective of VAE is to maximize the ELBO:

$$\max_{\phi, \theta} \sum_{x \in \mathcal{X}} \mathcal{L}(\phi, \theta; x)\tag{1.16}$$

where $\mathcal{X} = \{\mathbf{x}^{(\ell)} \mid \ell = 1, \dots, L\}$ is the training dataset.

Gradient intractability. For θ , the gradient $\nabla_{\theta} \mathcal{L} = \mathbb{E}_{q_{\phi}}[\nabla_{\theta} \log p_{\theta}(x, z)]$ is a standard Monte Carlo estimator: sample $z \sim q_{\phi}(z|x)$, differentiate $\log p_{\theta}$. For ϕ , the gradient

$$\nabla_{\phi} \mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x, z) - \log q_{\phi}(z|x)] = \nabla_{\phi} \int [\log p_{\theta}(x, z) - \log q_{\phi}(z|x)] q_{\phi}(z|x) dz$$

requires differentiating *through the sampling distribution* q_{ϕ} , which is not directly possible by automatic differentiation. The reparameterization trick resolves this.

Proposition 1.9 (Reparameterization trick). *Let $z = g_{\phi}(\varepsilon, x)$ be a differentiable function of ϕ , with $\varepsilon \sim p(\varepsilon)$ independent of ϕ . Then*

$$\nabla_{\phi} \mathbb{E}_{q_{\phi}(z|x)}[f(z)] = \mathbb{E}_{p(\varepsilon)}[\nabla_{\phi} f(g_{\phi}(\varepsilon, x))]. \quad (1.17)$$

Proof. For notational simplicity, we write $g(\varepsilon)$ instead of $g_{\phi}(\varepsilon, x)$. It follows from change of variables that

$$\begin{aligned} \mathbb{E}_{q_{\phi}(z|x)}[f(z)] &= \int f(z) \cdot q_{\phi}(z|x) dz \\ &= \int f(g(\varepsilon)) \cdot q_{\phi}(g(\varepsilon)|x) dg(\varepsilon) \\ &= \int f(g(\varepsilon)) \cdot q_{\phi}(g(\varepsilon)|x) \cdot \left| \det \left(\frac{\partial g(\varepsilon)}{\partial \varepsilon} \right) \right| d\varepsilon \\ &= \int f(g(\varepsilon)) \cdot p(\varepsilon) d\varepsilon \\ &= \mathbb{E}_{p(\varepsilon)}[f(z)] \end{aligned} \quad (1.18)$$

Taking the gradient with respect to ϕ , one verifies that

$$\begin{aligned} \nabla_{\phi} \mathbb{E}_{q_{\phi}(z|x)}[f(z)] &= \nabla_{\phi} \mathbb{E}_{(\varepsilon)}[f(z)] = \nabla_{\phi} \left\{ \int f(z) \cdot p(\varepsilon) d\varepsilon \right\} \\ &= \int \nabla_{\phi} \{f(z) \cdot p(\varepsilon)\} d\varepsilon \\ &= \int \{\nabla_{\phi} f(z)\} \cdot p(\varepsilon) d\varepsilon \\ &= \mathbb{E}_{p(\varepsilon)}[\nabla_{\phi} f(z)] \end{aligned} \quad (1.19)$$

□

Now, substituting $f(z) = -\log q_{\phi}(z|x)$, we can show that

$$\nabla_{\phi} \mathbb{E}_{q_{\phi}(z|x)}[-\log q_{\phi}(z|x)] = \mathbb{E}_{p(\varepsilon)}[-\nabla_{\phi} \log q_{\phi}(z|x)] \approx -\frac{1}{L} \sum_{l=1}^L \nabla_{\phi} \log q_{\phi}(z^{(l)}|x) \quad (1.20)$$

where $z^{(l)} = g_{\phi}(\varepsilon^{(l)}, x)$.

Definition 1.10 (Gaussian VAE). Prior $p(z) = \mathcal{N}(0, I_k)$; encoder $q_{\phi}(z|x) = \mathcal{N}(\mu_{\phi}(x), \text{diag}(\sigma_{\phi}^2(x)))$ with $(\mu_{\phi}, \log \sigma_{\phi}^2)$ output by an encoder network as illustrated in [Figure 1.2](#); decoder $p_{\theta}(x|z) = \mathcal{N}(\hat{x}_{\theta}(z), \sigma_{\text{out}}^2 I)$. The reparameterization $z_{\varepsilon} := \mu_{\phi}(x) + \sigma_{\phi}(x) \odot \varepsilon$, $\varepsilon \sim \mathcal{N}(0, I_k)$, makes z_{ε} a differentiable function of ϕ .

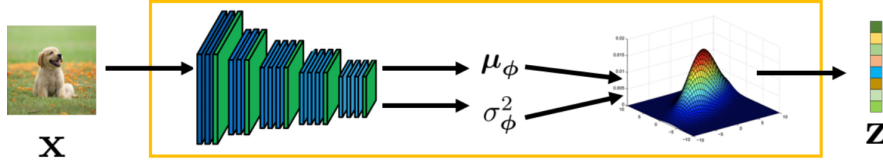


Figure 1.2: Implementation of a VAE encoder

Theorem 1.11 (Explicit ELBO for Gaussian VAE). Under [Definition 1.10](#),

$$\mathcal{L}(\phi, \theta; x) = \mathbb{E}_\epsilon \left[-\frac{\|x - \hat{x}_\theta(z_\epsilon)\|^2}{2\sigma_{\text{out}}^2} \right] - \frac{d}{2} \log(2\pi\sigma_{\text{out}}^2) - \frac{1}{2} \sum_{j=1}^k (\sigma_{\phi,j}^2 + \mu_{\phi,j}^2 - 1 - \log \sigma_{\phi,j}^2). \quad (1.21)$$

The gradient $\nabla_{\phi, \theta} \mathcal{L}$ is estimated without bias by a single sample $\epsilon \sim \mathcal{N}(0, I_k)$ via [Proposition 1.9](#).

Proof. Expanding the Gaussian log-decoder: $\log p_\theta(x | z_\epsilon) = -\|x - \hat{x}_\theta(z_\epsilon)\|^2 / (2\sigma_{\text{out}}^2) - \frac{d}{2} \log(2\pi\sigma_{\text{out}}^2)$. Applying [Proposition 1.9](#) to move ∇_ϕ inside the reconstruction expectation gives [Equation \(1.21\)](#). $\square$

In practice σ_{out}^2 is absorbed into the learning rate, and the training objective simplifies to minimizing $\text{MSE} + \beta \cdot \text{KL}$ (“ β -VAE” when $\beta \neq 1$).

1.5 Hierarchical VAE and the bridge to DDPM

Definition 1.12 (Top-down hierarchical VAE). With L stochastic layers $z_1, \dots, z_L$, we represent the joint distribution and the posterior of a Markovian Hierarchical VAE (HVAE) as:

$$p_\theta(x, z_{1:L}) = p_\theta(x | z_1) \prod_{\ell=1}^{L-1} p_\theta(z_\ell | z_{\ell+1}) \cdot p(z_L), \quad (1.22)$$

$$q_\phi(z_{1:L} | x) = q_\phi(z_L | x) \prod_{\ell=1}^{L-1} q_\phi(z_\ell | z_{\ell+1}, x). \quad (1.23)$$

Theorem 1.13 (Hierarchical ELBO).

$$\mathcal{L}(\phi, \theta; x) = \mathbb{E}_q[\log p_\theta(x | z_1)] - \text{KL}(q_\phi(z_L | x) \| p(z_L)) - \sum_{\ell=1}^{L-1} \mathbb{E}_{q(z_{\ell+1:L} | x)} [\text{KL}(q_\phi(z_\ell | z_{\ell+1}, x) \| p_\theta(z_\ell | z_{\ell+1}))]. \quad (1.24)$$

Proof. Apply [Theorem 1.7](#) with joint latent $(z_1, \dots, z_L)$ and expand via the generative and inference factorizations: the z_L pair yields $-\text{KL}(q(z_L | x) \| p(z_L))$; for each $\ell < L$ integrating over z_ℓ under the factored encoder yields the corresponding KL term. $\square$

Remark 1.14 (Bridge to DDPM). When all conditionals are Gaussian with linear means, every KL in [Equation \(1.24\)](#) has the closed form of [Proposition 1.4](#). DDPM fixes the inference model to a pre-specified Gaussian Markov noising chain, removing ϕ ; every KL term reduces to a weighted squared denoising error. This identification opens [Chapter 2](#).

2. DDPM and DDIM

Ho et al. [16] introduced denoising diffusion probabilistic models (DDPM) as a hierarchical variational autoencoder whose encoder is a *fixed* Gaussian Markov chain that gradually destroys structure in data. The denoising posterior is tractable in closed form, yielding a principled ELBO and a remarkably simple training objective. Song et al. [33] later showed that the forward marginals are compatible with an entire family of non-Markovian inference processes parametrized by a free schedule $\{\sigma_t\}$, which contains the stochastic DDPM sampler and a fully deterministic map—the probability flow ODE revisited in [Chapter 3](#).

2.1 Gaussian tools for diffusion

Every closed-form calculation in DDPM rests on three elementary Gaussian identities.

Lemma 2.1 (Gaussian score and affine closure).

(a) *Score*

If $p(x) = \mathcal{N}(x; \mu, \sigma^2 I)$ then $\nabla_x \log p(x) = -(x - \mu)/\sigma^2$.

(b) *Affine transport*

If $x \sim \mathcal{N}(\mu, \Sigma)$ and $y = Ax + b + \eta$, $\eta \sim \mathcal{N}(0, \Lambda)$ independent of x , then $y \sim \mathcal{N}(A\mu + b, A\Sigma A^\top + \Lambda)$.

(c) *Gaussian product*

$\mathcal{N}(x; \mu_1, \Sigma_1) \cdot \mathcal{N}(x; \mu_2, \Sigma_2) \propto \mathcal{N}(x; \mu, \Sigma)$ where

$$\Sigma^{-1} = \Sigma_1^{-1} + \Sigma_2^{-1}, \quad \mu = \Sigma(\Sigma_1^{-1}\mu_1 + \Sigma_2^{-1}\mu_2). \quad (2.1)$$

Proof. (a) Differentiate $-\|x - \mu\|^2/(2\sigma^2)$.

(b) Linearity of expectation and independence give the mean and covariance; Gaussianity follows from the characteristic function.

(c) Expand $-\frac{1}{2}(x - \mu_1)^\top \Sigma_1^{-1}(x - \mu_1) - \frac{1}{2}(x - \mu_2)^\top \Sigma_2^{-1}(x - \mu_2)$ and complete the square using the precision sum $\Sigma^{-1} = \Sigma_1^{-1} + \Sigma_2^{-1}$; all remaining terms are independent of x , see [Appendix A.1](#) for the complete proof.

□

2.1.1 Tweedie's identity

Let $\tilde{X} = \alpha X_0 + \sigma \epsilon$, $\epsilon \sim \mathcal{N}(0, I)$, $X_0 \sim p_{\text{data}}$, $\alpha, \sigma > 0$. The marginal $p_{\alpha, \sigma}(\tilde{x}) := \int \mathcal{N}(\tilde{x}; \alpha x_0, \sigma^2 I) p_{\text{data}}(x_0) dx_0$ is C^∞ and strictly positive.

Theorem 2.2 (Tweedie's identity).

$$\mathbb{E}[X_0 | \tilde{X} = \tilde{x}] = \frac{\tilde{x} + \sigma^2 \nabla_{\tilde{x}} \log p_{\alpha, \sigma}(\tilde{x})}{\alpha}. \quad (2.2)$$

Setting $\hat{\epsilon} := \mathbb{E}[\epsilon | \tilde{X} = \tilde{x}]$, the score, optimal x_0 -denoiser, and optimal ϵ -denoiser satisfy

$$\nabla_{\tilde{x}} \log p_{\alpha, \sigma}(\tilde{x}) = \frac{\alpha \mathbb{E}[X_0 | \tilde{X}] - \tilde{x}}{\sigma^2} = -\frac{\hat{\epsilon}}{\sigma}. \quad (2.3)$$

Proof. Differentiate $\log p_{\alpha, \sigma}(\tilde{x})$ under the integral (dominated convergence) and apply [Lemma 2.1\(a\)](#) to each kernel:

$$\nabla_{\tilde{x}} p_{\alpha, \sigma}(\tilde{x}) = \int \mathcal{N}(\tilde{x}; \alpha x_0, \sigma^2 I) \left(-\frac{\tilde{x} - \alpha x_0}{\sigma^2} \right) p_{\text{data}}(x_0) dx_0. \quad (2.4)$$

Dividing by $p_{\alpha, \sigma}(\tilde{x})$ and recognising the posterior $p(x_0 | \tilde{x}) = \mathcal{N}(\tilde{x}; \alpha x_0, \sigma^2 I) p_{\text{data}}(x_0) / p_{\alpha, \sigma}(\tilde{x})$ by Bayes' theorem:

$$\nabla_{\tilde{x}} \log p_{\alpha, \sigma}(\tilde{x}) = -\frac{\tilde{x}}{\sigma^2} \underbrace{\int p(x_0 | \tilde{x}) dx_0}_{=1} + \frac{\alpha}{\sigma^2} \underbrace{\int x_0 p(x_0 | \tilde{x}) dx_0}_{=\mathbb{E}[X_0 | \tilde{X} = \tilde{x}]} \quad (2.5)$$

Rearranging gives [Equation \(2.2\)](#). Substituting $\tilde{x} = \alpha X_0 + \sigma \epsilon$ and taking conditional expectation yields the $\hat{\epsilon}$ -form in [Equation \(2.3\)](#). $\square$

Key Idea

Every diffusion model learns to estimate one member of the trio ($\nabla \log p_t$, $\mathbb{E}[X_0 | \tilde{X}]$, $\mathbb{E}[\epsilon | \tilde{X}]$), which are in bijection via [Equation \(2.3\)](#). Every training objective (ϵ -, x_0 -, v -, score-prediction) and every sampler is a different way to parameterize or discretize this trio.

2.2 Forward diffusion chain

Fix a variance schedule $\{\beta_t\}_{t=1}^T \subset (0, 1)$; set $\alpha_t := 1 - \beta_t$ and $\bar{\alpha}_t := \prod_{s=1}^t \alpha_s$. The DDPM forward chain is

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t} x_{t-1}, \beta_t I). \quad (2.6)$$

Theorem 2.3 (Variance-preserving marginal). For every $t \geq 1$,

$$q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) I), \quad (2.7)$$

equivalently $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$, $\epsilon \sim \mathcal{N}(0, I)$. As $t \rightarrow T$ with $\bar{\alpha}_T \approx 0$, the marginal converges to $\mathcal{N}(0, I)$.

Proof. Induction using [Lemma 2.1\(b\)](#).

- *Base:* $q(x_1 | x_0) = \mathcal{N}(\sqrt{\alpha_1} x_0, \beta_1 I) = \mathcal{N}(\sqrt{\bar{\alpha}_1} x_0, (1 - \bar{\alpha}_1) I)$.
- *Step:* assume [Equation \(2.7\)](#) holds at t . Write $x_{t+1} = \sqrt{\alpha_{t+1}} x_t + \sqrt{\beta_{t+1}} \epsilon'$, $\epsilon' \perp x_t$. By [Lemma 2.1\(b\)](#),

$$q(x_{t+1} | x_0) = \mathcal{N}(\sqrt{\alpha_{t+1} \bar{\alpha}_t} x_0, (\alpha_{t+1}(1 - \bar{\alpha}_t) + \beta_{t+1}) I).$$

Variance coefficient: $\alpha_{t+1}(1 - \bar{\alpha}_t) + \beta_{t+1} = 1 - \alpha_{t+1}\bar{\alpha}_t = 1 - \bar{\alpha}_{t+1}$.

□

The label *variance-preserving* reflects $\text{Var}(x_t) = \bar{\alpha}_t \text{Var}(x_0) + (1 - \bar{\alpha}_t) = \bar{\alpha}_t + (1 - \bar{\alpha}_t) = 1$: for unit-variance data, the total variance of x_t is 1 at every t .

2.2.1 Noise schedules

The schedule governs how quickly $\text{SNR}(t) := \bar{\alpha}_t/(1 - \bar{\alpha}_t)$ decays from $\text{SNR}(0) = \infty$ to $\text{SNR}(T) \approx 0$.

Linear schedule [16]. $\beta_t = \beta_{\min} + \frac{t-1}{T-1}(\beta_{\max} - \beta_{\min})$ with $\beta_{\min} = 10^{-4}$, $\beta_{\max} = 0.02$, $T = 1000$. Simple and effective for images, though $\bar{\alpha}_T \approx 0.0002$ causes mild over-noising at the end of the chain.

Cosine schedule [29]. Directly specifies

$$\bar{\alpha}_t = \frac{f(t)}{f(0)}, \quad f(t) = \cos^2\left(\frac{\pi}{2} \cdot \frac{t/T + s}{1 + s}\right), \quad (2.8)$$

with offset $s = 0.008$ and $\beta_t = 1 - \bar{\alpha}_t/\bar{\alpha}_{t-1}$ clipped at 0.999. The smoother SNR trajectory improves sample quality on image tasks.

2.3 Denoising posterior

Although $q(x_{t-1} | x_t)$ is intractable, conditioning on x_0 makes the reverse transition Gaussian via Bayes' rule.

Proposition 2.4 (Denoising posterior). $q(x_{t-1} | x_t, x_0) = \mathcal{N}(\tilde{\mu}_t(x_t, x_0), \tilde{\beta}_t I)$ where

$$\tilde{\mu}_t(x_t, x_0) = \frac{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}} \beta_t}{1 - \bar{\alpha}_t} x_0, \quad (2.9)$$

$$= \frac{1}{\sqrt{\bar{\alpha}_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon \right), \quad (2.10)$$

$$\tilde{\beta}_t = \frac{(1 - \bar{\alpha}_{t-1}) \beta_t}{1 - \bar{\alpha}_t}, \quad (2.11)$$

where in Equation (2.10), ϵ is the realization in $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$.

Proof. By Bayes' rule, $q(x_{t-1} | x_t, x_0) \propto q(x_t | x_{t-1})q(x_{t-1} | x_0)$. Viewing the forward kernel as a Gaussian in x_{t-1} , $q(x_t | x_{t-1}) = \mathcal{N}(x_{t-1}; x_t/\sqrt{\bar{\alpha}_t}, (\beta_t/\alpha_t)I)$, and applying the Gaussian product formula Equation (2.1) with $q(x_{t-1} | x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_{t-1}}x_0, (1 - \bar{\alpha}_{t-1})I)$:

$$\tilde{\beta}_t^{-1} = \frac{\alpha_t}{\beta_t} + \frac{1}{1 - \bar{\alpha}_{t-1}} = \frac{\alpha_t(1 - \bar{\alpha}_{t-1}) + \beta_t}{\beta_t(1 - \bar{\alpha}_{t-1})} = \frac{1 - \bar{\alpha}_t}{\beta_t(1 - \bar{\alpha}_{t-1})},$$

confirming Equation (2.11). The mean Equation (2.9) follows from the precision-weighted formula Equation (2.1); substituting $x_0 = (x_t - \sqrt{1 - \bar{\alpha}_t}\epsilon)/\sqrt{\bar{\alpha}_t}$ yields Equation (2.10). □

Remark 2.5 (Two canonical variance choices). The posterior variance $\tilde{\beta}_t$ is the MMSE lower bound on the reverse-step variance; the upper bound is β_t . Ho et al. [16] used $\tilde{\beta}_t$; Nichol and Dhariwal [29] learned a scalar interpolation between both.

2.4 Reverse model and ELBO

Define the reverse model $p_\theta(x_T) = \mathcal{N}(0, I)$ and $p_\theta(x_{t-1} | x_t) = \mathcal{N}(\mu_\theta(x_t, t), \tilde{\beta}_t I)$.

ELBO via Jensen's inequality. By Jensen's inequality,

$$\log p_\theta(x_0) = \log \int p_\theta(x_{0:T}) dx_{1:T} \geq \mathbb{E}_{q(x_{1:T}|x_0)} \left[\log \frac{p_\theta(x_{0:T})}{q(x_{1:T}|x_0)} \right] =: \mathcal{L}(x_0; \theta). \quad (2.12)$$

Three-term decomposition. Write $\log q(x_{1:T}|x_0) = \sum_{t=1}^T \log q(x_t|x_{t-1})$. For $t \geq 2$, apply Bayes' rule in the forward direction, $q(x_t|x_{t-1}) = q(x_{t-1}|x_t, x_0) q(x_t|x_0)/q(x_{t-1}|x_0)$, and telescope the ratio $q(x_t|x_0)/q(x_{t-1}|x_0)$ over $t = 2, \dots, T$:

$$\sum_{t=1}^T \log q(x_t|x_{t-1}) = \sum_{t=2}^T \log q(x_{t-1}|x_t, x_0) + \log q(x_T|x_0).$$

Substituting into Equation (2.12) and grouping:

$$\begin{aligned} \mathcal{L}(x_0; \theta) &= \underbrace{-\text{KL}(q(x_T|x_0) \parallel p(x_T))}_{\approx 0 \text{ by schedule design}} + \underbrace{\mathbb{E}_q[\log p_\theta(x_0|x_1)]}_{\mathcal{L}_0} \\ &\quad - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(x_t|x_0)}[\text{KL}(q(x_{t-1}|x_t, x_0) \parallel p_\theta(x_{t-1}|x_t))]}_{\mathcal{L}_{t-1}}. \end{aligned} \quad (2.13)$$

Closed-form KL. Both Gaussians in each $\mathcal{L}_{t-1}$ share the same variance $\tilde{\beta}_t I$, so

$$\text{KL}(\mathcal{N}(\mu_1, \sigma^2 I) \parallel \mathcal{N}(\mu_2, \sigma^2 I)) = \frac{\|\mu_1 - \mu_2\|^2}{2\sigma^2}, \quad (2.14)$$

giving $\mathcal{L}_{t-1} = \mathbb{E}_{q(x_t|x_0)}[\|\tilde{\mu}_t - \mu_\theta\|^2]/(2\tilde{\beta}_t)$.

Noise-prediction parameterization. Match the structure of Equation (2.10):

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right). \quad (2.15)$$

Then $\|\tilde{\mu}_t - \mu_\theta\|^2 = \frac{\beta_t^2}{\alpha_t(1-\bar{\alpha}_t)} \|\epsilon - \epsilon_\theta\|^2$, so $\mathcal{L}_{t-1} = \frac{\beta_t^2}{2\alpha_t(1-\bar{\alpha}_t)\tilde{\beta}_t} \mathbb{E}[\|\epsilon - \epsilon_\theta(x_t, t)\|^2]$. Ho et al. found that dropping this time-dependent weight improves sample quality:

$$\mathcal{L}_{\text{simple}}(\theta) = \mathbb{E}_{t, x_0, \epsilon} [\|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t)\|^2]. \quad (2.16)$$

2.5 Parameterizations, SNR, and v -prediction

2.5.1 Three equivalent prediction targets

Under $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$, define the *signal-to-noise ratio* $\text{SNR}(t) := \bar{\alpha}_t/(1 - \bar{\alpha}_t)$. Tweedie Equation (2.3) gives a bijection among three prediction targets:

$$\nabla \log p_t(x_t) = -\frac{\epsilon_\theta(x_t, t)}{\sqrt{1 - \bar{\alpha}_t}} = \frac{\sqrt{\bar{\alpha}_t} x_{0,\theta}(x_t, t) - x_t}{1 - \bar{\alpha}_t}, \quad (2.17)$$

with conversion identities

$$\hat{x}_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \hat{\epsilon}}{\sqrt{\bar{\alpha}_t}}, \quad \hat{\epsilon} = \frac{x_t - \sqrt{\bar{\alpha}_t} \hat{x}_0}{\sqrt{1 - \bar{\alpha}_t}}. \quad (2.18)$$

2.5.2 SNR equivalence of the denoising ELBO

Theorem 2.6 (SNR equivalence). *Each denoising term in the ELBO satisfies*

$$\mathcal{L}_{t-1}(\theta) = \frac{1}{2}(\text{SNR}(t-1) - \text{SNR}(t)) \mathbb{E}_{q(x_t|x_0)}[\|\hat{x}_{0,\theta}(x_t) - x_0\|^2]. \quad (2.19)$$

Proof. Express $\mathcal{L}_{t-1}$ via the x_0 -prediction parameterization. From Equation (2.9), the mean difference is

$$\tilde{\mu}_t - \mu_\theta = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}(x_0 - \hat{x}_{0,\theta}).$$

Substituting into Equation (2.14) with $\sigma^2 = \tilde{\beta}_t$:

$$\mathcal{L}_{t-1} = \frac{\bar{\alpha}_{t-1}\beta_t^2}{2(1 - \bar{\alpha}_t)^2\tilde{\beta}_t} \mathbb{E}[\|\hat{x}_{0,\theta} - x_0\|^2] = \frac{\bar{\alpha}_{t-1}\beta_t}{2(1 - \bar{\alpha}_t)(1 - \bar{\alpha}_{t-1})} \mathbb{E}[\|\hat{x}_{0,\theta} - x_0\|^2],$$

using $\tilde{\beta}_t = (1 - \bar{\alpha}_{t-1})\beta_t/(1 - \bar{\alpha}_t)$ in the last step. It remains to show $\text{SNR}(t-1) - \text{SNR}(t) = \bar{\alpha}_{t-1}\beta_t/[(1 - \bar{\alpha}_{t-1})(1 - \bar{\alpha}_t)]$. Using $\bar{\alpha}_t = \alpha_t\bar{\alpha}_{t-1}$:

$$\begin{aligned} \text{SNR}(t-1) - \text{SNR}(t) &= \frac{\bar{\alpha}_{t-1}}{1 - \bar{\alpha}_{t-1}} - \frac{\alpha_t\bar{\alpha}_{t-1}}{1 - \alpha_t\bar{\alpha}_{t-1}} = \bar{\alpha}_{t-1} \cdot \frac{(1 - \alpha_t\bar{\alpha}_{t-1}) - \alpha_t(1 - \bar{\alpha}_{t-1})}{(1 - \bar{\alpha}_{t-1})(1 - \bar{\alpha}_t)} \\ &= \bar{\alpha}_{t-1} \cdot \frac{1 - \alpha_t}{(1 - \bar{\alpha}_{t-1})(1 - \bar{\alpha}_t)} = \frac{\bar{\alpha}_{t-1}\beta_t}{(1 - \bar{\alpha}_{t-1})(1 - \bar{\alpha}_t)}. \quad \square \end{aligned}$$

Key Idea

The denoising ELBO is a time-weighted x_0 -prediction MSE. The weight $\text{SNR}(t-1) - \text{SNR}(t)$ equals the local SNR drop: steps near the data (small t , high SNR) receive large weight; steps near pure noise (large t) receive small weight. $\mathcal{L}_{\text{simple}}$ uniformly upweights the noisy end.

2.5.3 v -prediction and the rotation structure

Proposition 2.7 (v -parameterization [32]). *Define the velocity target*

$$v_t := \sqrt{\bar{\alpha}_t}\epsilon - \sqrt{1 - \bar{\alpha}_t}x_0. \quad (2.20)$$

Then (x_t, v_t) and (x_0, ϵ) are related by the orthogonal rotation

$$\begin{pmatrix} x_t \\ v_t \end{pmatrix} = \underbrace{\begin{pmatrix} \sqrt{\bar{\alpha}_t} & \sqrt{1 - \bar{\alpha}_t} \\ -\sqrt{1 - \bar{\alpha}_t} & \sqrt{\bar{\alpha}_t} \end{pmatrix}}_{R_t \in O(2)} \begin{pmatrix} x_0 \\ \epsilon \end{pmatrix}, \quad (2.21)$$

with inverse $R_t^{-1} = R_t^\top$ giving

$$x_0 = \sqrt{\bar{\alpha}_t} x_t - \sqrt{1 - \bar{\alpha}_t} v_t, \quad \epsilon = \sqrt{1 - \bar{\alpha}_t} x_t + \sqrt{\bar{\alpha}_t} v_t. \quad (2.22)$$

If $v_\theta \approx v_t$ is a model prediction and $\hat{x}_0, \hat{\epsilon}$ are recovered via Equation (2.22), then

$$\|x_0 - \hat{x}_0\|^2 = (1 - \bar{\alpha}_t) \|v_t - v_\theta\|^2, \quad \|\epsilon - \hat{\epsilon}\|^2 = \bar{\alpha}_t \|v_t - v_\theta\|^2. \quad (2.23)$$

Proof. Orthogonality: $R_t^\top R_t = (\bar{\alpha}_t + (1 - \bar{\alpha}_t))I_2 = I_2$, so $R_t \in O(2)$. *Inverse:* $R_t^{-1} = R_t^\top$ gives Equation (2.22) directly. *Error decomposition:* from Equation (2.22), $x_0 - \hat{x}_0 = -\sqrt{1 - \bar{\alpha}_t}(v_t - v_\theta)$ and $\epsilon - \hat{\epsilon} = \sqrt{\bar{\alpha}_t}(v_t - v_\theta)$; squaring yields Equation (2.23). $\square$

Remark 2.8 (v -prediction as SNR-adaptive conditioning). By Equation (2.23), training on $\|v_t - v_\theta\|^2$ is equivalent to training on x_0 - and ϵ -prediction with SNR-dependent weights $(1 - \bar{\alpha}_t)$ and $\bar{\alpha}_t$ respectively. At high SNR ($\bar{\alpha}_t \approx 1$), x_0 -errors are suppressed and v -loss tracks ϵ -loss; at low SNR ($\bar{\alpha}_t \approx 0$), the roles reverse. This smooth interpolation avoids the numerical instabilities of either pure ϵ - or pure x_0 -prediction at extreme noise levels.

2.6 DDIM: generalized non-Markovian samplers

The DDPM derivation used only the marginals $q(x_t | x_0)$, not the full Markov structure. Song et al. [33] asked: which non-Markovian inference processes share these marginals, and what reverse samplers do they imply?

2.6.1 Underdetermined system and the generalized update

Fix consecutive steps $s < t$ and seek a Gaussian reverse kernel

$$q_\sigma(x_s | x_t, x_0) = \mathcal{N}(\kappa x_0 + \lambda x_t, \sigma^2 I) \quad (2.24)$$

for unknowns $(\kappa, \lambda, \sigma)$. Requiring consistency with the forward marginal, i.e., $q_\sigma(x_s | x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_s} x_0, (1 - \bar{\alpha}_s)I)$, yields (using $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \eta$, $\eta \sim \mathcal{N}(0, I)$):

$$\kappa + \lambda \sqrt{\bar{\alpha}_t} = \sqrt{\bar{\alpha}_s}, \quad (2.25)$$

$$\lambda^2(1 - \bar{\alpha}_t) + \sigma^2 = 1 - \bar{\alpha}_s. \quad (2.26)$$

Two equations, three unknowns: σ is a free parameter. Solving:

$$\lambda = \sqrt{\frac{1 - \bar{\alpha}_s - \sigma^2}{1 - \bar{\alpha}_t}}, \quad \kappa = \sqrt{\bar{\alpha}_s} - \lambda \sqrt{\bar{\alpha}_t}. \quad (2.27)$$

Replacing x_0 with the model prediction $\hat{x}_0 := (x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(x_t, t)) / \sqrt{\bar{\alpha}_t}$ in Equation (2.24) gives the DDIM update:

$$x_s = \sqrt{\bar{\alpha}_s} \hat{x}_0 + \sqrt{1 - \bar{\alpha}_s - \sigma^2} \epsilon_\theta(x_t, t) + \sigma z, \quad z \sim \mathcal{N}(0, I). \quad (2.28)$$

Proposition 2.9 (Special cases). (i) Deterministic DDIM ($\sigma = 0$): $x_s = \sqrt{\bar{\alpha}_s} \hat{x}_0 + \sqrt{1 - \bar{\alpha}_s} \epsilon_\theta(x_t, t)$, a deterministic map $x_t \mapsto x_s$ for any $s < t$.

(ii) DDPM stochastic sampler ($\sigma = \sqrt{\tilde{\beta}_t}, s = t-1$): the update Equation (2.28) reduces to sampling from the denoising posterior $q(x_{t-1} | x_t, x_0 = \hat{x}_0)$, recovering the DDPM reverse step exactly.

Proof. (i) Set $\sigma = 0$ in Equation (2.27): $\lambda = \sqrt{(1 - \bar{\alpha}_s)/(1 - \bar{\alpha}_t)}$, $\kappa = \sqrt{\bar{\alpha}_s} - \lambda \sqrt{\bar{\alpha}_t}$; Equation (2.28) simplifies directly.

(ii) With $\sigma^2 = \tilde{\beta}_t = (1 - \bar{\alpha}_{t-1})\beta_t/(1 - \bar{\alpha}_t)$ and $s = t-1$:

$$1 - \bar{\alpha}_{t-1} - \sigma^2 = (1 - \bar{\alpha}_{t-1}) \left(1 - \frac{\beta_t}{1 - \bar{\alpha}_t} \right) = \frac{\alpha_t(1 - \bar{\alpha}_{t-1})^2}{1 - \bar{\alpha}_t},$$

so $\lambda = \sqrt{\alpha_t(1 - \bar{\alpha}_{t-1})/(1 - \bar{\alpha}_t)}$ and $\kappa = \sqrt{\bar{\alpha}_{t-1}}\beta_t/(1 - \bar{\alpha}_t)$. These match the coefficients of x_t and x_0 in $\tilde{\mu}_t$ from Equation (2.9), and the stochastic term has variance $\tilde{\beta}_t I$, recovering $q(x_{t-1} | x_t, x_0)$. $\square$

2.6.2 Sub-sequence sampling and the probability flow ODE

Since Equation (2.28) depends only on $\bar{\alpha}_s$ and $\bar{\alpha}_t$, it is valid for any sub-sequence $\tau_1 < \dots < \tau_S \subset \{1, \dots, T\}$: a DDPM-trained model can generate samples with only $S \ll T$ network evaluations without retraining.

In the deterministic case $\sigma = 0$, refining the sub-sequence to $S \rightarrow \infty$ yields the *probability flow ODE*

$$\frac{dx_t}{dt} = \frac{1}{2} \frac{d \log \bar{\alpha}_t}{dt} \left(x_t + \frac{1 - \bar{\alpha}_t}{\bar{\alpha}_t} \cdot \frac{-\epsilon_\theta(x_t, t)}{\sqrt{(1 - \bar{\alpha}_t)/\bar{\alpha}_t}} \right), \quad (2.29)$$

whose derivation as the zero-diffusion limit of the reverse SDE is developed in Chapter 3. The ODE formulation enables exact likelihood evaluation via the instantaneous change-of-variables formula.

Historical Note

DDPM [16] popularized diffusion-based generation; DDIM [33] revealed the ODE structure and enabled fast sampling. The SNR reformulation (Theorem 2.6) was systematized by Luo [27] and Chan [6]. The v -parameterization [32] was later adopted as the default in several large-scale text-to-image systems.

3. Continuous-Time Score-Based Models

Song and Ermon [36] introduced score-based generative modeling: instead of learning a likelihood, train a neural network to estimate the *score* $\nabla_x \log p(x)$ of the data distribution, then run Langevin dynamics to draw samples. A critical obstacle—score estimation fails in low-density regions—was overcome by training a single network conditioned on multiple noise scales, yielding the *Noise Conditional Score Network* (NCSN). Song et al. [39] later lifted both DDPM and NCSN into a unified continuous-time SDE framework: a forward SDE gradually destroys data, and its analytically derived reverse-time SDE generates samples using only the time-dependent score function.

3.1 Itô calculus prerequisites

All continuous-time diffusion models rest on two fundamental results from stochastic calculus: the Itô change-of-variables formula and the Fokker–Planck equation.

Definition 3.1 (Standard Brownian motion). A stochastic process $\{W_t\}_{t \geq 0}$ on $\mathbb{R}^d$ is a *standard Brownian motion* if: $W_0 = 0$ a.s.; increments $W_t - W_s \sim \mathcal{N}(0, (t-s)I)$ for all $0 \leq s < t$; increments over disjoint intervals are independent; paths are continuous a.s.

Definition 3.2 (Itô SDE). An *Itô SDE* with drift $f : \mathbb{R}^d \times [0, T] \rightarrow \mathbb{R}^d$ and scalar diffusion coefficient $g : [0, T] \rightarrow \mathbb{R}$ is

$$dX_t = f(X_t, t) dt + g(t) dW_t, \quad (3.1)$$

understood as the integral equation $X_t = X_0 + \int_0^t f(X_s, s) ds + \int_0^t g(s) dW_s$. The stochastic integral $\int g dW$ is defined as the L^2 -limit of left-endpoint Riemann sums, inheriting the non-anticipativity ($g(s)$ depends only on $\{W_r : r \leq s\}$).

Theorem 3.3 (Itô’s formula). For $\phi \in C^{1,2}(\mathbb{R}^d \times [0, T])$ and X_t satisfying Equation (3.1):

$$d\phi(X_t, t) = \left(\partial_t \phi + \langle \nabla_x \phi, f \rangle + \frac{g^2}{2} \Delta_x \phi \right) dt + \langle \nabla_x \phi, g dW_t \rangle. \quad (3.2)$$

The correction term $\frac{g^2}{2} \Delta_x \phi dt$ has no deterministic analogue; it arises from the quadratic variation $(dW_t)^2 = I dt$.

A complete proof via Taylor expansion and quadratic variation is in [Appendix A.2](#).

Theorem 3.4 (Fokker–Planck equation). If X_t satisfies Equation (3.1) with initial density p_0 , the

marginal density $p_t(x) = \frac{d}{dx} \Pr(X_t \leq x)$ satisfies the PDE

$$\partial_t p_t = -\nabla \cdot (f p_t) + \frac{g^2}{2} \Delta p_t. \quad (3.3)$$

A self-contained proof with full justification of each step is in [Appendix A.2](#).

Remark 3.5 (Physical interpretation). The two terms in [Equation \(3.3\)](#) have complementary roles: $-\nabla \cdot (f p_t)$ is advection (deterministic drift moves probability mass), and $\frac{g^2}{2} \Delta p_t$ is diffusion (the Brownian term spreads mass).

3.2 Score function and Langevin dynamics

Definition 3.6 (Stein score function). For a differentiable density $p(x)$ on $\mathbb{R}^d$, the (Stein) score function is

$$s(x) := \nabla_x \log p(x). \quad (3.4)$$

It is a vector field pointing in the direction of steepest increase of $\log p$. For a Gaussian $p(x) = \mathcal{N}(x; \mu, \sigma^2 I)$, the score is $(\mu - x)/\sigma^2$, which is the signed displacement toward the mean scaled by σ^{-2} .

The score is computable without the normalizing constant: if $p(x) = Z^{-1} e^{-E(x)}$ for energy E and partition function Z , then $s(x) = -\nabla E(x)$, independent of Z . This makes score learning attractive for unnormalized energy-based models.

From score to samples: Langevin dynamics. Given access to $s(\cdot)$, one can sample from p by iterating

$$x_{k+1} = x_k + \tau s(x_k) + \sqrt{2\tau} z_k, \quad z_k \sim \mathcal{N}(0, I), \quad (3.5)$$

for a small step size $\tau > 0$, starting from any x_0 . The deterministic part $x + \tau \nabla \log p(x)$ is gradient ascent on $\log p$, pushing the iterate toward high-probability regions. The noise term $\sqrt{2\tau} z$ prevents collapse to a mode and ensures the chain explores the full distribution.

Theorem 3.7 (Langevin dynamics has stationary distribution p). *The continuous-time Langevin SDE*

$$dX_t = \nabla_x \log p(X_t) dt + \sqrt{2} dW_t \quad (3.6)$$

has p as its unique stationary density (under standard regularity on p).

Proof. Apply [Theorem 3.4](#) with $f = \nabla \log p$ and $g = \sqrt{2}$: the right-hand side at $p_t = p$ is

$$-\nabla \cdot (p \nabla \log p) + \Delta p = -\nabla \cdot (\nabla p) + \Delta p = -\Delta p + \Delta p = 0,$$

using $p \nabla \log p = \nabla p$. Hence $\partial_t p_t|_{p_t=p} = 0$, confirming p is stationary. $\square$

A complete proof covering existence, detailed balance, and uniqueness is in [Appendix A.3](#).

Key Idea

Learning the score $s_\theta(x) \approx \nabla_x \log p(x)$ gives a *generator* (plug into Langevin dynamics) and a *probabilistic model* (the score encodes all of p up to a constant). The central challenge of this chapter is how to learn s_θ from data.

3.3 Score matching

The ideal training objective would directly compare s_θ with $\nabla \log p$. But $\nabla \log p$ is unknown. Three techniques circumvent this: explicit score matching, implicit score matching, and denoising score matching—each reaching the same optimum.

Explicit score matching (ESM).

Definition 3.8 (ESM loss).

$$J_{\text{ESM}}(\theta) := \frac{1}{2} \mathbb{E}_{x \sim p} [\|s_\theta(x) - \nabla_x \log p(x)\|^2]. \quad (3.7)$$

$J_{\text{ESM}}(\theta) \geq 0$ with equality iff $s_\theta = \nabla \log p$ a.e. under p . The difficulty: computing $\nabla \log p$ requires the true density, not the dataset.

Implicit score matching (ISM). Expanding the square in Equation (3.7) and integrating by parts on the cross term $\mathbb{E}[\langle s_\theta, \nabla \log p \rangle] = \int s_\theta(x) \cdot \nabla p(x) \, dx = - \int (\nabla \cdot s_\theta(x)) p(x) \, dx = -\mathbb{E}[\nabla \cdot s_\theta]$ (assuming $p(x) \rightarrow 0$ fast enough at infinity) yields the following.

Theorem 3.9 (ISM = ESM up to a constant, Hyvärinen 19).

$$J_{\text{ISM}}(\theta) := \mathbb{E}[\nabla \cdot s_\theta(x) + \frac{1}{2} \|s_\theta(x)\|^2] = J_{\text{ESM}}(\theta) + C, \quad (3.8)$$

where $C = \frac{1}{2} \mathbb{E}[\|\nabla \log p\|^2]$ is independent of θ .

Proof. Expand $J_{\text{ESM}} = \frac{1}{2} \mathbb{E}[\|s_\theta\|^2] - \mathbb{E}[\langle s_\theta, \nabla \log p \rangle] + \frac{1}{2} \mathbb{E}[\|\nabla \log p\|^2]$. The cross term equals $-\mathbb{E}[\nabla \cdot s_\theta]$ by the integration-by-parts identity above. Thus $J_{\text{ESM}} = \mathbb{E}[\nabla \cdot s_\theta + \frac{1}{2} \|s_\theta\|^2] + C$. $\square$

J_{ISM} requires only samples from p (no density evaluation), but computing $\nabla \cdot s_\theta$ costs d backward passes for a d -dimensional score network, which is prohibitive in high dimensions.

Denoising score matching (DSM). The practical alternative replaces the intractable $\nabla \log p$ with the tractable conditional score under Gaussian corruption $\tilde{x} = x_0 + \sigma\epsilon$, $\epsilon \sim \mathcal{N}(0, I)$, $x_0 \sim p$.

Theorem 3.10 (DSM = ESM up to a constant, Vincent 42). Define the ESM and DSM losses at noise level σ :

$$J_{\text{ESM}}^\sigma(\theta) := \frac{1}{2} \mathbb{E}_{\tilde{x} \sim p_\sigma} [\|s_\theta(\tilde{x}) - \nabla_{\tilde{x}} \log p_\sigma(\tilde{x})\|^2], \quad (3.9)$$

$$J_{\text{DSM}}^\sigma(\theta) := \frac{1}{2} \mathbb{E}_{x_0 \sim p, \epsilon \sim \mathcal{N}(0, I)} [\|s_\theta(x_0 + \sigma\epsilon) + \epsilon/\sigma\|^2], \quad (3.10)$$

where $p_\sigma(\tilde{x}) = \int \mathcal{N}(\tilde{x}; x_0, \sigma^2 I) p(x_0) \, dx_0$ is the noise-convolved density and $\nabla_{\tilde{x}} \log p(\tilde{x}|x_0) = -\epsilon/\sigma$. Then

$$J_{\text{ESM}}^\sigma(\theta) = J_{\text{DSM}}^\sigma(\theta) + C, \quad (3.11)$$

with $C = \frac{1}{2} \mathbb{E}[\|\nabla \log p_\sigma\|^2] - \frac{d}{2\sigma^2}$ independent of θ .

Proof sketch. Expand J_{ESM}^σ via $\|a - b\|^2$ into terms (I) $\frac{1}{2} \mathbb{E}[\|s_\theta\|^2]$, (II) $-\mathbb{E}[\langle s_\theta, \nabla \log p_\sigma \rangle]$, and (III) $\frac{1}{2} \mathbb{E}[\|\nabla \log p_\sigma\|^2]$. For the cross term, differentiating $p_\sigma = \int p(\tilde{x}|x_0) p(x_0) \, dx_0$ gives $\nabla \log p_\sigma(\tilde{x}) = \mathbb{E}_{x_0|\tilde{x}}[\nabla \log p(\tilde{x}|x_0)]$, so (II) = $\mathbb{E}[\langle s_\theta, \epsilon/\sigma \rangle]$ after computing the Gaussian conditional score $\nabla \log p(\tilde{x}|x_0) =$

$-\epsilon/\sigma$. Expanding J_{DSM}^σ via $\|a + b\|^2$ yields the same (I) and (II) plus the constant $\frac{1}{2}\mathbb{E}[\|\epsilon/\sigma\|^2] = d/(2\sigma^2)$. The θ -dependent parts of both objectives are therefore identical. A complete proof is in [Appendix A.4](#). $\square$

Remark 3.11 (DSM target and the Tweedie trio). The DSM target $\nabla_{\tilde{x}} \log p(\tilde{x}|x_0) = -\epsilon/\sigma$ is the conditional score under Gaussian noise, identical to $-\hat{\epsilon}/\sigma$ in [Equation \(2.3\)](#). Hence minimizing J_{DSM} is equivalent to learning the denoising trio established in [Chapter 2](#): the score, the optimal denoiser $\mathbb{E}[x_0|\tilde{x}]$, and the noise predictor $\mathbb{E}[\epsilon|\tilde{x}]$ are all in bijection.

3.4 Noise Conditional Score Networks

3.4.1 Three failure modes of vanilla score matching

Despite the elegance of [Theorem 3.10](#), training a score network with a single fixed σ fails in practice for three interconnected reasons identified by Song and Ermon [36].

Manifold degeneracy. High-dimensional data (images, audio) concentrates near a low-dimensional manifold $\mathcal{M} \subset \mathbb{R}^d$. At points $x \notin \mathcal{M}$, the density $p(x) = 0$, so $\nabla \log p(x)$ is undefined. Even near $\mathcal{M}$ but off it, p is so small that the score estimate has unbounded variance. A small σ perturbs data only slightly and leaves most of $\mathbb{R}^d$ unvisited during training.

Inaccuracy in low-density regions. Langevin dynamics must traverse low-density regions between modes. If s_θ is inaccurate there (few training samples, high noise-to-signal ratio), the chain gets stuck between modes or mixes exponentially slowly.

Mode isolation. For a mixture distribution with well-separated modes, the chain initialized in one mode's basin has negligible probability of crossing to another mode, regardless of step size. Without a global "map" of the energy landscape, mixing fails.

Solution: multiscale noise. Training with a sufficiently large σ resolves all three: it smears the manifold, fills low-density gaps, and blurs between modes. The trade-off is that a large σ corrupts the data severely, biasing the generated samples. NCSN resolves this trade-off by using a *geometric sequence* of noise levels that transitions from coarse to fine.

3.4.2 NCSN training objective

Fix L noise levels $\sigma_1 > \sigma_2 > \dots > \sigma_L > 0$ arranged in a geometric sequence with ratio $r = \sigma_i/\sigma_{i+1} > 1$ (typical values: $\sigma_1 \sim 10$, $\sigma_L \sim 0.01$, $L = 10$).

Definition 3.12 (Noise-conditional DSM loss, NCSN). Train a single score network $s_\theta : \mathbb{R}^d \times \{\sigma_1, \dots, \sigma_L\} \rightarrow \mathbb{R}^d$ by minimizing

$$\mathcal{L}_{\text{NCSN}}(\theta) = \frac{1}{L} \sum_{i=1}^L \lambda(\sigma_i) \ell(\theta; \sigma_i), \quad (3.12)$$

where the single-scale loss is

$$\ell(\theta; \sigma) = \frac{1}{2} \mathbb{E}_{x_0 \sim p, \epsilon \sim \mathcal{N}(0, I)} \left[\left\| s_\theta(x_0 + \sigma\epsilon, \sigma) + \epsilon/\sigma \right\|^2 \right]. \quad (3.13)$$

Choice of weight $\lambda(\sigma) = \sigma^2$. The optimal score at scale σ satisfies $s^*(x, \sigma) \approx -\epsilon/\sigma$, so $\|s^*(x, \sigma)\|^2 = O(\sigma^{-2})$. Without weighting, the loss at small σ would dominate and drive training. Setting $\lambda(\sigma) = \sigma^2$ makes every term in Equation (3.12) dimensionally $O(1)$, balancing gradient signals across all scales.

Proposition 3.13 (Optimal noise-conditional score). *The minimizer of $\ell(\theta; \sigma_i)$ over all measurable s_θ is*

$$s^*(x, \sigma_i) = \nabla_x \log p_{\sigma_i}(x) = \mathbb{E}[-\epsilon/\sigma_i \mid x_0 + \sigma_i \epsilon = x] = \frac{\mathbb{E}[x_0 \mid x_0 + \sigma_i \epsilon = x] - x}{\sigma_i^2}. \quad (3.14)$$

Proof. The last two equalities follow from Tweedie’s identity Equation (2.2) applied to $\tilde{x} = x$ with $\alpha = 1$: $\mathbb{E}[X_0 \mid \tilde{X} = x] = (x + \sigma_i^2 \nabla_x \log p_{\sigma_i}(x))$, rearranging gives $\nabla_x \log p_{\sigma_i}(x) = (\mathbb{E}[x_0 \mid x] - x)/\sigma_i^2$. That this is the minimizer of $\ell(\theta; \sigma_i)$ follows from Theorem 3.10 (DSM=ESM) and the fact that ESM is minimized by the true score. $\square$

3.4.3 Annealed Langevin dynamics

At inference, NCSN generates samples by running Langevin dynamics sequentially at decreasing noise levels. At the largest scale σ_1 , the perturbed distribution is nearly Gaussian, so Langevin mixes rapidly. As σ_i decreases, the chain refines toward the data manifold, using the previous iterate as a warm start.

Definition 3.14 (Annealed Langevin dynamics algorithm). Given step-size parameter $\varepsilon > 0$, number of Langevin steps K per scale, and scale-dependent step size $\alpha_i = \varepsilon \sigma_i^2 / \sigma_L^2$:

1. Sample $x \sim \mathcal{N}(0, \sigma_1^2 I)$.
2. For $i = 1, \dots, L$:
 - (a) For $k = 1, \dots, K$: $x \leftarrow x + \frac{\alpha_i}{2} s_\theta(x, \sigma_i) + \sqrt{\alpha_i} z$, $z \sim \mathcal{N}(0, I)$.
3. Return x .

The step size $\alpha_i \propto \sigma_i^2$ rescales the Langevin update to account for the score magnitude $\|s^*\| \sim \sigma_i^{-1}$, keeping the effective displacement $\frac{\alpha_i}{2} \|s^*\| \sim \frac{\varepsilon \sigma_i}{2 \sigma_L^2}$ comparable across scales.

Remark 3.15 (NCSN++, Song and Ermon 37). The improved NCSN adopts an *exponential* schedule $\sigma_i = \sigma_1(\sigma_L/\sigma_1)^{(i-1)/(L-1)}$ and trains with an EMA (exponential moving average) of θ . A larger architecture (U-Net with attention) and a more careful score parameterization push FID on CIFAR-10 below 3. The continuous-time limit of this schedule becomes the VE-SDE of Section 3.5.

Common Pitfall

The step size α_i must be small enough for Langevin to be in its diffusive regime (i.e., $\alpha_i \ll \sigma_i^2$), but large enough for K steps to mix at scale σ_i . Setting K too small at coarse scales causes the chain to carry artifacts into finer scales.

3.5 The SDE framework

Song et al. [39] unified DDPM and NCSN by lifting both to a continuous-time SDE. The discrete T -step DDPM chain and the L -level NCSN are different discretizations of two families of SDEs—VP-SDE and VE-SDE—whose reverse processes require only the score function.

3.5.1 From discrete chains to continuous SDEs

DDPM $\rightarrow$ VP-SDE. Recall the DDPM forward kernel $q(x_t|x_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}x_{t-1}, \beta_t I)$. Introduce a continuous index $t \in [0, 1]$ by letting $\beta_t = \beta(t)\Delta t$ as $T \rightarrow \infty, \Delta t = 1/T$. Then

$$x_{t+\Delta t} - x_t = (\sqrt{1-\beta(t)\Delta t} - 1)x_t + \sqrt{\beta(t)\Delta t}\epsilon \approx -\frac{\beta(t)}{2}x_t \Delta t + \sqrt{\beta(t)\Delta t}\epsilon. \quad (3.15)$$

Dividing by Δt and sending $\Delta t \rightarrow 0$ yields the *Variance-Preserving SDE* (VP-SDE):

$$dx = -\frac{\beta(t)}{2}x dt + \sqrt{\beta(t)} dW_t. \quad (3.16)$$

NCSN $\rightarrow$ VE-SDE. In NCSN, the perturbed sample is $x_t = x_0 + \sigma(t)\epsilon$ at noise level $\sigma(t)$. Between consecutive levels, $x_{t+\Delta t} - x_t \approx \sqrt{\sigma(t+\Delta t)^2 - \sigma(t)^2}\epsilon$. Sending $\Delta t \rightarrow 0$ yields the *Variance-Exploding SDE* (VE-SDE):

$$dx = \sqrt{\frac{d[\sigma^2(t)]}{dt}} dW_t. \quad (3.17)$$

3.5.2 Transition kernels and boundary conditions

Both SDEs have analytically tractable transition kernels $p(x_t|x_0)$, established by solving the corresponding Fokker–Planck equations.

Proposition 3.16 (Transition kernels for VP and VE SDEs). Let $\bar{\beta}(t) := \int_0^t \beta(s) ds$.

- (a) **VP-SDE:** $p(x_t|x_0) = \mathcal{N}(e^{-\bar{\beta}(t)/2}x_0, (1 - e^{-\bar{\beta}(t)})I)$. As $t \rightarrow T$ with $\bar{\beta}(T) \gg 1$, $p(x_T|x_0) \approx \mathcal{N}(0, I)$.
- (b) **VE-SDE:** $p(x_t|x_0) = \mathcal{N}(x_0, \sigma(t)^2 I)$. As $\sigma(T) \rightarrow \infty$, the marginal $p(x_T) \approx \mathcal{N}(0, \sigma(T)^2 I)$.

Proof. (a) The VP-SDE is a linear (Ornstein–Uhlenbeck) SDE. Setting $h(t) = e^{-\bar{\beta}(t)/2}$ and applying variation of parameters: $x_t = h(t)x_0 + \int_0^t h(t)/h(s) \sqrt{\beta(s)} dW_s$. The stochastic integral is Gaussian with mean zero and variance $\int_0^t e^{-(\bar{\beta}(t)-\bar{\beta}(s))} \beta(s) ds = 1 - e^{-\bar{\beta}(t)}$.

(b) The VE-SDE has zero drift; by Itô isometry, $x_t - x_0 = \int_0^t \sqrt{\dot{\sigma}^2(s)} dW_s \sim \mathcal{N}(0, \sigma(t)^2 I)$ since $\int_0^t \dot{\sigma}^2(s) ds = \sigma(t)^2 - \sigma(0)^2 \approx \sigma(t)^2$. $\square$

Sub-VP SDE. The sub-VP SDE uses the VP drift but a reduced diffusion coefficient:

$$dx = -\frac{\beta(t)}{2}x dt + \sqrt{\beta(t)(1 - e^{-2\bar{\beta}(t)})} dW_t. \quad (3.18)$$

Its transition kernel has the same mean as VP-SDE but a smaller variance, yielding tighter upper bounds on the score norm $\|\nabla_x \log p_t\|$ and more stable score matching at early times.

3.6 Reverse-time SDE and predictor–corrector samplers

3.6.1 Anderson’s reverse-time SDE

Theorem 3.17 (Reverse-time SDE, Anderson 2). Let $\{X_t\}_{t \in [0, T]}$ satisfy the forward SDE $dx = f(x, t) dt + g(t) dW_t$ with marginal densities $\{p_t\}$. Define the time-reversed process $\bar{X}_s := X_{T-s}$. Then $\bar{X}_s$ satisfies

$$d\bar{X}_s = \left[-f(\bar{X}_s, T-s) + g(T-s)^2 \nabla_x \log p_{T-s}(\bar{X}_s) \right] ds + g(T-s) d\bar{W}_s, \quad (3.19)$$

where $\bar{W}_s$ is a Brownian motion running forward in s . The reversed process has the same marginals: $\bar{X}_s \sim p_{T-s}$.

Proof sketch. Let $\bar{p}_s := p_{T-s}$ and $\bar{f} := -f + g^2 \nabla \log p_{T-s}$. Because $\partial_s p_{T-s} = -\partial_t p_t|_{t=T-s}$, the forward Fokker–Planck implies $\partial_s \bar{p}_s = \nabla \cdot (f p_{T-s}) - \frac{g^2}{2} \Delta p_{T-s}$. The Fokker–Planck of the candidate reverse SDE simplifies (using $p \nabla \log p = \nabla p$) to the same expression, confirming $\bar{p}_s = p_{T-s}$ for all s . A complete proof with discrete-time motivation and full Fokker–Planck verification is in [Appendix A.5](#). $\square$

Key Idea

The reverse drift $-f + g^2 \nabla \log p_{T-s}$ differs from the forward drift f by the score correction $g^2 \nabla \log p_t$. In all three SDE families (VP, VE, sub-VP), f and g are known analytically; the score $\nabla \log p_t$ is the *only unknown*. Replacing it with the learned network $s_\theta(x, t) \approx \nabla \log p_t(x)$ gives a practical generative sampler.

3.6.2 Continuous-time score matching loss

Combining [Proposition 3.16](#) and [Theorem 3.10](#), the score network is trained by minimizing the continuous-time DSM objective:

$$\mathcal{L}(\theta) = \mathbb{E}_{t \sim \mathcal{U}[0, T], x_0 \sim p, x_t \sim p(\cdot | x_0)} \left[\lambda(t) \|s_\theta(x_t, t) - \nabla_{x_t} \log p(x_t | x_0)\|^2 \right] + C \quad (3.20)$$

where C is θ -free and $\nabla_{x_t} \log p(x_t | x_0)$ has the tractable closed forms given in [Proposition 3.16](#). Under [Theorem 3.10](#), the minimizer of [Equation \(3.20\)](#) is $s^* = \nabla \log p_t$, which plugged into [Equation \(3.19\)](#) gives the exact reverse process.

The weight $\lambda(t)$ is chosen to equalize the loss scale across t : for the VP-SDE, $\lambda(t) = 1 - e^{-\bar{\beta}(t)}$ (matching L_{simple} from [Chapter 2](#)); for the VE-SDE, $\lambda(t) = \sigma(t)^2$ (matching the NCSN weight).

3.6.3 Predictor–corrector samplers

Numerical SDE solvers introduce discretization error: a single Euler–Maruyama step from T to 0 does not follow the exact marginals $\{p_t\}$. Predictor–corrector (PC) samplers correct this by interleaving two operations:

Definition 3.18 (PC sampler). At each time step $t \rightarrow t - \Delta t$:

1. *Predictor step*: advance the sample using a numerical solver for [Equation \(3.19\)](#), e.g. Euler–Maruyama $x_{t-\Delta t} \leftarrow x_t + [-f(x_t, t) + g(t)^2 s_\theta(x_t, t)] \Delta t + g(t) \sqrt{\Delta t} z$, $z \sim \mathcal{N}(0, I)$.
2. *Corrector step*: run M steps of Langevin dynamics at the *current* time $t - \Delta t$ $x \leftarrow x + \eta s_\theta(x, t - \Delta t) + \sqrt{2\eta} z$, targeting $p_{t-\Delta t}$.

The corrector acts as an MCMC refinement: even if the predictor lands off the correct marginal, the Langevin corrector—which has $p_{t-\Delta t}$ as its stationary distribution by [Theorem 3.7](#)—pulls the sample back. With $M \rightarrow \infty$ corrector steps, the chain achieves exact samples from $p_{t-\Delta t}$ at every time, and the predictor merely proposes the next time step. In practice, $M = 1$ or $M = 5$ already removes most discretization artifacts.

Remark 3.19 (Choice of predictor and corrector). The predictor can be any numerical SDE/ODE solver: Euler–Maruyama, reverse diffusion (exact DDPM update), or the probability flow ODE solver (introduced in [Chapter 4](#)). The corrector can be annealed Langevin or any MCMC

kernel that leaves p_t invariant. The probability flow ODE as predictor with Langevin corrector typically gives the best quality–speed trade-off.

Historical Note

Score matching was introduced by Hyvärinen [19] as a principled estimator for unnormalized models, and the connection to denoising was established by Vincent [42]. Song and Ermon [36] first combined multi-scale score networks with annealed Langevin sampling to produce competitive image samples without adversarial training. Their follow-up NCSN++ [37] matched GAN quality. Anderson’s 1982 result [2]—that every Itô diffusion has a reverse-time SDE—lay dormant for nearly four decades before Song et al. [39] recognized it as the theoretical foundation for score-based generation, unifying DDPM and NCSN into the single framework of this chapter. The probability flow ODE derivation, which provides a deterministic sampler with exact likelihoods, is taken up in [Chapter 4](#).

4. Probability Flow ODEs and Flow Matching

We develop the theory of deterministic generative modelling in two stages. First, we establish the *flow model* framework: a velocity field u_t determines a diffeomorphic flow ψ_t transporting p_0 to p_1 , and the *mass conservation* theorem characterizes this transport equivalently via the continuity equation. Second, we apply this framework to the SDE setting of [Chapter 3](#): setting $\sigma_t = 0$ in the Fokker–Planck equivalence class yields the *probability flow ODE*, whose deterministic trajectories enable exact likelihood computation via the *instantaneous change-of-variables* formula. Stepping back from the SDE scaffold, we develop *flow matching* (FM): a simulation-free training objective whose intractable marginal form is gradient-equivalent to a tractable conditional form by the *marginalization trick*. Closing the circle, affine conditional flows recover the Gaussian processes of DDPM and SMLD as special cases, and DDIM emerges as the Euler discretization of the OT–Gaussian probability flow ODE.

Common Pitfall

[title=Convention: X_0 = noise, X_1 = data in this chapter] Throughout this chapter, *the subscript 0 denotes noise and the subscript 1 denotes data*: $X_0 \sim \mathcal{N}(0, I)$ is the source noise distribution and $X_1 \sim p_{\text{data}}$ is the target. An interpolant takes the form $X_t = \alpha_t X_1 + \sigma_t X_0$. This is the reverse of the DDPM/SMLD convention ([Chapters 2 and 3](#)), where x_0 denotes the clean data sample and ϵ denotes noise. The dictionary is: $X_0^{\text{FM}} = \epsilon$, $X_1^{\text{FM}} = x_0^{\text{DDPM}}$.

4.1 Flow models and the continuity equation

All results in this chapter hinge on a single equivalence: a velocity field u_t transports p_0 to p_t *if and only if* p_t satisfies the *continuity equation*—a first-order PDE encoding conservation of probability mass under deterministic transport. We establish this equivalence here; every subsequent section applies it.

Definition 4.1 (Flow and velocity field). A C^r *flow* is a map $\psi : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ such that $\psi_t := \psi(t, \cdot)$ is a C^r diffeomorphism for every t , determined by the ODE

$$\frac{d}{dt}\psi_t(x) = u_t(\psi_t(x)), \quad \psi_0(x) = x, \tag{4.1}$$

for a *velocity field* $u : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$. A *flow model* is a continuous-time process $X_t = \psi_t(X_0)$ with $X_0 \sim p_0$.

By the Picard–Lindelöf theorem, if u_t is C^r with $r \geq 1$ and globally Lipschitz in x , a unique

C^r flow exists. The marginal density of the flow model at time t is the push-forward

$$p_t = [\psi_t]_{\#} p_0, \quad p_t(x) = p_0(\psi_t^{-1}(x)) |\det \partial_x \psi_t^{-1}(x)|.$$

Theorem 4.2 (Mass Conservation / Continuity Equation). *Let p_t be a probability path and u_t a locally Lipschitz, integrable velocity field. The following are equivalent:*

- (i) $\partial_t p_t + \nabla \cdot (p_t u_t) = 0$ (continuity equation) for $t \in [0, 1]$.
- (ii) u_t generates p_t : the flow model $X_t = \psi_t(X_0)$ satisfies $X_t \sim p_t$ for all t .

The proof, via the Dirac delta representation of push-forward densities (mirroring the Fokker–Planck derivation of [Chapter 3](#)), is given in [Appendix A.6](#).

Remark 4.3. [Theorem 4.2](#) is the ODE analogue of the Fokker–Planck equation ([Theorem 3.4](#)): it is precisely the $g_t = 0$ case of [Equation \(4.3\)](#). The probability flow ODE corresponds to the special choice $u_t = f_t - \frac{g_t^2}{2} \nabla \log p_t$; see [Section 4.2](#).

4.2 The probability flow ODE

4.2.1 An equivalence class of forward processes

Recall from [Chapter 3](#) the Itô forward SDE

$$dx = f_t(x) dt + g_t dW_t, \tag{4.2}$$

whose marginal densities $\{p_t\}$ satisfy the Fokker–Planck equation

$$\partial_t p_t = -\nabla \cdot (f_t p_t) + \frac{g_t^2}{2} \Delta p_t. \tag{4.3}$$

A natural question is whether one can alter the diffusion coefficient while keeping $\{p_t\}$ unchanged. Rewrite [Equation \(4.3\)](#) as

$$\partial_t p_t = -\nabla \cdot \left[\left(f_t - \frac{g_t^2 - \sigma_t^2}{2} \nabla \log p_t \right) p_t \right] + \frac{\sigma_t^2}{2} \Delta p_t,$$

which holds for any scalar function $\sigma_t^2 \leq g_t^2$ (the identity uses $\nabla \log p_t \cdot p_t = \nabla p_t$). This is the Fokker–Planck equation for the SDE

$$dx = \left(f_t(x) - \frac{g_t^2 - \sigma_t^2}{2} \nabla_x \log p_t(x) \right) dt + \sigma_t dW_t. \tag{4.4}$$

Hence *every* SDE of the form [Equation \(4.4\)](#) with $\sigma_t^2 \leq g_t^2$ shares the same marginal path $\{p_t\}$. Setting $\sigma_t = 0$ yields the noise-free probability flow ODE.

Definition 4.4 (Probability flow ODE). Given a forward SDE [Equation \(4.2\)](#), the associated *probability flow ODE* is

$$\frac{dx}{dt} = u_t(x), \quad u_t(x) := f_t(x) - \frac{g_t^2}{2} \nabla_x \log p_t(x). \tag{4.5}$$

Theorem 4.5 (Shared marginals). *Let $\{p_t\}$ be the marginal densities of the forward SDE [Equation \(4.2\)](#). If x_t follows the probability flow ODE [Equation \(4.5\)](#) with $x_0 \sim p_0$, then $x_t \sim p_t$ for all t .*

Proof. By [Theorem 4.2](#), it suffices to verify the continuity equation $\partial_t p_t + \nabla \cdot (u_t p_t) = 0$ for the velocity u_t of [Equation \(4.5\)](#). Compute

$$\nabla \cdot (u_t p_t) = \nabla \cdot (f_t p_t) - \frac{g_t^2}{2} \nabla \cdot (\nabla \log p_t \cdot p_t) = \nabla \cdot (f_t p_t) - \frac{g_t^2}{2} \Delta p_t,$$

where we used $\nabla \log p_t \cdot p_t = \nabla p_t$. Hence $\partial_t p_t + \nabla \cdot (u_t p_t) = \partial_t p_t - (-\nabla \cdot (f_t p_t) + \frac{g_t^2}{2} \Delta p_t) = 0$ by the Fokker–Planck equation [Equation \(4.3\)](#). $\square$

Remark 4.6 (Deterministic sampling and connection to normalizing flows). Since [Equation \(4.5\)](#) is an ODE, the map $x_T \mapsto x_0$ obtained by integrating backwards is a deterministic bijection. This connects diffusion models to *continuous normalizing flows* [7]: the probability flow ODE is a CNF with drift determined by the score, allowing higher-order ODE solvers (Runge–Kutta, Adams methods) for faster sampling than SDE-based approaches.

4.3 Instantaneous change of variables

The deterministic character of the probability flow ODE unlocks exact likelihood computation, a capability unavailable to SDE-based samplers without variational bounds.

Theorem 4.7 (Instantaneous change of variables). *Let $\dot{\psi}_t(x_0) = u_t(\psi_t(x_0))$ be the flow of the ODE [Equation \(4.5\)](#), and suppose (u_t, p_t) satisfies the continuity equation ([Theorem 4.2](#)). Then along every trajectory,*

$$\frac{d}{dt} \log p_t(\psi_t(x_0)) = -\text{div}(u_t)(\psi_t(x_0)). \quad (4.6)$$

Proof. Set $y = \psi_t(x_0)$. By the chain rule and the flow ODE $\dot{\psi}_t = u_t(\psi_t)$,

$$\frac{d}{dt} \log p_t(y) = \frac{\partial_t p_t(y) + \langle \nabla p_t(y), u_t(y) \rangle}{p_t(y)}.$$

Expanding the continuity equation: $\partial_t p_t = -\nabla \cdot (p_t u_t) = -p_t \text{div}(u_t) - \langle \nabla p_t, u_t \rangle$. Substituting into the numerator:

$$\partial_t p_t(y) + \langle \nabla p_t(y), u_t(y) \rangle = -p_t(y) \text{div}(u_t)(y) - \langle \nabla p_t(y), u_t(y) \rangle + \langle \nabla p_t(y), u_t(y) \rangle = -p_t(y) \text{div}(u_t)(y).$$

Dividing by $p_t(y)$ gives [Equation \(4.6\)](#). $\square$

Integrating [Equation \(4.6\)](#) from 0 to T yields the log-likelihood formula:

Corollary 4.8 (Exact log-likelihood).

$$\log p_0(x_0) = \log p_T(\psi_T(x_0)) + \int_0^T \text{div}(u_t)(\psi_t(x_0)) \, dt.$$

Remark 4.9 (Hutchinson trace estimator). Computing $\text{div}(u_t)(x) = \text{tr}(\partial_x u_t(x))$ exactly costs $O(d)$ backward passes. The *Hutchinson estimator* [18]

$$\text{div}(u_t)(x) = \mathbb{E}_z \left[z^\top \partial_x u_t(x) z \right], \quad z \sim \mathcal{N}(0, I),$$

replaces this with a single vector–Jacobian product per sample. In practice [Corollary 4.8](#) is evaluated by integrating the augmented ODE $(u_t(\psi_t(x_0)), -\text{div}(u_t)(\psi_t(x_0)))$ simultaneously.

Key Idea

The probability flow ODE and the reverse-time SDE generate the same $\{p_t\}$, but only the ODE admits *exact* likelihoods: the divergence of the velocity field, integrated along deterministic trajectories, gives $\log p_0(x_0)$ without any variational approximation.

4.4 Flow matching

Training a flow model by maximizing $\log p_0(x)$ via [Corollary 4.8](#) requires solving the augmented ODE at each training step—a costly simulation. *Flow matching* [1, 25, 26] bypasses this entirely by directly regressing the velocity field against a tractable target.

4.4.1 The flow matching loss

Fix a source $p_0 = \mathcal{N}(0, I)$ and a target $p_1 = q_{\text{data}}$.

Definition 4.10 (Flow matching loss). Let u_t be any velocity field known to generate p_t (with $p_0 = p, p_1 = q$). The *flow matching* (FM) loss is

$$\mathcal{L}_{\text{FM}}(\theta) := \mathbb{E}_{t \sim U[0,1], X_t \sim p_t} [\|u_t^\theta(X_t) - u_t(X_t)\|^2]. \quad (4.7)$$

The marginal velocity $u_t(x) = \mathbb{E}[u_t(X_t|X_1) \mid X_t = x]$ involves the intractable density p_t . We resolve this by constructing p_t and u_t *conditionally*, via a target sample $X_1 \sim q$.

4.4.2 Conditional probability paths

Definition 4.11 (Conditional probability path). Given a source coupling $\pi_{0|1}(\cdot|x_1)$ and $X_1 \sim q$, a *conditional probability path* $p_{t|1}(x|x_1)$ satisfies

$$p_{0|1}(\cdot|x_1) = \pi_{0|1}(\cdot|x_1), \quad p_{1|1}(\cdot|x_1) = \delta_{x_1}(\cdot). \quad (4.8)$$

The *marginal path* is

$$p_t(x) = \int p_{t|1}(x|x_1) q(x_1) dx_1. \quad (4.9)$$

With independent coupling $\pi_{0|1}(\cdot|x_1) = p_0$, the boundary conditions [Equation \(4.8\)](#) guarantee that [Equation \(4.9\)](#) interpolates $p_0 \rightarrow q$. For each x_1 , let $\psi_t(\cdot|x_1)$ be the flow of a conditional ODE with marginals $p_{t|1}(\cdot|x_1)$, and define the *conditional velocity field*

$$u_t(x|x_1) := \dot{\psi}_t \left(\psi_t^{-1}(x|x_1) \mid x_1 \right). \quad (4.10)$$

By [Theorem 4.2](#), $u_t(\cdot|x_1)$ generates $p_{t|1}(\cdot|x_1)$.

4.4.3 The marginalization trick

Theorem 4.12 (Marginalization trick). Suppose for each x_1 , the velocity $u_t(\cdot|x_1)$ generates $p_{t|1}(\cdot|x_1)$. Then the marginal velocity field

$$u_t(x) := \mathbb{E} \left[u_t(X_t|X_1) \mid X_t = x \right] = \int u_t(x|x_1) p_{1|t}(x_1|x) dx_1, \quad (4.11)$$

where $p_{1|t}(x_1|x) = p_{t|1}(x|x_1)q(x_1)/p_t(x)$ is the posterior, generates the marginal path p_t .

The proof—differentiating p_t under the integral, applying the Leibniz rule, and using Bayes' theorem—is given in [Appendix A.7](#).

Remark 4.13. [Equation \(4.11\)](#) expresses $u_t(x)$ as the mean-square-optimal predictor of $u_t(X_t|X_1)$ given $X_t = x$. This immediately suggests regressing u_t^θ against the conditional target $u_t(X_t|X_1)$, sampled jointly over (X_1, X_t) .

4.4.4 Conditional flow matching and gradient equivalence

Definition 4.14 (Conditional flow matching loss).

$$\mathcal{L}_{\text{CFM}}(\theta) := \mathbb{E}_{t, X_1 \sim q, X_t \sim p_{t|1}(\cdot|X_1)} [\|u_t^\theta(X_t) - u_t(X_t|X_1)\|^2]. \quad (4.12)$$

The loss [Equation \(4.12\)](#) is fully tractable: sample $t \sim U[0, 1]$, $X_1 \sim q$, $X_0 \sim \pi_{0|1}(\cdot|X_1)$, $X_t = \psi_t(X_0|X_1)$ from the known conditional flow, and evaluate $u_t(X_t|X_1)$ in closed form.

Theorem 4.15 (CFM = FM). $\nabla_\theta \mathcal{L}_{\text{FM}}(\theta) = \nabla_\theta \mathcal{L}_{\text{CFM}}(\theta)$.

Proof. Expand the squared norm in the CFM loss [Equation \(4.12\)](#):

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}[\|u_t^\theta(X_t)\|^2] - 2 \mathbb{E}[\langle u_t^\theta(X_t), u_t(X_t|X_1) \rangle] + \mathbb{E}[\|u_t(X_t|X_1)\|^2].$$

For the middle term, apply the tower property (conditioning on X_t):

$$\mathbb{E}[\langle u_t^\theta(X_t), u_t(X_t|X_1) \rangle] = \mathbb{E}_{t, X_t} \left[\left\langle u_t^\theta(X_t), \underbrace{\mathbb{E}[u_t(X_t|X_1) | X_t]}_{= u_t(X_t)} \right\rangle \right] = \mathbb{E}_{t, X_t} [\langle u_t^\theta(X_t), u_t(X_t) \rangle], \quad (4.13)$$

where the underbraced equality is the marginalization trick ([Theorem 4.12](#)). Hence

$$\mathcal{L}_{\text{CFM}}(\theta) = \underbrace{\mathbb{E}[\|u_t^\theta(X_t)\|^2] - 2 \mathbb{E}_{t, X_t} [\langle u_t^\theta(X_t), u_t(X_t) \rangle] + \mathbb{E}_{t, X_t} [\|u_t(X_t)\|^2]}_{\mathcal{L}_{\text{FM}}(\theta)} + \underbrace{\mathbb{E}[\|u_t(X_t|X_1)\|^2] - \mathbb{E}_{t, X_t} [\|u_t(X_t)\|^2]}_{C, \text{ independent of } \theta}.$$

Therefore $\mathcal{L}_{\text{CFM}}(\theta) = \mathcal{L}_{\text{FM}}(\theta) + C$, and differentiating in θ gives the result. $\square$

Key Idea

The pivotal step is [Equation \(4.13\)](#): the cross-term between the network output and the conditional target collapses to the cross-term against the *marginal* target, because $\mathbb{E}[u_t(X_t|X_1)|X_t] = u_t(X_t)$ by the marginalization trick. The remaining difference $\mathbb{E}\|u_t(\cdot|X_1)\|^2 - \mathbb{E}\|u_t\|^2$ is a variance-like constant that vanishes from the gradient.

4.5 Affine conditional flows and connections to diffusion

4.5.1 Affine paths and their velocities

Definition 4.16 (Affine conditional flow). Let $\alpha_t, \sigma_t : [0, 1] \rightarrow \mathbb{R}_{\geq 0}$ be smooth with $\alpha_0 = 0, \sigma_0 = 1, \alpha_1 = 1, \sigma_1 = 0$. The *affine conditional flow* with $X_0 \sim \mathcal{N}(0, I)$ independent of X_1 is

$$\psi_t(x_0|x_1) = \alpha_t x_1 + \sigma_t x_0, \quad p_{t|1}(x|x_1) = \mathcal{N}(x; \alpha_t x_1, \sigma_t^2 I). \quad (4.14)$$

Proposition 4.17 (Affine conditional velocity). Under [Definition 4.16](#), for $X_t = \alpha_t X_1 + \sigma_t X_0$,

$$u_t(x|x_1) = \dot{\alpha}_t x_1 + \frac{\dot{\sigma}_t}{\sigma_t} (x - \alpha_t x_1), \quad (4.15)$$

where $\dot{\alpha}_t = d\alpha_t/dt, \dot{\sigma}_t = d\sigma_t/dt$.

Proof. Differentiate $\psi_t(x_0|x_1)$ in t : $\dot{\psi}_t = \dot{\alpha}_t x_1 + \dot{\sigma}_t x_0$. Substitute $x_0 = (x - \alpha_t x_1)/\sigma_t$ to obtain [Equation \(4.15\)](#). $\square$

The CFM training objective [Equation \(4.12\)](#) therefore becomes

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, X_0, X_1} \left[\left\| u_t^\theta(X_t, t) - \dot{\alpha}_t X_1 - \dot{\sigma}_t X_0 \right\|^2 \right], \quad X_t = \alpha_t X_1 + \sigma_t X_0. \quad (4.16)$$

No ODE simulation is required during training.

4.5.2 The OT–Gaussian (linear) path

Definition 4.18 (OT–Gaussian / linear path). Set $\alpha_t = t, \sigma_t = 1 - t$. Then $X_t = tX_1 + (1 - t)X_0$ and

$$u_t(x|x_1) = x_1 - x_0 = \frac{x_1 - x}{1 - t}, \quad \mathcal{L}_{\text{OT}}(\theta) = \mathbb{E}_{t, X_0, X_1} \left[\left\| u_t^\theta(X_t, t) - (X_1 - X_0) \right\|^2 \right]. \quad (4.17)$$

The constant target $(X_1 - X_0)$ is the direction of the straight-line trajectory from X_0 to X_1 , independent of t . This *straight-line* construction coincides with Rectified Flow [\[26\]](#) and the independent-coupling instance of stochastic interpolants [\[1\]](#).

4.5.3 Connection to diffusion models: recovering the score

Proposition 4.19 (Velocity–score equivalence under affine paths). Under [Definition 4.16](#), by Tweedie’s formula ([Theorem 2.2](#)),

$$\nabla \log p_t(x) = -\frac{x - \alpha_t \mathbb{E}[X_1|X_t = x]}{\sigma_t^2}.$$

Substituting into [Equation \(4.15\)](#), the marginal velocity becomes

$$u_t(x) = \frac{\dot{\alpha}_t \sigma_t - \dot{\sigma}_t \alpha_t}{\sigma_t} \mathbb{E}[X_1|X_t = x] + \dot{\sigma}_t \frac{x}{\sigma_t}. \quad (4.18)$$

For the VP parametrization $\alpha_t = \sqrt{\bar{\alpha}_t}, \sigma_t = \bar{\beta}_t = \sqrt{1 - \bar{\alpha}_t}$, this recovers the probability flow ODE velocity [Equation \(4.5\)](#).

[Proposition 4.19](#) shows that affine flow matching is a simulation-free reparametrization of

score-based modelling: one trains on the simple objective [Equation \(4.16\)](#) without computing a score or simulating an SDE.

Table 4.1: Parametrization conversion for affine paths $X_t = \alpha_t X_1 + \sigma_t X_0$.

Parametrization	Network output	Relation to x_0, x_1, x_t
Velocity	$u_t^\theta(x, t)$	$\dot{\alpha}_t x_1 + \dot{\sigma}_t x_0$
Noise (score)	$\epsilon_t^\theta(x, t)$	$x_0 = (x_t - \alpha_t x_1) / \sigma_t$
Denoiser	$D_t^\theta(x, t)$	x_1
v -prediction	$v_t^\theta(x, t)$	$\alpha_t x_0 - \sigma_t x_1$ [†]

[†] In the FM convention of this chapter, x_0 denotes *noise* and x_1 denotes *data* (cf. the convention box above). Under the substitution $x_0^{\text{FM}} = \epsilon$, $x_1^{\text{FM}} = x_0^{\text{DDPM}}$, this becomes $v = \sqrt{\bar{\alpha}_t} \epsilon - \sqrt{1 - \bar{\alpha}_t} x_0^{\text{DDPM}}$, matching the definition in Salimans and Ho [32].

Remark 4.20 (DDIM as OT–FM with VP scheduling). Under the VP scheduling $\alpha_t = \bar{\alpha}_t, \sigma_t = \bar{\beta}_t$, the OT–Gaussian conditional velocity [Equation \(4.17\)](#) with corresponding $\dot{\sigma}_t / \sigma_t$ reproduces the DDIM update rule upon Euler discretization. DDIM is thus an Euler method on the probability flow ODE, which is itself the zero-variance member of the SDE equivalence class—linking DDPM, SMLD, and flow matching into a single framework.

5. Unification

The frameworks developed in [Chapters 2 to 4](#) appear to speak different mathematical languages—Markov chains, SDEs, continuity equations, velocity fields. This chapter shows they are all descriptions of a single underlying object: a marginal path $\{p_t\}$ together with its score field $\nabla \log p_t$. Four equivalences are established: (i) DDPM is the Euler–Maruyama discretization of the VP-SDE; (ii) every SDE sharing marginals $\{p_t\}$ belongs to a one-parameter equivalence class, with the probability flow ODE as its unique deterministic representative; (iii) all training objectives (ϵ -, x_0 -, v -prediction, score matching) minimize the same denoising loss up to reweighting; (iv) the flow matching velocity field and the score field are linearly related under affine paths, so SMLD and FM train and sample the same probability flow ODE.

5.1 The discrete-to-continuous limit: DDPM $\rightarrow$ VP-SDE

The DDPM forward process ([Chapter 2](#)) is a discrete-time Markov chain with transition

$$q(x_t | x_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} x_{t-1}, \beta_t I), \quad t = 1, \dots, T.$$

As $T \rightarrow \infty$, this chain converges to the VP-SDE ([Equation \(3.16\)](#)). The argument is a standard Euler–Maruyama identification.

Proposition 5.1 (DDPM $\rightarrow$ VP-SDE in the continuum limit). *Let $\beta: [0, 1] \rightarrow (0, \infty)$ be a smooth noise schedule and set $\beta_t^{(T)} := \beta(t/T)/T$ for $t = 1, \dots, T$. With continuous-time index $s = t/T \in [0, 1]$ and step size $\Delta s = 1/T$, the DDPM transition kernel satisfies*

$$x_{s+\Delta s} = \sqrt{1 - \beta(s)\Delta s} x_s + \sqrt{\beta(s)\Delta s} \xi, \quad \xi \sim \mathcal{N}(0, I),$$

which, upon Taylor-expanding $\sqrt{1 - \beta(s)\Delta s} = 1 - \frac{\beta(s)}{2}\Delta s + O(\Delta s^2)$, becomes

$$x_{s+\Delta s} - x_s = -\frac{\beta(s)}{2} x_s \Delta s + \sqrt{\beta(s)\Delta s} \xi + O(\Delta s^{3/2}).$$

This is precisely the Euler–Maruyama step of the VP-SDE

$$dx = -\frac{\beta(t)}{2} x dt + \sqrt{\beta(t)} dW_t. \tag{5.1}$$

Consequently, the marginal distribution $p_t^{(T)}$ of the T -step chain converges weakly to p_t of the VP-SDE as $T \rightarrow \infty$ (see [Appendix A.8](#) for the rigorous statement).

Remark 5.2. The DDIM sampler ([Chapter 2](#)) and the probability flow ODE ([Chapter 4](#)) are already continuous-time objects defined on the same VP-SDE marginals. [Proposition 5.1](#) makes

precise the sense in which the discrete reverse chain is an approximation: it is a first-order (Euler–Maruyama) discretization of the reverse VP-SDE (Equation (3.19)). Every sampler improvement—DDIM, DPM-Solver, higher-order Runge–Kutta—is therefore a more accurate numerical ODE/SDE integrator applied to the same continuous-time equation.

5.2 The SDE equivalence class and the probability flow ODE

Fix a family of marginals $\{p_t\}_{t \in [0, T]}$. Theorem 3.4 shows that any SDE $dx = f_t dt + g_t dW_t$ producing $\{p_t\}$ must have its drift and diffusion satisfy the Fokker–Planck equation. A key observation is that the Fokker–Planck equation constrains only the *divergence* of the probability current $J_t = f_t p_t - \frac{\tilde{g}_t^2}{2} \nabla p_t$, leaving a large family of valid SDEs.

Theorem 5.3 (SDE equivalence class). *Let (f_t, g_t) generate marginals $\{p_t\}$ via the Fokker–Planck equation. For any $\tilde{g}_t \geq 0$, define*

$$\tilde{f}_t(x) := f_t(x) + \frac{\tilde{g}_t^2 - g_t^2}{2} \nabla \log p_t(x). \quad (5.2)$$

Then the SDE $dx = \tilde{f}_t dt + \tilde{g}_t dW_t$ also has marginals $\{p_t\}$.

Proof. Write the Fokker–Planck equation for the new SDE:

$$\partial_t p_t = -\nabla \cdot (\tilde{f}_t p_t) + \frac{\tilde{g}_t^2}{2} \Delta p_t = -\nabla \cdot \left(\tilde{f}_t p_t - \frac{\tilde{g}_t^2}{2} \nabla p_t \right).$$

Substituting Equation (5.2) and using $\nabla p_t = p_t \nabla \log p_t$:

$$\begin{aligned} \tilde{f}_t p_t - \frac{\tilde{g}_t^2}{2} \nabla p_t &= \left(f_t + \frac{\tilde{g}_t^2 - g_t^2}{2} \nabla \log p_t \right) p_t - \frac{\tilde{g}_t^2}{2} p_t \nabla \log p_t \\ &= f_t p_t - \frac{g_t^2}{2} \nabla p_t, \end{aligned}$$

which equals the probability current of the original SDE. Hence $\partial_t p_t$ is unchanged, so both SDEs share the same Fokker–Planck equation and therefore the same marginals $\{p_t\}$. $\square$

Corollary 5.4 (Probability flow ODE as the deterministic representative). *Taking $\tilde{g}_t = 0$ in Theorem 5.3 yields the unique deterministic flow generating $\{p_t\}$:*

$$dx = \underbrace{\left(f_t(x) - \frac{g_t^2}{2} \nabla \log p_t(x) \right)}_{=: u_t(x)} dt, \quad (5.3)$$

which is precisely the probability flow ODE (Equation (4.5)). The velocity $u_t = f_t - \frac{g_t^2}{2} \nabla \log p_t$ is the same for every member of the equivalence class—it is the invariant deterministic backbone shared by all SDEs with marginals $\{p_t\}$.

Remark 5.5 (Named instances of the equivalence class). The main frameworks developed in Chapter 3 are all instances of Theorem 5.3 with the same marginals $\{p_t\}$:

- **Forward VP-SDE** (Equation (3.16)): $f_t = -\frac{\beta_t}{2}x$, $g_t = \sqrt{\beta_t}$; $u_t = -\frac{\beta_t}{2}x - \frac{\beta_t}{2} \nabla \log p_t$.
- **Forward VE-SDE** (Equation (3.17)): $f_t = 0$, $g_t = \sqrt{2\sigma_t \dot{\sigma}_t}$; $u_t = -\dot{\sigma}_t \sigma_t \nabla \log p_t$.

- **Reverse-time SDE** (Theorem 3.17): $\tilde{g}_t = g_t$ (same diffusion magnitude as forward, run backward in time).
- **Probability flow ODE** (Equation (4.5)): $\tilde{g}_t = 0$ (unique deterministic element of the class).

5.3 Objective equivalence

All practical training objectives— ϵ -prediction, x_0 -prediction, v -prediction, and score matching—are reweighted instances of the continuous denoising score matching loss (Equation (3.20)).

Proposition 5.6 (All objectives minimize the same score). *Under the VP marginal $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$, $\epsilon \sim \mathcal{N}(0, I)$, define $\sigma_t := \sqrt{1 - \bar{\alpha}_t}$. The four objectives in the table below all share the unique minimizer $\nabla \log p_t$, and are related by the linear identities*

$$s_\theta^* = \nabla \log p_t, \quad \epsilon_\theta^* = -\sigma_t \nabla \log p_t, \quad x_{0,\theta}^* = \mathbb{E}[x_0 | x_t], \quad v_\theta^* = \sqrt{\bar{\alpha}_t} \epsilon - \sigma_t x_0.$$

Parameterization	Training target	Effective $\lambda(t)$	Reference
Score matching	$\nabla \log p_t$	$g(t)^2$	Equation (3.20)
ϵ -prediction	ϵ	$1 - \bar{\alpha}_t$	Equation (2.16)
x_0 -prediction	x_0	$(1 - \bar{\alpha}_t)^2 / \bar{\alpha}_t$	Theorem 2.2
v -prediction	$\sqrt{\bar{\alpha}_t} \epsilon - \sigma_t x_0$	$(1 - \bar{\alpha}_t) / \bar{\alpha}_t$	[32]

Proof. Let $\alpha_t := \sqrt{\bar{\alpha}_t}$, so $x_t = \alpha_t x_0 + \sigma_t \epsilon$ and $\alpha_t^2 + \sigma_t^2 = 1$. By the conditional score formula (Theorem 2.2),

$$\nabla_{x_t} \log p(x_t | x_0) = -\frac{\epsilon}{\sigma_t}, \quad \nabla_{x_t} \log p_t(x_t) = -\frac{\mathbb{E}[\epsilon | x_t]}{\sigma_t}.$$

Each objective is a quadratic in its prediction variable, so its minimizer is determined by the Bayes-optimal predictor. It suffices to show that each prediction residual is a scalar multiple of the score residual $\delta := s_\theta + \epsilon / \sigma_t$ (which vanishes iff $s_\theta = \nabla \log p_t$), giving effective weight $\lambda(t) = c(t)^2$ where the prediction residual equals $c(t) \delta$.

ϵ -prediction. Set $\epsilon_\theta := -\sigma_t s_\theta$. Then

$$\epsilon_\theta - \epsilon = -\sigma_t s_\theta - \epsilon = -\sigma_t \left(s_\theta + \frac{\epsilon}{\sigma_t} \right) = -\sigma_t \delta,$$

so $\|\epsilon_\theta - \epsilon\|^2 = \sigma_t^2 \|\delta\|^2$ and $\lambda_\epsilon(t) = \sigma_t^2 = 1 - \bar{\alpha}_t$. The minimizer is $\epsilon_\theta^* = \mathbb{E}[\epsilon | x_t]$.

x_0 -prediction. Set $x_{0,\theta} := (x_t + \sigma_t^2 s_\theta) / \alpha_t$. Since $x_t - \alpha_t x_0 = \sigma_t \epsilon$,

$$x_{0,\theta} - x_0 = \frac{\sigma_t \epsilon + \sigma_t^2 s_\theta}{\alpha_t} = \frac{\sigma_t^2}{\alpha_t} \left(s_\theta + \frac{\epsilon}{\sigma_t} \right) = \frac{\sigma_t^2}{\alpha_t} \delta,$$

so $\|x_{0,\theta} - x_0\|^2 = (\sigma_t^2 / \alpha_t^2) \|\delta\|^2$ and $\lambda_{x_0}(t) = \sigma_t^4 / \alpha_t^2 = (1 - \bar{\alpha}_t)^2 / \bar{\alpha}_t$. The minimizer $x_{0,\theta}^* = \mathbb{E}[x_0 | x_t]$ is Tweedie's formula (Theorem 2.2).

v-prediction. Define $v := \alpha_t \epsilon - \sigma_t x_0$ and set $v_\theta := \alpha_t \epsilon_\theta - \sigma_t x_{0,\theta}$. Using the residuals computed above,

$$v_\theta - v = \alpha_t(\epsilon_\theta - \epsilon) - \sigma_t(x_{0,\theta} - x_0) = -\alpha_t \sigma_t \delta - \frac{\sigma_t^3}{\alpha_t} \delta = -\frac{\sigma_t(\alpha_t^2 + \sigma_t^2)}{\alpha_t} \delta = -\frac{\sigma_t}{\alpha_t} \delta,$$

where the last step uses $\alpha_t^2 + \sigma_t^2 = 1$. Hence $\lambda_v(t) = \sigma_t^2/\alpha_t^2 = (1 - \bar{\alpha}_t)/\bar{\alpha}_t$. The minimizer is $v_\theta^* = \alpha_t \mathbb{E}[\epsilon | x_t] - \sigma_t \mathbb{E}[x_0 | x_t]$.

All four objectives therefore share the minimizer $s_\theta^* = \nabla \log p_t$, with the stated linear identities between optimal predictors. $\square$

Key Idea

The choice of parameterization (ϵ , x_0 , or v) affects the *effective weighting* $\lambda(t)$ in the score metric across noise levels, and hence which SNR regime the network is trained hardest on. It does not change the theoretical optimum. In terms of signal-to-noise ratio $\text{SNR}(t) := \bar{\alpha}_t/(1 - \bar{\alpha}_t)$, the three weights are $\lambda_\epsilon = \text{SNR}^{-1}$, $\lambda_{x_0} = \text{SNR}^{-2}$, and $\lambda_v = \text{SNR}^{-1}/(1 + \text{SNR}^{-1})$; the v -parameterization concentrates less heavily on low-SNR (high-noise) time steps and is preferred for cascaded or high-resolution generation [32].

5.4 Flow matching and score-based models

Flow matching (Chapter 4) and score-based models (Chapter 3) appear to speak entirely different languages: one trains a velocity field u_θ via the CFM objective, the other trains a score s_θ via DSM. Proposition 4.19 (proved in Chapter 4) resolves this: under affine conditional paths, the marginal velocity field u_t is a known linear function of $\nabla \log p_t$.

Proposition 5.7 (FM $\leftrightarrow$ Score: explicit equivalence, restatement of Proposition 4.19). *Under the affine path $x_t = \alpha_t x_1 + \sigma_t x_0$, $x_0 \sim \mathcal{N}(0, I)$, $x_1 \sim p_{\text{data}}$, the marginal velocity field satisfies*

$$u_t(x) = \frac{\dot{\alpha}_t}{\alpha_t} x + \left(\frac{\dot{\alpha}_t \sigma_t^2}{\alpha_t} - \dot{\sigma}_t \sigma_t \right) \nabla \log p_t(x). \quad (5.4)$$

Conversely, the score is recovered from the velocity by inverting Equation (5.4):

$$\nabla \log p_t(x) = \frac{\alpha_t \left(u_t(x) - \frac{\dot{\alpha}_t}{\alpha_t} x \right)}{\dot{\alpha}_t \sigma_t^2 - \dot{\sigma}_t \alpha_t \sigma_t}. \quad (5.5)$$

Remark 5.8 (Consequences). Proposition 5.7 has three immediate consequences.

- (i) **Same ODE, different parameterizations.** A model trained with CFM and one trained with DSM both implicitly define the *same* probability flow ODE (Equation (4.5)). Switching from FM to SDE sampling (or vice versa) requires only the linear conversion in Equations (5.4) and (5.5).
- (ii) **Rectified Flow / OT-FM as a special case of VE-SDE.** The OT-Gaussian path $\alpha_t = t$, $\sigma_t = 1 - t$ (Chapter 4) yields $u_t = (x_1 - x_0)/(1 - t + t) = x_1 - x_0$ on the straight-line trajectory. Substituting into Equation (5.4) recovers the VE-type PF-ODE velocity. The straight path of FM is thus the OT-optimal coupling of the VE-SDE.

- (iii) **CFM loss = DSM loss.** Combining the gradient equivalence $\nabla_{\theta} \mathcal{L}_{\text{FM}} = \nabla_{\theta} \mathcal{L}_{\text{CFM}}$ (Theorem 4.15) with Proposition 5.7: training either the CFM objective or the DSM objective minimizes the same functional up to a σ_t -dependent reweighting.

5.5 Sampling as numerical integration

Every diffusion sampler is a numerical method applied to one of two objects: the reverse-time SDE (Theorem 3.17) or the probability flow ODE (Corollary 5.4). The choice of object (stochastic vs. deterministic) and numerical scheme (order, step size) fully determines the sampler.

Table 5.1: All major samplers as instances of SDE/ODE integration. N = number of function evaluations (NFE).

Sampler	Target equation	Numerical scheme	NFE/step
DDPM sampler	Reverse-time SDE	Euler–Maruyama	1
DDIM	Prob. flow ODE	Euler (1st-order)	1
PLMS / DPM-Solver	Prob. flow ODE	Adams (2nd/3rd-order)	1–2
Heun (EDM)	Prob. flow ODE	Heun (2nd-order)	2
PC sampler	Reverse-time SDE	EM + Langevin correction	≥ 2

The accuracy hierarchy follows directly from classical ODE/SDE theory: Euler–Maruyama is $O(h^{1/2})$ (SDE) or $O(h)$ (ODE); Heun and Adams-type methods are $O(h^2)$ or higher. All samplers converge to the same distribution as step size $h \rightarrow 0$, since they integrate the same equation.

5.6 The unified picture

The four equivalences above crystallize into a single diagram. The central object is $\{p_t\}$ together with $\nabla \log p_t$; every framework is a different arrow into or out of this core.

Key Idea

The score field $\nabla \log p_t$ is the *lingua franca* of diffusion-based generative modelling. Every method learns it (via DSM, ESM, CFM, or their equivalents) and every sampler uses it (via the reverse SDE, probability flow ODE, or FM velocity field). The choice of framework is a choice of *language*, not of underlying mathematics.

5.7 Stochastic Interpolants

5.7.1 Motivation

The frameworks developed so far—VP-SDE, VE-SDE, and flow matching—share a common structural assumption: the source and target random variables $X_0 \sim p_0$ and $X_1 \sim p_1$ are drawn

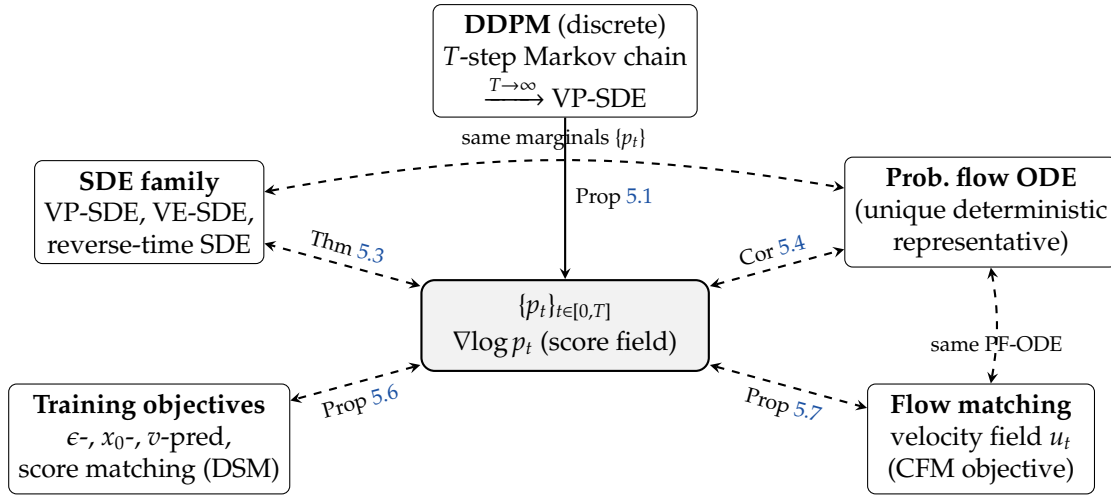


Figure 5.1: The unified diffusion landscape. Dashed double-headed arrows indicate equivalences (different descriptions of the same object); the single solid arrow from DDPM indicates a limiting relationship ($T \rightarrow \infty$). The score field $\nabla \log p_t$ is the invariant core from which every framework is derived.

independently. This independence makes the forward process tractable (the noise schedule is closed-form) but it is not a theoretical necessity—it is merely a convenient default.

Why would one ever want to depart from it? There are at least two compelling reasons.

- (i) **Leveraging known structure between p_0 and p_1 .** In image restoration, super-resolution, or medical imaging, X_0 (degraded observation) and X_1 (clean image) are *not* independent—they share substantial mutual information. An interpolant built from the joint distribution $\pi(X_0, X_1)$ can exploit this structure, producing straighter interpolating paths and more efficient transport.
- (ii) **Connecting to optimal transport and Schrödinger bridges.** The choice of coupling π determines the *geometry* of the interpolating family $\{p_t\}_{t \in [0,1]}$. The *entropy-regularised optimal transport* coupling (the Schrödinger Bridge) minimises kinetic energy along the interpolating path, yielding dynamics that are “as straight as possible” in the stochastic sense. This cannot be achieved with an independent coupling.

Stochastic Interpolants (SI), introduced by Albergo et al. [1], provide a unified framework that captures all of these cases. The additional $\gamma(t)Z$ term in the interpolant (see Equation (5.7) below) plays a dual role: it injects stochasticity into the interpolating path even when $\gamma \neq 0$, and it allows the *marginal* p_t to differ from the deterministic mixture $\alpha(t)p_0 + \beta(t)p_1$, thereby recovering the Gaussian noise schedule of diffusion models as a special case.

Key Idea

The SI framework unifies flow matching (arbitrary coupling, $\gamma \equiv 0$), VP/VE diffusion (independent coupling, specific α, β, γ), and Schrödinger bridges (entropy-regularised OT coupling) under a single interpolant $X_t = \alpha(t)X_0 + \beta(t)X_1 + \gamma(t)Z$. The *coupling* π encodes the geometric structure of the transport; the *coefficient triple* (α, β, γ) encodes the temporal interpolation and the degree of injected stochasticity.

5.7.2 Construction

Let $(X_0, X_1) \sim \pi$ be a joint distribution on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals $X_0 \sim p_0$ (source) and $X_1 \sim p_1$ (target). Let $Z \sim \mathcal{N}(0, I_d)$ be independent of (X_0, X_1) . Choose smooth coefficient functions $\alpha, \beta, \gamma: [0, 1] \rightarrow \mathbb{R}_{\geq 0}$ satisfying the boundary conditions

$$\alpha(0) = 1, \quad \alpha(1) = 0, \quad \beta(0) = 0, \quad \beta(1) = 1, \quad \gamma(0) = 0, \quad \gamma(1) = 0. \quad (5.6)$$

The *stochastic interpolant* is then defined by

$$X_t := \alpha(t) X_0 + \beta(t) X_1 + \gamma(t) Z, \quad t \in [0, 1]. \quad (5.7)$$

The boundary conditions [Equation \(5.6\)](#) guarantee that $X_0 \sim p_0$ and $X_1 \sim p_1$ exactly. The intermediate marginal p_t of X_t is determined by the triple (α, β, γ) . Conditioned on $(X_0, X_1) = (x_0, x_1)$, the interpolant is a Gaussian:

$$X_t \mid (X_0, X_1) = (x_0, x_1) \sim \mathcal{N}(\alpha(t)x_0 + \beta(t)x_1, \gamma(t)^2 I_d). \quad (5.8)$$

Marginalising over the coupling gives the interpolating density

$$p_t(x) = \int \int \mathcal{N}(x; \alpha x_0 + \beta x_1, \gamma^2 I) \pi(dx_0, dx_1). \quad (5.9)$$

The continuity equation. Since $(X_t)_{t \in [0, 1]}$ is a smooth stochastic process (in t), its marginal density p_t satisfies the *continuity equation*

$$\partial_t p_t(x) + \nabla_x \cdot [b_t(x) p_t(x)] = 0, \quad (5.10)$$

where $b_t(x)$ is a *marginal velocity field* transporting p_t . There are infinitely many velocity fields that satisfy [Equation \(5.10\)](#) for a given p_t —they differ by divergence-free perturbations. The natural choice is the one of minimum kinetic energy, which turns out to be the conditional expectation $b_t(x) = \mathbb{E}[\dot{X}_t \mid X_t = x]$, and is exactly the velocity learned in (stochastic) flow matching.

5.7.3 Velocity and Score as Quadratic Minimisers

We now state and prove the two central identities of the SI framework.

Proposition 5.9 (SI velocity and score). *Let X_t be given by [Equation \(5.7\)](#) with $\gamma(t) > 0$ for $t \in (0, 1)$. Define the marginal velocity field $b_t(x) := \mathbb{E}[\dot{X}_t \mid X_t = x]$ and the score $s_t(x) := \nabla_x \log p_t(x)$. Then:*

$$b_t = \arg \min_{v: \mathbb{R}^d \rightarrow \mathbb{R}^d} \mathbb{E}[\|v(X_t) - \dot{X}_t\|^2], \quad (5.11)$$

$$s_t(x) = -\frac{1}{\gamma(t)} \mathbb{E}[Z \mid X_t = x]. \quad (5.12)$$

Moreover, when $\alpha(t) > 0$, the two fields satisfy the velocity-score relation

$$b_t(x) = \frac{\dot{\alpha}(t)}{\alpha(t)} x + \left(\dot{\beta}(t) - \frac{\dot{\alpha}(t)\beta(t)}{\alpha(t)} \right) \mathbb{E}[X_1 \mid X_t = x] - \gamma(t) \left(\dot{\gamma}(t) - \frac{\dot{\alpha}(t)\gamma(t)}{\alpha(t)} \right) s_t(x). \quad (5.13)$$

Proof. We prove each part in turn.

Part 1 (Velocity as L^2 minimiser). For any measurable $v: \mathbb{R}^d \rightarrow \mathbb{R}^d$, expand the squared loss:

$$\mathbb{E}[\|v(X_t) - \dot{X}_t\|^2] = \mathbb{E}[\|v(X_t) - \mathbb{E}[\dot{X}_t | X_t]\|^2] + \mathbb{E}[\|\dot{X}_t - \mathbb{E}[\dot{X}_t | X_t]\|^2].$$

The second term does not depend on v ; the first is non-negative and vanishes if and only if $v(x) = \mathbb{E}[\dot{X}_t | X_t = x]$ for p_t -a.e. x . Hence $b_t(x) = \mathbb{E}[\dot{X}_t | X_t = x]$ is the unique minimiser.

Part 2 (Score identity). Fix t and condition on $(X_0, X_1) = (x_0, x_1)$. From [Equation \(5.8\)](#), $X_t | (X_0, X_1) \sim \mathcal{N}(\mu, \gamma^2 I)$ where $\mu = \alpha x_0 + \beta x_1$. The conditional score of this Gaussian is

$$\nabla_x \log \mathcal{N}(x; \mu, \gamma^2 I) = -\frac{x - \mu}{\gamma^2} = -\frac{x - \alpha x_0 - \beta x_1}{\gamma^2}.$$

Marginalise over (X_0, X_1) using the log-derivative identity. For a mixture $p_t(x) = \int p_t(x | x_0, x_1) \pi(dx_0, dx_1)$,

$$\nabla_x \log p_t(x) = \frac{\int \nabla_x p_t(x | x_0, x_1) \pi(dx_0, dx_1)}{p_t(x)} = \mathbb{E} \left[\nabla_x \log p_t(X_t | X_0, X_1) \mid X_t = x \right].$$

Substituting the Gaussian score:

$$s_t(x) = \mathbb{E} \left[-\frac{x - \alpha X_0 - \beta X_1}{\gamma^2} \mid X_t = x \right] = -\frac{1}{\gamma^2} \mathbb{E}[x - \alpha X_0 - \beta X_1 \mid X_t = x].$$

On the support of p_t , the residual $x - \alpha X_0 - \beta X_1 = \gamma Z$ (from [Equation \(5.7\)](#)), so

$$s_t(x) = -\frac{1}{\gamma^2} \mathbb{E}[\gamma Z \mid X_t = x] = -\frac{1}{\gamma} \mathbb{E}[Z \mid X_t = x].$$

Part 3 (Velocity-score relation). We derive [Equation \(5.13\)](#) by combining Parts 1 and 2 via two eliminations.

Step 3a. By Part 1, $b_t(x) = \mathbb{E}[\dot{X}_t | X_t = x]$. Since $\dot{X}_t = \dot{\alpha} X_0 + \dot{\beta} X_1 + \dot{\gamma} Z$ (differentiating [Equation \(5.7\)](#)), linearity of conditional expectation gives

$$b_t(x) = \dot{\alpha} \mathbb{E}[X_0 | X_t = x] + \dot{\beta} \mathbb{E}[X_1 | X_t = x] + \dot{\gamma} \mathbb{E}[Z | X_t = x]. \quad (5.14)$$

By Part 2, $\mathbb{E}[Z | X_t = x] = -\gamma s_t(x)$. Substituting:

$$b_t(x) = \dot{\alpha} \mathbb{E}[X_0 | X_t = x] + \dot{\beta} \mathbb{E}[X_1 | X_t = x] - \dot{\gamma} \gamma s_t(x). \quad (5.15)$$

Step 3b. We eliminate $\mathbb{E}[X_0 | X_t = x]$ in favour of the observable x , $\mathbb{E}[X_1 | X_t = x]$, and $s_t(x)$. Take conditional expectations of both sides of [Equation \(5.7\)](#):

$$\mathbb{E}[X_t | X_t = x] = \alpha \mathbb{E}[X_0 | X_t = x] + \beta \mathbb{E}[X_1 | X_t = x] + \gamma \mathbb{E}[Z | X_t = x].$$

The left side is x ; substituting $\mathbb{E}[Z | X_t = x] = -\gamma s_t(x)$:

$$x = \alpha \mathbb{E}[X_0 | X_t = x] + \beta \mathbb{E}[X_1 | X_t = x] - \gamma^2 s_t(x).$$

Solving for $\mathbb{E}[X_0 | X_t = x]$ (valid since $\alpha(t) > 0$ by assumption):

$$\mathbb{E}[X_0 | X_t = x] = \frac{1}{\alpha} (x - \beta \mathbb{E}[X_1 | X_t = x] + \gamma^2 s_t(x)). \quad (5.16)$$

Step 3c. Substitute Equation (5.16) into Equation (5.15):

$$\begin{aligned} b_t(x) &= \frac{\dot{\alpha}}{\alpha} \left(x - \beta \mathbb{E}[X_1 | X_t = x] + \gamma^2 s_t(x) \right) + \dot{\beta} \mathbb{E}[X_1 | X_t = x] - \dot{\gamma} \gamma s_t(x) \\ &= \frac{\dot{\alpha}}{\alpha} x + \left(\dot{\beta} - \frac{\dot{\alpha}\beta}{\alpha} \right) \mathbb{E}[X_1 | X_t = x] + \frac{\dot{\alpha}\gamma^2}{\alpha} s_t(x) - \dot{\gamma} \gamma s_t(x) \\ &= \frac{\dot{\alpha}}{\alpha} x + \left(\dot{\beta} - \frac{\dot{\alpha}\beta}{\alpha} \right) \mathbb{E}[X_1 | X_t = x] + \gamma \left(\frac{\dot{\alpha}\gamma}{\alpha} - \dot{\gamma} \right) s_t(x). \end{aligned}$$

Rewriting the last coefficient as $-\gamma(\dot{\gamma} - \frac{\dot{\alpha}\gamma}{\alpha})$ yields Equation (5.13). $\square$

Remark 5.10 (Practical significance of Equation (5.13)). Equation (5.13) shows that the velocity field b_t can be *fully expressed* in terms of the score s_t and the conditional mean $\mathbb{E}[X_1 | X_t = x]$. When $\gamma \equiv 0$ (deterministic interpolant, as in standard FM), the relation reduces to $b_t(x) = \dot{\alpha}/\alpha \cdot x + (\dot{\beta} - \dot{\alpha}\beta/\alpha) \mathbb{E}[X_1 | X_t = x]$, a pure regression target. When $\gamma > 0$, an additional *score correction* $-\gamma(\dot{\gamma} - \dot{\alpha}\gamma/\alpha) s_t(x)$ appears, reflecting the stochastic spread of the interpolating distribution. This correction vanishes at $t = 0$ and $t = 1$ because $\gamma(0) = \gamma(1) = 0$.

5.7.4 Special Cases

- (i) **Flow matching (independent coupling, $\gamma \equiv 0$):** Setting $\gamma \equiv 0$ and $\pi = p_0 \otimes p_1$ reduces the SI to the (conditional) flow matching framework of Section 4.4. The score correction vanishes, and Equation (5.13) reduces to the FM regression objective. OT-FM corresponds to the special coupling obtained by Brenier's theorem.
- (ii) **VP diffusion (independent coupling, specific α, β, γ):** With $\pi = p_0 \otimes p_1$, $X_0 \sim p_{\text{data}}$, $\alpha(t) = \sqrt{\bar{\alpha}_t}$, $\beta(t) = 0$, $\gamma(t) = \sqrt{1 - \bar{\alpha}_t}$, the SI marginals $\{p_t\}$ coincide with the VP-SDE forward marginals. The score identity Equation (5.12) then gives $s_t(x) = -\mathbb{E}[Z | X_t = x] / \sqrt{1 - \bar{\alpha}_t}$, recovering the standard DDPM denoising objective.
- (iii) **General coupling (Schrödinger Bridge):** See Remark 5.11.

Remark 5.11 (Schrödinger Bridge). The *Schrödinger Bridge* (SB) [4] is the SI with coupling π^* chosen to minimise the KL divergence from the reference Wiener measure W^{01} , subject to the marginal constraints $\pi^*(\cdot, \mathbb{R}^d) = p_0$ and $\pi^*(\mathbb{R}^d, \cdot) = p_1$:

$$\pi^* = \arg \min_{\pi : \text{marginals} = p_0, p_1} \text{KL}(P_\pi^{01} \parallel W^{01}).$$

The entropy-regularised OT problem reduces to the SB as a special case: as the temperature $\varepsilon \rightarrow 0$, the SB coupling converges to the deterministic OT coupling (Brenier map). From the SI viewpoint, π^* is the unique coupling for which Equation (5.7) produces a process that is itself a Brownian bridge. Iterative Proportional Fitting (IPF) [4] alternately marginalises P^{01} to match p_0 and p_1 , converging to π^* . Learning the SB velocity reduces, via Equation (5.13), to learning both $\mathbb{E}[X_1 | X_t = x]$ and $s_t(x)$ simultaneously.

6. Extensions

The preceding chapters establish the theoretical foundations of diffusion-based generative modelling: the VP/VE-SDE family, score matching, flow matching, and their unification. This chapter examines four axes along which the core framework is extended: (i) **conditional generation**—how external information guides the reverse process; (ii) **representation space**—running diffusion in a compressed latent space rather than data space; (iii) **sampling acceleration**—reducing the number of function evaluations at inference; (iv) **data type**—extending diffusion to discrete state spaces.

6.1 Conditional Generation

Let y denote conditioning information (class label, text, measurement). The goal is to sample from the posterior $p(x_0 | y)$. All conditioning strategies rest on the same Bayesian decomposition of the conditional score.

Proposition 6.1 (Bayesian score decomposition). *Assume $p_t(x_t) > 0$ and that $p_t(y | x_t)$ is differentiable. Then*

$$\nabla_{x_t} \log p_t(x_t | y) = \underbrace{\nabla_{x_t} \log p_t(x_t)}_{\text{unconditional score}} + \underbrace{\nabla_{x_t} \log p_t(y | x_t)}_{\text{likelihood score}}. \quad (6.1)$$

Proof. Immediate from $\log p_t(x_t | y) = \log p_t(x_t) + \log p_t(y | x_t) - \log p_t(y)$ and differentiating with respect to x_t (the last term is constant in x_t). $\square$

The strategies below differ in how they approximate the intractable time-dependent likelihood score $\nabla_{x_t} \log p_t(y | x_t)$.

6.1.1 Classifier guidance

Dhariwal and Nichol [11] train a separate *noisy classifier* $p_\phi(y | x_t, t)$ on noisy data and approximate $\nabla_{x_t} \log p_t(y | x_t) \approx \nabla_{x_t} \log p_\phi(y | x_t, t)$. The modified score is

$$\tilde{s}_\theta(x_t, t, y) = s_\theta(x_t, t) + \gamma \nabla_{x_t} \log p_\phi(y | x_t, t) \quad (6.2)$$

with guidance scale $\gamma > 0$. Plugging Equation (6.2) into the reverse SDE or PF-ODE samples from the γ -tilted posterior

$$p_t^\gamma(x_t | y) \propto p_t(x_t) p_\phi(y | x_t, t)^\gamma, \quad (6.3)$$

trading diversity (small γ) for class fidelity (large γ). The main drawback is the need for a separate classifier trained on all noise levels.

6.1.2 Classifier-free guidance

Ho and Salimans [15] eliminate the external classifier by jointly training a single model $s_\theta(x_t, t, c)$ for both conditional ($c = y$) and unconditional ($c = \emptyset$, via random conditioning dropout) generation. At inference the guidance extrapolates beyond the conditional score:

$$\tilde{s}_\theta(x_t, t, y) = s_\theta(x_t, t, \emptyset) + \gamma[s_\theta(x_t, t, y) - s_\theta(x_t, t, \emptyset)], \quad (6.4)$$

equivalent to Equation (6.2) with the *implicit classifier* $\log p_\theta^{\text{impl}}(y | x_t) \propto \log p_\theta(x_t | y) - \log p_\theta(x_t | \emptyset)$. CFG requires no auxiliary model and achieves state-of-the-art FID/CLIP tradeoffs in text-to-image generation.

6.1.3 Energy-based and loss-guided conditioning

When y is provided through a differentiable constraint or loss $\mathcal{E}(x_0, y)$ (e.g., a perceptual loss or a measurement residual), the likelihood score is approximated using the *Tweedie denoiser* $\hat{x}_0(x_t, t) := \mathbb{E}[x_0 | x_t]$ (Theorem 2.2):

$$\nabla_{x_t} \log p_t(y | x_t) \approx -\lambda \left(\frac{\partial \hat{x}_0}{\partial x_t} \right)^\top \nabla_{x_0} \mathcal{E}(\hat{x}_0, y). \quad (6.5)$$

For linear inverse problems $\mathcal{E}(x_0, y) = \frac{1}{2\sigma_y^2} \|Ax_0 - y\|^2$, Equation (6.5) recovers the DPS update [9] (see Chapter 7). The energy-guided view unifies classifier guidance, DPS, and related methods as special cases of Equation (6.1) under different approximations of $p_t(y | x_t)$.

6.2 Latent Diffusion

Running diffusion directly in pixel or token space is computationally expensive. *Latent diffusion models* (LDM) [31] factor the generative process into a perceptual compression stage and a generative stage.

6.2.1 Setup

Let $\mathcal{E}: \mathcal{X} \rightarrow \mathcal{Z}$ be a pre-trained VAE encoder and $\mathcal{D}: \mathcal{Z} \rightarrow \mathcal{X}$ its decoder, with $\mathcal{D}(\mathcal{E}(x)) \approx x$. Define the *latent variable* $z = \mathcal{E}(x)$. LDM trains the diffusion process on the latent distribution $q_1(z) := (\mathcal{E})_{\#} p_{\text{data}}$, producing latent samples $\hat{z} \sim p_0(z)$ that are decoded to $\hat{x} = \mathcal{D}(\hat{z})$.

6.2.2 Score in latent vs. data space

The score in data space relates to the score in latent space via the encoder Jacobian.

Proposition 6.2 (Score pullback). *Assume $d_z = d_x$ and $\mathcal{E}$ is a C^1 diffeomorphism. Let $J_{\mathcal{E}}(x) = \partial \mathcal{E}(x) / \partial x \in \mathbb{R}^{d_z \times d_x}$ be the Jacobian of the encoder. If $z = \mathcal{E}(x)$ and $q_t(z)$ are the latent marginals, then*

$$\nabla_x \log p_t(x) = J_{\mathcal{E}}(x)^\top \nabla_z \log q_t(z) \Big|_{z=\mathcal{E}(x)}. \quad (6.6)$$

Proof. Since $\mathcal{E}$ is a diffeomorphism, the change-of-variables formula gives $p_t(x) = q_t(\mathcal{E}(x)) |\det J_{\mathcal{E}}(x)|$. Taking $\nabla_x \log$ and applying the chain rule:

$$\nabla_x \log p_t(x) = J_{\mathcal{E}}(x)^\top \nabla_z \log q_t(z) \Big|_{z=\mathcal{E}(x)} + \nabla_x \log |\det J_{\mathcal{E}}(x)|.$$

The second term depends only on the fixed encoder geometry, not on the learned score, so it is constant with respect to score-matching training and can be absorbed into the parametrization or treated as negligible in practice. $\square$

Remark 6.3 (LDM regime $d_z < d_x$). In the standard LDM setting (e.g. $d_z = 64^2 \times 4$, $d_x = 512^2 \times 3$ in Stable Diffusion), $J_{\mathcal{E}}(x) \in \mathbb{R}^{d_z \times d_x}$ is non-square and $|\det J_{\mathcal{E}}|$ is undefined. Equation (6.6) therefore does not follow from a rigorous change-of-variables argument in this regime; it serves as first-order motivation for the latent score formulation. In practice the score network $s_{\theta}(z_t, t)$ is trained entirely in $\mathcal{Z}$ and Equation (6.6) is never evaluated explicitly.

In practice, the score network $s_{\theta}(z_t, t)$ is trained entirely in $\mathcal{Z}$, so Equation (6.6) is never evaluated explicitly. The key benefit is computational: if $d_z \ll d_x$ (e.g., $d_z = 64 \times 64 \times 4$ vs. $d_x = 512 \times 512 \times 3$ in Stable Diffusion), the diffusion model operates on a 4–8 $\times$ spatially compressed representation, reducing memory and compute by roughly d_x/d_z .

6.3 Sampling Acceleration

The reverse SDE and PF-ODE require $T = 100$ – 1000 network function evaluations (NFE). Two complementary strategies reduce this to 1–4 NFE.

6.3.1 Consistency models

Song et al. [38] define a *consistency function* $f: \mathbb{R}^d \times [0, T] \rightarrow \mathbb{R}^d$ as a function that maps any point (x_t, t) on a PF-ODE trajectory to the trajectory’s starting point x_0 :

$$f(x_t, t) = x_0 \quad \forall t \in [0, T] \text{ along the PF-ODE.} \quad (6.7)$$

This *self-consistency* condition characterizes f : if x_t and $x_{t'}$ lie on the same PF-ODE trajectory, then $f(x_t, t) = f(x_{t'}, t')$. The self-consistency conditional Equation (6.7) combined with the boundary condition $f(x_0, 0) = x_0$ uniquely determines f along each PF-ODE trajectory.

Theorem 6.4 (Consistency training loss). *Let $\{t_1 < t_2 < \dots < t_N = T\}$ be a discretization of $[0, T]$ and let $\hat{x}_{t_n}^{\phi}$ denote one step of a numerical ODE solver from $x_{t_{n+1}}$ to t_n using a pre-trained score s^{ϕ} . For any metric $d: \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}_{\geq 0}$, the consistency training objective is*

$$\mathcal{L}_{CT}(\theta, \theta^-) = \mathbb{E} \left[d \left(f_{\theta}(x_{t_{n+1}}, t_{n+1}), f_{\theta^-}(\hat{x}_{t_n}^{\phi}, t_n) \right) \right], \quad (6.8)$$

where θ^- is an exponential moving average (EMA) or stop-gradient target. As the ODE solver step size $\Delta t = t_{n+1} - t_n \rightarrow 0$ and score error $\rightarrow 0$, minimizers of Equation (6.8) converge to a consistent f_{θ} satisfying Equation (6.7).

Consistency models enable one-step generation ($f_{\theta}(x_T, T)$ with $x_T \sim \mathcal{N}(0, I)$) and iterative refinement by alternating denoising and renoising steps.

6.3.2 Progressive distillation

Salimans and Ho [32] directly distill a $2N$ -step DDIM sampler into an N -step student. For each student step ($x_t \rightarrow x_{t-\Delta}$), two consecutive teacher steps ($x_t \rightarrow x_{t-\Delta/2} \rightarrow x_{t-\Delta}$) provide the training target. Iterating this halving procedure $\log_2 T$ times compresses T steps to 1 with minimal quality loss in practice.

Remark 6.5 (Equivalence of consistency-type models). Lai et al. [24] show that consistency distillation and progressive distillation are equivalent under mild regularity conditions: both minimize a weighted integral of the PF-ODE truncation error, and their limiting objectives coincide as the step size $\Delta t \rightarrow 0$. This unifies the two approaches under a single theoretical framework.

6.4 Discrete Diffusion

Diffusion in continuous $\mathbb{R}^d$ relies on Gaussian perturbations. For discrete data—tokens, graphs, molecular structures—we require a diffusion process on a *finite* state space $\mathcal{V} = \{1, \dots, K\}$. The D3PM framework [3] provides the natural analogue.

6.4.1 Forward process: continuous-time Markov chain

Fix a finite state space $\mathcal{V}$ (e.g., a vocabulary of tokens). For a single coordinate $x \in \mathcal{V}$, the *forward process* is a continuous-time Markov chain (CTMC) governed by a *rate matrix* $Q_t \in \mathbb{R}^{K \times K}$, with entries:

- $Q_t[j, i] \geq 0$ for $j \neq i$: the instantaneous rate of transitioning from state i to state j ,
- $Q_t[i, i] = -\sum_{j \neq i} Q_t[j, i]$: each column sums to zero.

The marginal distribution $\mathbf{p}_t \in \Delta^{K-1}$ (probability simplex) satisfies the *Kolmogorov forward equation* (the discrete analogue of the Fokker–Planck equation):

$$\frac{d\mathbf{p}_t}{dt} = Q_t \mathbf{p}_t. \quad (6.9)$$

6.4.2 Transition matrices and marginals

Define the *cumulative transition matrix* $\bar{Q}_t := \mathcal{T} \exp\left(\int_0^t Q_s ds\right)$, the time-ordered matrix exponential. For the common case of *time-homogeneous* rates or piecewise-constant Q_t , $\bar{Q}_t = e^{Q_t}$. The conditional marginal distribution given x_0 is:

$$q(x_t | x_0) = \text{Cat}(x_t; \bar{Q}_t \mathbf{e}_{x_0}), \quad (6.10)$$

where $\mathbf{e}_{x_0} \in \{0, 1\}^K$ is the one-hot encoding of x_0 . The discrete-time version of Equation (6.10) uses $\bar{Q}_{t_n} = Q_{t_n} Q_{t_{n-1}} \cdots Q_{t_1}$ with per-step transition matrices $Q_{t_i} = I + \Delta t \cdot Q$ (Euler) or exact matrix exponentials.

Specific choices of the rate matrix Q yield interpretable forward processes:

- (i) **Uniform (masking)**: $Q = \beta_t(\frac{1}{K}\mathbf{1}\mathbf{1}^\top - I)$, converging to the uniform distribution.
- (ii) **Absorbing (MASK token)**: $Q[K, i] = \beta_t$ for all $i \neq K$, $Q[K, K] = 0$, so tokens are irreversibly masked at rate β_t .
- (iii) **Nearest-neighbour (on ordered states)**: $Q[i+1, i] = Q[i-1, i] = \beta_t/2$, inducing a discrete random walk.

6.4.3 Closed-form reverse kernel

By Bayes' theorem, the reverse kernel conditioned on x_0 is tractable:

$$q(x_{t-1} | x_t, x_0) = \frac{q(x_t | x_{t-1}) q(x_{t-1} | x_0)}{q(x_t | x_0)}. \quad (6.11)$$

In one-hot notation with Q_t as the per-step (discrete-time) transition matrix and $\bar{Q}_t$ as the cumulative matrix:

$$q(x_{t-1} | x_t, x_0) = \text{Cat}\left(x_{t-1}; \frac{Q_t^\top \mathbf{e}_{x_t} \odot \bar{Q}_{t-1} \mathbf{e}_{x_0}}{\mathbf{e}_{x_t}^\top \bar{Q}_t \mathbf{e}_{x_0}}\right), \quad (6.12)$$

where $\odot$ is the elementwise (Hadamard) product. As in the Gaussian case ([Proposition 2.4](#)), this posterior is *analytically tractable* given x_0 , allowing the model to be trained by predicting $\hat{x}_0$ from x_t and substituting into [Equation \(6.12\)](#).

6.4.4 Training objective: discrete ELBO

The D3PM generative model $p_\theta(x_{t-1} | x_t)$ is trained by minimizing the negative ELBO. Following the hierarchical VAE derivation ([Chapter 1](#)), the ELBO decomposes into per-step KL divergences:

$$\mathcal{L}_{\text{D3PM}}(\theta) = \mathbb{E}_q \left[\underbrace{-\log p_\theta(x_0 | x_1)}_{L_0} + \sum_{t=2}^T \underbrace{D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \parallel p_\theta(x_{t-1} | x_t))}_{L_{t-1}} + \underbrace{D_{\text{KL}}(q(x_T | x_0) \parallel p(x_T))}_{L_T} \right]. \quad (6.13)$$

The prior L_T is a constant once the forward process is fixed. Each L_{t-1} involves the *categorical* KL between two distributions over $\mathcal{V}$, which has the closed form $D_{\text{KL}}(p \parallel q) = \sum_{k=1}^K p_k \log(p_k/q_k)$. In practice, the model $p_\theta(x_{t-1}|x_t)$ is parameterized by predicting $\hat{x}_0 = \tilde{x}_{0,\theta}(x_t, t)$ and computing $q(x_{t-1}|x_t, \hat{x}_0)$ via [Equation \(6.12\)](#). An auxiliary cross-entropy loss $-\log p_\theta(x_0|x_t)$ is added to encourage accurate x_0 prediction.

Table 6.1: Structural parallel between continuous DDPM and discrete D3PM.

	DDPM (continuous)	D3PM (discrete)
State space	$\mathbb{R}^d$	$\mathcal{V}^d$ (finite)
Forward process	$q(x_t x_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t} x_{t-1}, \beta_t I)$	$q(x_t x_{t-1}) = \text{Cat}(x_t; Q_t \mathbf{e}_{x_{t-1}})$
Marginal	$q(x_t x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t)I)$	$q(x_t x_0) = \text{Cat}(x_t; \bar{Q}_t \mathbf{e}_{x_0})$
Posterior	Gaussian (Proposition 2.4)	Categorical (Equation (6.12))
Forward eq.	Fokker–Planck (cont.)	Kolmogorov forward (Equation (6.9))
Training loss	$\mathbb{E} \ \epsilon - \epsilon_\theta\ ^2$	KL per step (Equation (6.13))
Model output	$\hat{\epsilon}_\theta(x_t, t)$	$\hat{x}_{0,\theta}(x_t, t) \in \Delta^{K-1}$

7. Diffusion for Inverse Problems

Diffusion models provide a pre-trained prior over the data distribution $p_{\text{data}}(x_0)$. This chapter shows how to combine that prior with a likelihood model $p(y | x_0)$ to draw samples from the posterior $p(x_0 | y)$ —the fundamental goal of *Bayesian inverse problems*. All methods hinge on a single exact identity, the *posterior score decomposition*, and differ only in how they approximate the intractable likelihood score. We survey the four major approximation families (spectral/projection, gradient/Tweedie, plug-and-play, and variational) and the asymptotically exact SMC approach.

7.1 Problem Setup

Definition 7.1 (Forward measurement model). Let $x_0 \in \mathbb{R}^d$ be the unknown signal and $y \in \mathbb{R}^m$ the observation. We consider two cases:

$$\text{(linear): } y = Ax_0 + n, \quad A \in \mathbb{R}^{m \times d}, \quad (7.1)$$

$$\text{(nonlinear): } y = h(x_0) + n, \quad h : \mathbb{R}^d \rightarrow \mathbb{R}^m \text{ differentiable}, \quad (7.2)$$

with measurement noise $n \sim \mathcal{N}(0, \sigma_y^2 I)$.

The Bayesian goal is to sample from the posterior

$$p(x_0 | y) \propto p_{\text{data}}(x_0) p(y | x_0), \quad p(y | x_0) = \mathcal{N}(y; Ax_0, \sigma_y^2 I). \quad (7.3)$$

Direct sampling from [Equation \(7.3\)](#) is intractable for high-dimensional x_0 because p_{data} has no closed form. Diffusion models replace p_{data} by a learned score $s_\theta \approx \nabla \log p_t$ and run a reverse process to produce approximate samples.

Proposition 7.2 (Posterior score decomposition). *For any noise level t , let $p_t(x_t | y)$ denote the t -marginal of the posterior process. Then*

$$\nabla_{x_t} \log p_t(x_t | y) = \underbrace{\nabla_{x_t} \log p_t(x_t)}_{\approx s_\theta(x_t, t)} + \underbrace{\nabla_{x_t} \log p_t(y | x_t)}_{\text{intractable likelihood score}}. \quad (7.4)$$

Proof. From Bayes' theorem, $p_t(x_t | y) \propto p_t(x_t) p_t(y | x_t)$. Taking $\nabla_{x_t} \log$ of both sides gives [Equation \(7.4\)](#); the normalizing constant $p_t(y)$ vanishes under differentiation. $\square$

Why the likelihood score is intractable. The marginal likelihood at noise level t requires integrating out x_0 :

$$p_t(y | x_t) = \int p(y | x_0) p_t(x_0 | x_t) dx_0. \quad (7.5)$$

Even for linear A and Gaussian noise, the integrand is non-Gaussian in x_0 (because $p_t(x_0 | x_t)$ is a Gaussian mixture over all data points). Every method below is a different approximation of Equation (7.5).

7.2 Approximation Taxonomy

Proposition 7.3 (Likelihood score via posterior expectation). *Let $\mathcal{E}(x_0, y) := -\log p(y | x_0)$ denote the negative log-likelihood. For the Gaussian model Equation (7.1), an exact identity holds:*

$$\nabla_{x_t} \log p_t(y | x_t) = \mathbb{E}_{x_0 \sim p_t(x_0 | x_t)} \left[J_{\hat{x}_0}^\top \nabla_{\hat{x}_0} \log p(y | x_0) \right]_{\hat{x}_0=x_0} + \text{Cov-correction}, \quad (7.6)$$

where $J_{\hat{x}_0}(x_t) = \partial \hat{x}_0(x_t) / \partial x_t$ is the Jacobian of the Tweedie denoiser and the covariance-correction term is of order $O(r_t^2)$ with $r_t^2 = \text{Var}(x_0 | x_t)$.

The covariance-correction term vanishes as $t \rightarrow 0$ (no noise, $r_t \rightarrow 0$) and dominates at large t (pure noise). All practical methods drop or approximate this term.

Table 7.1: Taxonomy of approximation strategies for $\nabla_{x_t} \log p_t(y | x_t)$.

Branch	Approx. of $p_t(x_0 x_t)$	Key methods	Exact?
Spectral/projection	$\mathcal{N}(\hat{x}_0, r_t^2 I)$	DDRM, Π GdM, DDNM	Linear A only
Gradient (Tweedie)	$\delta_{\hat{x}_0(x_t)}$	DPS, loss-guided	No ($O(r_t^2)$ bias)
Variable splitting	Implicit (denoiser)	DiffPIR, DAPS	No
Variational inference	Parametric q_ϕ	VDPS, Mardani et al.	In limit
Sequential MC	Empirical (particles)	MCGDiff, TDS	As $N \rightarrow \infty$

7.3 Linear and Spectral Methods

When A is linear and n is Gaussian, the Gaussian approximation $p_t(x_0 | x_t) \approx \mathcal{N}(\hat{x}_0, r_t^2 I)$ enables *closed-form* likelihood score computation via the Gaussian product formula (Lemma 2.1).

7.3.1 DDRM: Denoising diffusion restoration models

Kawar et al. [21] decompose A via SVD as $A = U \Sigma V^\top$ and work in the rotated basis $\tilde{x}_0 = V^\top x_0$, $\tilde{y} = U^\top y$. In this basis the system decouples componentwise:

$$\tilde{y}_i = \sigma_i^A \tilde{x}_{0,i} + \tilde{n}_i, \quad \tilde{n}_i \sim \mathcal{N}(0, \sigma_y^2), \quad (7.7)$$

where σ_i^A is the i -th singular value of A .

Proposition 7.4 (DDRM posterior in SVD basis). *Approximate $p_t(\tilde{x}_{0,i} | \tilde{x}_{t,i}) \approx \mathcal{N}(\hat{\tilde{x}}_{0,i}, r_t^2)$. Combin-*

ing with the likelihood [Equation \(7.7\)](#) via Bayes:

$$p_t(\tilde{x}_{0,i} | \tilde{x}_{t,i}, \tilde{y}_i) \approx \mathcal{N}(\hat{\tilde{x}}_{0,i}^y, \tilde{r}_{t,i}^2),$$

$$\hat{\tilde{x}}_{0,i}^y = \frac{(\sigma_i^A)^2 r_t^2 (\tilde{y}_i / \sigma_i^A) + \sigma_y^2 \hat{\tilde{x}}_{0,i}}{(\sigma_i^A)^2 r_t^2 + \sigma_y^2}, \quad \tilde{r}_{t,i}^2 = \frac{\sigma_y^2 r_t^2}{(\sigma_i^A)^2 r_t^2 + \sigma_y^2}. \quad (7.8)$$

For null-space components ($\sigma_i^A = 0$) the likelihood is uninformative: $\hat{\tilde{x}}_{0,i}^y = \hat{\tilde{x}}_{0,i}$ (purely from the diffusion prior).

Proof. Direct application of [Lemma 2.1](#) (Gaussian product formula): prior $\mathcal{N}(\hat{\tilde{x}}_{0,i}, r_t^2)$ times likelihood $\mathcal{N}(\tilde{y}_i; \sigma_i^A \tilde{x}, \sigma_y^2)$ gives a Gaussian with the stated parameters. $\square$

The DDRM reverse step uses $\hat{\tilde{x}}_0^y$ (assembled from [Equation \(7.8\)](#)) as the corrected denoised estimate at each diffusion step. The SVD needs to be computed once; the per-step cost is $O(m)$ elementwise operations.

7.3.2 DDNM: Denoising diffusion null-space model

Wang et al. [43] handle the *noiseless* case $\sigma_y = 0$. In this limit the range-space components of $\hat{x}_0$ must satisfy $A\hat{x}_0 = y$ exactly, while the null-space components are unconstrained. The correction is a simple projection:

$$\hat{x}_0^y = A^+ y + (I - A^+ A) \hat{x}_0, \quad (7.9)$$

where $A^+ = V_r \Sigma_r^{-1} U_r^\top$ is the Moore–Penrose pseudo-inverse ($r = \text{rank}(A)$, subscript r denotes the top- r SVD factors). One verifies $A\hat{x}_0^y = y$ immediately: $AA^+ y = U_r U_r^\top y = y$ (since $y \in \text{range}(A)$ by assumption) and $A(I - A^+ A) = 0$.

Remark 7.5. [Equation \(7.9\)](#) is *gradient-free*: no backpropagation through s_θ is needed. For noisy measurements, a range-space noise term is added: $\hat{x}_0^y = A^+(y - \eta_t \tilde{n}) + (I - A^+ A) \hat{x}_0$ with $\tilde{n} \sim \mathcal{N}(0, I)$ and schedule $\eta_t \propto \sigma_y \sqrt{1 - \bar{\alpha}_t}$.

7.3.3 Π GdM: Pseudoinverse-guided diffusion

Song et al. [34] approximate the full Gaussian posterior for the noisy case. With $p_t(x_0 | x_t) \approx \mathcal{N}(\hat{x}_0, r_t^2 I)$, marginalizing x_0 :

$$p_t(y | x_t) \approx \mathcal{N}(y; A\hat{x}_0, \sigma_y^2 I + r_t^2 AA^\top). \quad (7.10)$$

Taking $\nabla_{x_t} \log$ of [Equation \(7.10\)](#) and applying the chain rule:

$$\nabla_{x_t} \log p_t(y | x_t) \approx J_{\hat{x}_0}^\top A^\top \underbrace{(\sigma_y^2 I + r_t^2 AA^\top)^{-1}}_{\text{covariance-corrected}} (y - A\hat{x}_0). \quad (7.11)$$

As $\sigma_y \rightarrow 0$, $(\sigma_y^2 I + r_t^2 AA^\top)^{-1} A \rightarrow (AA^\top)^+ A = (A^+)^\top$ and [Equation \(7.11\)](#) reduces to $J_{\hat{x}_0}^\top A^{\top+} (y - A\hat{x}_0)$, recovering the DDNM null-space structure but now *with* the Jacobian term that accounts for the dependence of $\hat{x}_0$ on x_t .

Remark 7.6 (Posterior variance schedule). For the VP-SDE, a natural choice is $r_t^2 = (1 - \bar{\alpha}_t)/\bar{\alpha}_t$,

which is the posterior variance of $x_0 \mid x_t$ if p_{data} were itself Gaussian. For general data distributions this is an approximation; the practical guidance is to tune r_t as a hyperparameter.

7.4 Gradient-Based Methods

7.4.1 DPS: Complete derivation

Chung et al. [9] approximate the posterior $p_t(x_0 \mid x_t)$ by a point mass at the Tweedie denoiser:

$$p_t(x_0 \mid x_t) \approx \delta_{x_0=\hat{x}_0(x_t)}, \quad \hat{x}_0(x_t) := \mathbb{E}[x_0 \mid x_t] = \frac{x_t + (1 - \bar{\alpha}_t) s_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}}. \quad (7.12)$$

The consequent likelihood approximation is

$$p_t(y \mid x_t) \approx p(y \mid \hat{x}_0(x_t)). \quad (7.13)$$

Step 1: Likelihood gradient via chain rule.

$$\begin{aligned} \nabla_{x_t} \log p_t(y \mid x_t) &\approx \nabla_{x_t} \log p(y \mid \hat{x}_0(x_t)) \\ &= J_{\hat{x}_0}(x_t)^\top \nabla_{\hat{x}_0} \log p(y \mid \hat{x}_0), \end{aligned} \quad (7.14)$$

where $J_{\hat{x}_0}(x_t) = \partial \hat{x}_0 / \partial x_t \in \mathbb{R}^{d \times d}$ is the Jacobian of the denoiser.

Step 2: Jacobian of the Tweedie denoiser. From Equation (7.12):

$$J_{\hat{x}_0}(x_t) = \frac{\partial \hat{x}_0}{\partial x_t} = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(I + (1 - \bar{\alpha}_t) \frac{\partial s_\theta(x_t, t)}{\partial x_t} \right). \quad (7.15)$$

Computing the full Hessian $\partial s_\theta / \partial x_t \in \mathbb{R}^{d \times d}$ is expensive ($O(d^2)$ backpropagation). DPS avoids this by using *automatic differentiation of the scalar loss*:

$$J_{\hat{x}_0}^\top \nabla_{\hat{x}_0} \log p(y \mid \hat{x}_0) = \nabla_{x_t} [\log p(y \mid \hat{x}_0(x_t))], \quad (7.16)$$

which costs a single backward pass through s_θ — $O(d)$ operations.

Step 3: Explicit gradient for the linear–Gaussian model. For $p(y \mid x_0) = \mathcal{N}(y; Ax_0, \sigma_y^2 I)$:

$$\nabla_{x_0} \log p(y \mid x_0) = -\frac{1}{\sigma_y^2} A^\top (Ax_0 - y). \quad (7.17)$$

Substituting into Equation (7.16):

$$\nabla_{x_t} \log p_t(y \mid x_t) \approx -\frac{1}{\sigma_y^2} \nabla_{x_t} \|y - A\hat{x}_0(x_t)\|^2. \quad (7.18)$$

For the nonlinear model Equation (7.2): $\nabla_{x_t} \log p_t(y \mid x_t) \approx -\frac{1}{\sigma_y^2} \nabla_{x_t} \|y - h(\hat{x}_0(x_t))\|^2$, which requires only that h be differentiable—no linearity assumed.

Step 4: Modified reverse process. Substituting Equation (7.18) into Equation (7.4) and running the DDPM reverse step (Equation (2.10)) with the posterior score gives the DPS update:

$$x_{t-1} \leftarrow \underbrace{\mu_\theta(x_t, t)}_{\text{DDPM reverse mean}} - \underbrace{\zeta_t \nabla_{x_t} \|y - A\hat{x}_0(x_t)\|^2}_{\text{likelihood gradient step}}, \quad (7.19)$$

where $\zeta_t > 0$ is a step-size schedule. The score term s_θ is evaluated once; the gradient in the second term is obtained by backpropagation through the same forward pass.

Approximation error analysis. The Dirac approximation introduces a bias proportional to the posterior variance r_t^2 . A Taylor expansion of $\log p_t(y | x_t)$ around $\hat{x}_0$ gives

$$\log p_t(y | x_t) = \log p(y | \hat{x}_0) - \frac{r_t^2}{2\sigma_y^2} \text{tr}(A^\top A) + O(r_t^4), \quad (7.20)$$

so the gradient error is $O(r_t^2)$. Since $r_t^2 \rightarrow 0$ as $t \rightarrow 0$, the approximation is *asymptotically exact* near the end of the reverse process—precisely where it matters most for sample quality.

Key Idea

DPS trades exactness for generality: the Tweedie gradient Equation (7.18) works for *any* differentiable h , requires only one backward pass per step, and needs no SVD precomputation. The cost is a bias of $O(r_t^2)$ that is large at high noise levels but vanishes as generation completes.

7.4.2 Loss-guided diffusion and extensions

Song et al. [35] generalize Equation (7.18) to an arbitrary energy $\mathcal{E}: \mathbb{R}^d \times \mathbb{R}^m \rightarrow \mathbb{R}$:

$$\nabla_{x_t} \log p_t(y | x_t) \approx -\lambda \nabla_{x_t} \mathcal{E}(\hat{x}_0(x_t), y), \quad (7.21)$$

with scale $\lambda > 0$. This subsumes DPS ($\mathcal{E} = \|y - A\hat{x}_0\|^2 / (2\sigma_y^2)$), classifier guidance (Section 6.1.1), and perceptual or style losses used in image editing. The choice of $\mathcal{E}$ encodes the task; the diffusion score provides the prior.

7.5 Plug-and-Play Diffusion Priors

Plug-and-play (P&P) methods [20, 41] reformulate the inverse problem as a constrained optimization and solve it by alternating between a data-consistency step and a denoising (prior) step. Crucially, the denoiser does *not* need to be trained jointly with the measurement operator A .

7.5.1 Half-quadratic splitting (HQS)

The MAP estimation problem $\hat{x}_0 = \arg \min_{x_0} \frac{1}{2\sigma_y^2} \|y - Ax_0\|^2 - \log p_{\text{data}}(x_0)$ is split by introducing an auxiliary variable v and a quadratic penalty:

$$\min_{x_0, v} \frac{1}{2\sigma_y^2} \|y - Ax_0\|^2 + \frac{\mu}{2} \|x_0 - v\|^2 - \log p_{\text{data}}(v), \quad (7.22)$$

which alternates between two closed-form subproblems as $\mu \rightarrow \infty$:

$$x_0^{(k+1)} = \arg \min_{x_0} \frac{1}{2\sigma_y^2} \|y - Ax_0\|^2 + \frac{\mu}{2} \|x_0 - v^{(k)}\|^2 = \left(A^\top A / \sigma_y^2 + \mu I \right)^{-1} \left(A^\top y / \sigma_y^2 + \mu v^{(k)} \right), \quad (7.23)$$

$$v^{(k+1)} = \arg \min_v \frac{\mu}{2} \|x_0^{(k+1)} - v\|^2 - \log p_{\text{data}}(v) = D_{\sigma_k}(x_0^{(k+1)}), \quad (7.24)$$

where D_{σ_k} is any denoiser with noise level $\sigma_k = 1/\sqrt{\mu}$.

7.5.2 Denoiser–score connection

The HQS denoising step [Equation \(7.24\)](#) implicitly defines a prior through the denoiser D_σ . The fundamental link to score-based models is:

$$D_\sigma(x) = x + \sigma^2 \nabla \log p_\sigma(x), \quad (7.25)$$

where $p_\sigma(x) = \int p_{\text{data}}(x_0) \mathcal{N}(x; x_0, \sigma^2 I) dx_0$ is the noise-convolved data distribution. Equivalently, $D_\sigma(x) - x = \sigma^2 \nabla \log p_\sigma(x)$ is the score scaled by σ^2 .

Proposition 7.7 (P&P as score-based method). *[30] The P&P prior implicit in the denoiser D_σ equals the score-based prior $\nabla \log p_\sigma$ via [Equation \(7.25\)](#). Consequently, the HQS iterates [Equation \(7.23\)](#)–[Equation \(7.24\)](#) are equivalent to proximal gradient descent on the objective $\frac{1}{2\sigma_y^2} \|y - Ax\|^2 - \log p_\sigma(x)$, with the score evaluated at each iterate.*

7.5.3 DiffPIR and DAPS

DiffPIR [45] integrates the diffusion reverse process with the HQS loop: at time step t , the diffusion denoiser $\hat{x}_0(x_t, t) = \mathbb{E}[x_0 | x_t]$ plays the role of D_{σ_k} in [Equation \(7.24\)](#). The data-consistency step [Equation \(7.23\)](#) is inserted between consecutive diffusion reverse steps. The noise level $\sigma_k = \sqrt{1 - \bar{\alpha}_t} / \sqrt{\bar{\alpha}_t}$ matches the diffusion posterior variance, ensuring the two loops are synchronized throughout the reverse trajectory.

DAPS [44] (Decoupled Annealing Posterior Sampling) decouples the diffusion noise schedule from the data-consistency updates. Rather than inserting one data-consistency step per diffusion step, DAPS runs multiple HQS iterations at each noise level, then *renoises* the result to feed the next diffusion step. This produces a more stable annealing trajectory and reduces the sensitivity to the choice of μ in [Equation \(7.22\)](#).

Key Idea

P&P methods reframe the inverse problem as a sequence of denoising and data-projection steps. Their key advantage over DPS is *no backpropagation through s_θ* : the gradient step [Equation \(7.23\)](#) involves only $A^\top A$, which is often available in closed form (e.g., for convolutional A , via FFT). The cost is that global convergence guarantees require convexity of $-\log p_{\text{data}}$, which holds approximately only for well-trained denoisers.

7.6 Variational Inference Perspective

The methods above all modify the *reverse trajectory* to steer sampling toward the posterior. A conceptually distinct paradigm directly *optimizes* a variational distribution q_ϕ to approximate $p(x_0 | y)$ [14, 28].

7.6.1 Variational ELBO with diffusion score

Let $q_\phi(x_0)$ be a tractable family (e.g., Gaussian, or implicit via a flow). The variational objective is

$$\min_{\phi} \text{KL}(q_\phi(x_0) \parallel p(x_0 | y)) = \underbrace{\text{KL}(q_\phi(x_0) \parallel p_{\text{data}}(x_0))}_{\text{prior KL: uses diffusion score}} - \mathbb{E}_{q_\phi}[\log p(y | x_0)] + \text{const.} \quad (7.26)$$

The first term is intractable in closed form, but it can be estimated via the *diffusion DSM loss*:

$$\text{KL}(q_\phi \parallel p_{\text{data}}) \approx \int_0^T \lambda_t \mathbb{E}_{q_\phi(x_0), \mathcal{N}(\epsilon)} \left[\|s_\theta(x_t, t) - \nabla_{x_t} \log q_t(x_t)\|^2 \right] dt, \quad (7.27)$$

where $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$ and q_t is the noisy marginal of q_ϕ .

Proposition 7.8 (Score as KL surrogate). *The DSM loss Equation (7.27) equals $\text{KL}(q_\phi \parallel p_{\text{data}})$ up to a constant independent of ϕ . Minimizing Equation (7.27) with respect to ϕ is therefore equivalent to minimizing $\text{KL}(q_\phi \parallel p_{\text{data}})$, and the pre-trained score s_θ provides gradients without retraining the diffusion model. Here s_θ is the pre-trained score and is treated as fixed, gradients are taken with respect to ϕ only.*

7.6.2 Comparison with trajectory-based methods

The key distinction is in *where the conditioning happens*: trajectory-based methods (DPS, P&P, SMC) incorporate y during the stochastic reverse process, perturbing each reverse step; variational methods incorporate y through the ELBO objective Equation (7.26) and optimize over the initial distribution q_ϕ .

Variational methods require optimization at test time (inference-time optimization), which adds compute but enables principled uncertainty quantification via the variational distribution q_ϕ . Feng et al. [14] show that, as the variational family becomes expressive enough, q_ϕ^* converges to the true posterior $p(x_0 | y)$.

7.7 Sequential Monte Carlo

Sequential Monte Carlo (SMC) methods provide *asymptotically exact* posterior sampling by approximating the target distribution with a weighted particle system.

7.7.1 State-space formulation

Regard the diffusion chain $(x_T, x_{T-1}, \dots, x_0)$ as a hidden Markov model (HMM) with:

$$\text{prior transitions: } x_{t-1} | x_t \sim p_\theta(x_{t-1} | x_t) \quad (\text{the learned DDPM reverse}), \quad (7.28)$$

$$\text{observation: } y | x_0 \sim p(y | x_0) \quad (\text{the likelihood}). \quad (7.29)$$

The joint distribution over the full chain is $p_T(x_T) \prod_{t=1}^T p_\theta(x_{t-1} | x_t)$, denoted as $p_\theta(x_{0:T})$, and the posterior is $p_\theta(x_{0:T} | y) \propto p_\theta(x_{0:T}) p(y | x_0)$.

7.7.2 Twisted proposals and MCGDiff

Naïve particle filtering uses the prior reverse Equation (7.28) as the proposal and weights particles at $t = 0$ by $p(y | x_0^{(i)})$. This is inefficient because most particles will be in low-likelihood

regions.

Twisted proposals pre-multiply the reverse kernel by a *potential* $\psi_t(x_t, y)$ that approximates $p(y | x_t)$:

$$\tilde{q}(x_{t-1} | x_t, y) \propto p_\theta(x_{t-1} | x_t) \psi_{t-1}(x_{t-1}, y). \quad (7.30)$$

Importance weights correct for the deviation from the true posterior.

MCGDiff [5] uses the Gaussian approximation $\psi_t(x_t, y) \approx \mathcal{N}(y; A\hat{x}_0(x_t), \sigma_y^2 I + r_t^2 AA^\top)$ as the potential (the same approximation as ΠGdM). The key result is:

Theorem 7.9 (MCGDiff consistency). *As the number of particles $N \rightarrow \infty$, the MCGDiff empirical measure converges in total variation to the true posterior $p_\theta(x_0 | y)$, provided the twist potentials $\{\psi_t\}$ are uniformly bounded away from zero and the score estimate satisfies $\|s_\theta - \nabla \log p_t\|_2 \leq \varepsilon$ uniformly.*

7.7.3 Filtering perspective

Dou and Song [12] observe that the posterior reverse process is itself a diffusion: the Doob h -transform with $h_t(x_t) = p_t(y | x_t)$ turns the unconditional reverse SDE into a conditional one. Specifically, adding the term $\nabla_{x_t} \log p_t(y | x_t)$ to the reverse drift (Equation (7.4)) gives an SDE whose marginals are exactly $p_t(x_t | y)$. The twisted SMC proposals Equation (7.30) are then viewed as *numerical approximations of this conditional SDE*, with the particle system serving as the discretization.

Key Idea

SMC methods are the most principled: they converge to the exact posterior as $N \rightarrow \infty$ without any approximation of the likelihood. Their practical limitation is cost: N score evaluations per reverse step versus $O(1)$ for DPS. Current work [10] explores adaptive resampling and learned potentials to reduce N while maintaining coverage.

7.8 Error Analysis

All trajectory-based methods compound three error sources:

1. Approximation error δ_{app} . Replacing the true likelihood score by an approximation introduces a systematic bias. For DPS this is $O(r_t^2)$ per step (Equation (7.20)); for ΠGdM it is of the same order; for DDRM the error is zero within the Gaussian approximation (only the Gaussian approximation of $p_t(x_0 | x_t)$ contributes).

2. Discretization error δ_{disc} . Using finite step sizes $h = 1/T$ introduces truncation error $O(h)$ for Euler–Maruyama and $O(h^2)$ for Heun/second-order solvers.

3. Score estimation error $\varepsilon_{\text{score}}$. The network $s_\theta \neq \nabla \log p_t$ introduces an irreducible error that propagates through all T steps.

Proposition 7.10 (TV bound for DPS-type methods). *Under mild regularity assumptions (Lipschitz score, bounded data distribution), the total variation distance between the DPS output distribution $\hat{p}$ and the true posterior $p(x_0 | y)$ satisfies*

$$\text{TV}(\hat{p}, p(\cdot | y)) \leq C(\varepsilon_{\text{score}} + \delta_{\text{app}} + T^{-1/2}), \quad (7.31)$$

for a constant C depending on the Lipschitz constants and the operator norm $\|A\|$. Tight constants for nonlinear h and non-Gaussian data remain open [10].

Table 7.2: Summary comparison of diffusion-based inverse problem solvers.

Method	Nonlinear h ?	Grad. through s_θ ?	SVD needed?	Exact?
DDRM	No	No	Yes	In Gaussian approx.
DDNM	No	No	Yes	Noiseless only
Π GdM	No	Yes	Yes	In Gaussian approx.
DPS	Yes	Yes	No	No ($O(r_t^2)$ bias)
DiffPIR	No	No	Optional	No
DAPS	No	No	Optional	No
MCGDiff	Yes	No	No	As $N \rightarrow \infty$

8. Conclusion

8.1 Mathematical Synthesis

These notes develop one principle from six vantage points. The principle is: *to generate a sample from p_{data} , learn to reverse a noise process that destroys it*. Six chapters formalize this in increasingly abstract language, yet every formalization touches the same three invariant objects.

The three invariant objects.

- (i) **The marginal path** $\{p_t\}_{t \in [0, T]}$: a curve in the space of probability distributions connecting p_{data} to a tractable prior $p_T \approx \mathcal{N}(0, I)$. Every diffusion framework is a choice of how to parameterize this path.
- (ii) **The score field** $\nabla \log p_t$: the gradient of the log-density along the path. It is the *lingua franca*: every model learns it, every sampler uses it, every extension modifies or conditions it. Tweedie’s formula ([Theorem 2.2](#)) links the score to the minimal-variance denoiser $\hat{x}_0(x_t) = \mathbb{E}[x_0 \mid x_t]$ and to the noise predictor $\hat{\epsilon}(x_t)$ via

$$\nabla \log p_t(x_t) = \frac{\sqrt{\bar{\alpha}_t} \hat{x}_0(x_t) - x_t}{1 - \bar{\alpha}_t} = -\frac{\hat{\epsilon}(x_t)}{\sqrt{1 - \bar{\alpha}_t}}. \quad (8.1)$$

- (iii) **The probability flow ODE** ([Corollary 5.4](#)): the unique *deterministic* transport with marginals $\{p_t\}$. It is the zero-noise member of the SDE equivalence class ([Theorem 5.3](#)), enabling exact likelihood computation ([Corollary 4.8](#)) and serving as the foundation for fast samplers.

How each chapter contributes.

- **Chapter 1 (VAE)**: establishes the variational framing. The ELBO bound and the reparametrization trick reappear verbatim in DDPM’s hierarchical VAE interpretation ([Section 2.4](#)).
- **Chapter 2 (DDPM)**: defines the path $\{p_t\}$ as a discrete Markov chain and derives the simplified objective $\mathcal{L}_{\text{simple}} = \mathbb{E} \|\epsilon - \epsilon_\theta\|^2$ ([Equation \(2.16\)](#)), which is a reweighted DSM loss targeting the score. DDIM ([Section 2.6](#)) replaces the stochastic reverse with the PF-ODE.
- **Chapter 3 (SMLD)**: takes the path to continuous time via the VP/VE-SDE family. Anderson’s reverse-time SDE ([Theorem 3.17](#)) generalizes the discrete reverse chain; score matching ([Section 3.3](#)) provides the training signal.

- **Chapter 4 (Flow Matching):** parameterizes the path through its *velocity field* rather than through noise. The FM loss (Equation (4.16)) and the DSM loss are equivalent up to a linear change of variables (Proposition 4.19), confirming that both train the same underlying object.
- **Chapter 5 (Unification):** makes all equivalences explicit. The DDPM Markov chain converges to the VP-SDE (Proposition 5.1); all SDEs with marginals $\{p_t\}$ form an equivalence class (Theorem 5.3); all training objectives minimize the same loss up to $\lambda(t)$ -reweighting (Proposition 5.6).
- **Chapter 6 (Extensions):** shows how to condition, compress, accelerate, and discretize. Each extension modifies exactly one of the three invariant objects: conditioning modifies the score (Equation (6.1)); latent diffusion moves the path to a lower-dimensional space; consistency models replace the PF-ODE with a self-consistent mapping; discrete diffusion replaces the Gaussian marginals with categorical ones.
- **Chapter 7 (Inverse Problems):** recasts posterior sampling as conditional generation. The posterior score decomposition (Proposition 7.2) is Equation (6.1) with a physically meaningful $p(y | x_0)$; all inverse solvers approximate the intractable term $\nabla_{x_t} \log p_t(y | x_t)$.

Key Idea

The entire edifice rests on a single formula:

$$dx = \left[f_t(x) - \frac{g_t^2}{2} \nabla \log p_t(x) \right] dt + g_t d\bar{W}_t \quad (\text{reverse-time SDE, Equation (3.19)}).$$

Training provides $\nabla \log p_t$; sampling integrates this SDE backward. Every other equation in these notes either derives this formula, trains the score, or extends it.

8.2 Open Problems

The field has matured rapidly, but several foundational questions remain open.

Convergence theory. Existing guarantees (e.g., Chen et al. 8) require Lipschitz scores and smooth data; real distributions are neither. Tight convergence rates for the empirical risk minimizer s_θ (not the population-level score) under finite training data are unknown. The gap between the *learned* score and the *true* score is the main open term in Equation (7.31).

Information-optimal sampling. Given a fixed NFE budget K , what is the optimal allocation of noise levels, step sizes, and stochastic vs. deterministic steps? Consistency models (Section 6.3.1) and progressive distillation (Section 6.3.2) give empirically good answers, but a theoretical characterization in terms of the spectrum of $A^\top A$ (for inverse problems) or the Fisher information of p_{data} is missing.

Exact posterior sampling for inverse problems. All trajectory-based methods (Sections 7.4 and 7.5) introduce an $O(r_t^2)$ approximation error. The SMC methods of Section 7.7 are asymptotically exact but computationally expensive. A central open question is whether there exist methods that are both *polynomial-time* and *exact in the limit* for nonlinear operators h (Equation (7.2)).

Nonlinear inverse problems. For linear A , DDRM and Π GdM exploit the SVD structure for closed-form posterior updates. For nonlinear h , DPS provides a tractable gradient but with an uncontrolled bias that may be large when h is highly curved. Tight error bounds analogous to [Proposition 7.10](#) for nonlinear h —in terms of the curvature of h and the posterior variance r_t^2 —remain open.

Optimal step-size schedules. The DPS step-size ζ_t ([Equation \(7.19\)](#)) is typically hand-tuned. A principled derivation of ζ_t^* as a function of t , σ_y , $\|A\|$, and the score accuracy would remove this hyperparameter. The connection to the covariance-correction term in [Equation \(7.6\)](#) suggests that $\zeta_t \propto r_t^2 / \sigma_y^2$ is a natural candidate, but this has not been rigorously established.

Discrete and structured state spaces. The D3PM framework ([Section 6.4](#)) provides the categorical analogue of the Gaussian theory. However, the continuous-time limit (CTMC, [Equation \(6.9\)](#)) and the analogue of the Fokker–Planck/reverse-time SDE duality for non-commutative rate matrices Q_t remain partially explored. Extending the unification results of [Chapter 5](#) to discrete state spaces is both mathematically nontrivial and practically important for language modeling.

Stochastic interpolants and optimal transport. The SI framework ([Section 5.7](#)) with arbitrary coupling π subsumes all existing methods. The Schrödinger Bridge coupling ([Remark 5.11](#)) minimizes the kinetic energy of the interpolant, but computing it requires solving a high-dimensional optimization problem. Scalable IPF algorithms that exploit the score network structure of u_t remain an active frontier.

8.3 Research Outlook: Diffusion for Inverse Problems

The inverse problems chapter ([Chapter 7](#)) represents both the application frontier and the theoretical frontier of diffusion-based methods. Several directions are particularly ripe for new contributions.

Hybrid methods. Current methods occupy two extremes: gradient methods (DPS, Π GdM) are fast but biased; SMC methods (MCGDiff) are exact but require $N \gg 1$ particles. A natural middle ground is *variance-reduced SMC*: use the gradient approximation as a proposal distribution for SMC, then correct with importance weights. The gradient proposal concentrates particles near the posterior mode, reducing the effective particle count needed for coverage.

Learning the measurement operator. In blind inverse problems (A unknown), the forward model must be estimated jointly with the reconstruction. The variational perspective ([Section 7.6](#)) suggests an EM-like algorithm: alternate between (E) sampling $x_0 \sim q_\phi(\cdot | y, A)$ using a fixed A , and (M) updating A to maximize $\mathbb{E}_{q_\phi}[\log p(y | x_0, A)]$. The score provides the E-step gradient; the M-step is problem-specific.

Connections to classical regularization. The MAP estimate under a diffusion prior is a form of *regularization by denoising* (RED), related to Tikhonov and total-variation regularization through the denoiser–score identity $D_\sigma(x) - x = \sigma^2 \nabla \log p_\sigma(x)$ ([Equation \(7.25\)](#)). Making this connection quantitative—bounding the difference between the diffusion-based MAP and the classical

Tikhonov solution in terms of the score approximation error $\varepsilon_{\text{score}}$ —would provide a clean justification for using neural denoisers as priors.

Uncertainty quantification. All generation-based methods (Chapter 7) can produce *multiple samples* from the posterior, providing an empirical posterior distribution. But the coverage of this empirical distribution—whether it faithfully represents the true posterior variance—depends on the quality of the likelihood approximation. Formal *credible-set* guarantees for diffusion-based posterior samplers (analogous to frequentist confidence intervals for classical MAP estimators) are almost entirely absent from the current literature.

Key Idea

The deepest open question spanning these notes is the following. *Given a pre-trained diffusion model and a physical forward operator A , what is the information-theoretically minimal number of score evaluations needed to sample from $p(x_0 \mid y)$ to within total-variation error ε ?* Answering this question would simultaneously determine the optimal DPS step-size schedule, the optimal SMC particle count, and the fundamental computational complexity of Bayesian inverse problems with diffusion priors.

A. Omitted Proofs

For readability, we will restate the corresponding theorems and then give full derivations or proofs.

A.1 Lemma: Gaussian tools

Lemma 2.1 (Gaussian score and affine closure).

(a) *Score*

If $p(x) = \mathcal{N}(x; \mu, \sigma^2 I)$ then $\nabla_x \log p(x) = -(x - \mu)/\sigma^2$.

(b) *Affine transport*

If $x \sim \mathcal{N}(\mu, \Sigma)$ and $y = Ax + b + \eta$, $\eta \sim \mathcal{N}(0, \Lambda)$ independent of x , then $y \sim \mathcal{N}(A\mu + b, A\Sigma A^\top + \Lambda)$.

(c) *Gaussian product*

$\mathcal{N}(x; \mu_1, \Sigma_1) \cdot \mathcal{N}(x; \mu_2, \Sigma_2) \propto \mathcal{N}(x; \mu, \Sigma)$ where

$$\Sigma^{-1} = \Sigma_1^{-1} + \Sigma_2^{-1}, \quad \mu = \Sigma(\Sigma_1^{-1}\mu_1 + \Sigma_2^{-1}\mu_2). \quad (2.1)$$

Proof. **Goal:** Show that

$$\mathcal{N}(x; \mu_1, \Sigma_1) \cdot \mathcal{N}(x; \mu_2, \Sigma_2) \propto \mathcal{N}(x; \mu, \Sigma)$$

where $\Sigma^{-1} = \Sigma_1^{-1} + \Sigma_2^{-1}$ and $\mu = \Sigma(\Sigma_1^{-1}\mu_1 + \Sigma_2^{-1}\mu_2)$.

Step 1: Take the log and isolate the exponent

$$\ln[\mathcal{N}(x; \mu_1, \Sigma_1) \cdot \mathcal{N}(x; \mu_2, \Sigma_2)] = -\frac{1}{2}(x - \mu_1)^\top \Sigma_1^{-1}(x - \mu_1) - \frac{1}{2}(x - \mu_2)^\top \Sigma_2^{-1}(x - \mu_2) + C_0$$

where $C_0 = -\frac{1}{2} \ln \det(2\pi\Sigma_1) - \frac{1}{2} \ln \det(2\pi\Sigma_2)$ is independent of x .

Step 2: Expand each quadratic form

$$= -\frac{1}{2}[x^\top \Sigma_1^{-1}x - 2\mu_1^\top \Sigma_1^{-1}x + \mu_1^\top \Sigma_1^{-1}\mu_1] - \frac{1}{2}[x^\top \Sigma_2^{-1}x - 2\mu_2^\top \Sigma_2^{-1}x + \mu_2^\top \Sigma_2^{-1}\mu_2] + C_0$$

Collecting terms by order in x :

$$= -\frac{1}{2} x^\top \underbrace{(\Sigma_1^{-1} + \Sigma_2^{-1})}_{:= \Sigma^{-1}} x + \underbrace{(\mu_1^\top \Sigma_1^{-1} + \mu_2^\top \Sigma_2^{-1})}_{:= b^\top} x + C_1$$

where $b := \Sigma_1^{-1} \mu_1 + \Sigma_2^{-1} \mu_2$, and C_1 absorbs all x -independent constants:

$$C_1 = C_0 - \frac{1}{2} (\mu_1^\top \Sigma_1^{-1} \mu_1 + \mu_2^\top \Sigma_2^{-1} \mu_2)$$

Step 3: Complete the square

We wish to rewrite $-\frac{1}{2} x^\top \Sigma^{-1} x + b^\top x$ as a shifted quadratic. Observe that

$$(x - \mu)^\top \Sigma^{-1} (x - \mu) = x^\top \Sigma^{-1} x - 2\mu^\top \Sigma^{-1} x + \mu^\top \Sigma^{-1} \mu$$

Setting $\mu := \Sigma b$ so that $\Sigma^{-1} \mu = b$, we obtain

$$\begin{aligned} -\frac{1}{2} x^\top \Sigma^{-1} x + b^\top x &= -\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu) + \frac{1}{2} \mu^\top \Sigma^{-1} \mu \\ &= -\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu) + \frac{1}{2} b^\top \Sigma b \end{aligned}$$

Step 4: Collect constant terms

Substituting back:

$$\ln[\mathcal{N}(x; \mu_1, \Sigma_1) \cdot \mathcal{N}(x; \mu_2, \Sigma_2)] = -\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu) + \underbrace{C_1 + \frac{1}{2} b^\top \Sigma b}_{:= C_2}$$

where

$$C_2 = -\frac{1}{2} \ln \det(2\pi \Sigma_1) - \frac{1}{2} \ln \det(2\pi \Sigma_2) - \frac{1}{2} (\mu_1^\top \Sigma_1^{-1} \mu_1 + \mu_2^\top \Sigma_2^{-1} \mu_2 - b^\top \Sigma b)$$

is entirely independent of x .

Step 5: Conclusion

Exponentiating both sides:

$$\mathcal{N}(x; \mu_1, \Sigma_1) \cdot \mathcal{N}(x; \mu_2, \Sigma_2) = e^{C_2} \cdot \exp\left(-\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu)\right) \propto \mathcal{N}(x; \mu, \Sigma)$$

where the proportionality constant e^{C_2} is independent of x , and

$$\boxed{\Sigma^{-1} = \Sigma_1^{-1} + \Sigma_2^{-1}, \quad \mu = \Sigma(\Sigma_1^{-1} \mu_1 + \Sigma_2^{-1} \mu_2)}$$

□

A.2 Chapter 3: Itô Calculus

Proof of Theorem 3.3 (Itô's formula)

Theorem 3.3 (Itô's formula). For $\phi \in C^{1,2}(\mathbb{R}^d \times [0, T])$ and X_t satisfying Equation (3.1):

$$d\phi(X_t, t) = \left(\partial_t \phi + \langle \nabla_x \phi, f \rangle + \frac{g^2}{2} \Delta_x \phi \right) dt + \langle \nabla_x \phi, g dW_t \rangle. \quad (3.2)$$

The correction term $\frac{g^2}{2} \Delta_x \phi dt$ has no deterministic analogue; it arises from the quadratic variation $(dW_t)^2 = I dt$.

Proof. Goal. Derive the stochastic differential $d\phi(X_t, t)$ for a smooth function $\phi \in C^{1,2}$ when X_t evolves by the Itô SDE $dX_t = f dt + g dW_t$.

Step 1: Taylor expansion. Fix a small time increment $h > 0$. Taylor's theorem applied to $\phi \in C^{1,2}$ gives

$$\phi(X_{t+h}, t+h) - \phi(X_t, t) = \partial_t \phi \cdot h + \langle \nabla_x \phi, \Delta X \rangle + \frac{1}{2} (\Delta X)^\top H_x \phi (\Delta X) + R, \quad (A.1)$$

where $\Delta X := X_{t+h} - X_t$ and $H_x \phi = \nabla_x^2 \phi$ is the Hessian. The remainder R involves terms of order $h \cdot |\Delta X|$ and $|\Delta X|^3$.

Step 2: Order-of-magnitude analysis. From the Itô SDE:

$$\Delta X = fh + g \Delta W, \quad \Delta W := W_{t+h} - W_t \sim \mathcal{N}(0, hI_d).$$

Since $\Delta W = O(h^{1/2})$ (in the L^2 sense), we have $|\Delta X| = O(h^{1/2})$ and

$$h \cdot |\Delta X| = O(h^{3/2}) = o(h), \quad |\Delta X|^3 = O(h^{3/2}) = o(h).$$

So the remainder $R = o(h)$ and vanishes as $h \rightarrow 0$. Expanding the quadratic term with $\Delta X = fh + g\Delta W$:

$$(\Delta X)^\top H_x \phi (\Delta X) = \underbrace{(fh)^\top H_x \phi (fh)}_{=O(h^2)} + \underbrace{2(fh)^\top H_x \phi (g\Delta W)}_{=O(h^{3/2})} + g^2 (\Delta W)^\top H_x \phi \Delta W.$$

The first two summands are $o(h)$; only the last survives in the limit.

Step 3: Quadratic variation via the trace identity. Write $\Delta W = \sqrt{h} \epsilon$ where $\epsilon \sim \mathcal{N}(0, I_d)$ is a standard Gaussian independent of $\mathcal{F}_t$, so the surviving quadratic term becomes

$$g^2 (\Delta W)^\top H_x \phi \Delta W = g^2 h \epsilon^\top H_x \phi \epsilon.$$

For any symmetric matrix $A \in \mathbb{R}^{d \times d}$ and $\epsilon \sim \mathcal{N}(0, I_d)$, the *trace identity* holds:

$$\mathbb{E}[\epsilon^\top A \epsilon] = \sum_{i,j} A_{ij} \underbrace{\mathbb{E}[\epsilon_i \epsilon_j]}_{=\delta_{ij}} = \text{tr}(A).$$

(Each ϵ_i has unit variance and different components are independent.) The quadratic variation $\llbracket W^i \rrbracket_t = t$ ensures that the random variable $\epsilon^\top H_x \phi \epsilon$ concentrates around $\text{tr}(H_x \phi)$ in L^2 as $h \rightarrow 0$.

With $\text{tr}(H_x \phi) = \Delta_x \phi$:

$$g^2(\Delta W)^\top H_x \phi \Delta W \longrightarrow g^2 \Delta_x \phi \cdot h \quad \text{in } L^2.$$

Written as a differential rule: $(dW)(dW)^\top = I \, dt$.

Step 4: Combine. Substituting Steps 2–3 into [Equation \(A.1\)](#) and dropping all $o(h)$ terms:

$$\phi(X_{t+h}, t+h) - \phi(X_t, t) = \partial_t \phi \cdot h + \langle \nabla_x \phi, fh + g \Delta W \rangle + \frac{g^2}{2} \Delta_x \phi \cdot h + o(h).$$

Collecting the dt coefficient and the dW coefficient and passing $h \rightarrow 0$:

$$d\phi(X_t, t) = \underbrace{\left(\partial_t \phi + \langle \nabla_x \phi, f \rangle + \frac{g^2}{2} \Delta_x \phi \right) dt}_{\text{drift correction}} + \underbrace{\langle \nabla_x \phi, g \, dW_t \rangle}_{\text{stochastic part}}. \quad \square$$

Proof of [Theorem 3.4](#) (Fokker–Planck equation)

Theorem 3.4 (Fokker–Planck equation). *If X_t satisfies [Equation \(3.1\)](#) with initial density p_0 , the marginal density $p_t(x) = \frac{d}{dx} \Pr(X_t \leq x)$ satisfies the PDE*

$$\partial_t p_t = -\nabla \cdot (f p_t) + \frac{g^2}{2} \Delta p_t. \quad (3.3)$$

Proof. Goal. Prove that $\partial_t p_t(x_0) = -\nabla \cdot (f p_t)(x_0) + \frac{g^2}{2} \Delta p_t(x_0)$ at each fixed point $x_0 \in \mathbb{R}^d$. The strategy: represent $p_t(x_0)$ as an expectation via a Gaussian kernel, differentiate using Itô's formula, then recover the PDE by integration by parts and a limiting argument.

Step 1: Represent $p_t(x_0)$ as an expectation. Define the Gaussian kernel

$$\eta_k(x) := \mathcal{N}(x; x_0, k^{-2} I_d), \quad k = 1, 2, 3, \dots$$

As $k \rightarrow \infty$, η_k narrows around x_0 and approximates the Dirac mass δ_{x_0} : for any continuous bounded F , $\int_{\mathbb{R}^d} \eta_k(x) F(x) \, dx \rightarrow F(x_0)$. Applying this to $F = p_t$:

$$p_t(x_0) = \lim_{k \rightarrow \infty} \int_{\mathbb{R}^d} \eta_k(x) p_t(x) \, dx = \lim_{k \rightarrow \infty} \mathbb{E}[\eta_k(X_t)]. \quad (*)$$

Step 2: Differentiate using Itô's formula. Since $\eta_k \in C^2$ and has no explicit time dependence, [Theorem 3.3](#) gives

$$d\eta_k(X_t) = \langle \nabla \eta_k(X_t), f \rangle \, dt + \frac{g^2}{2} \Delta \eta_k(X_t) \, dt + \langle \nabla \eta_k(X_t), g \, dW_t \rangle.$$

Since $\nabla \eta_k$ is bounded (Gaussian), the stochastic integral is a martingale with zero expectation. Differentiating in t and writing the expectation as an integral:

$$\frac{d}{dt} \mathbb{E}[\eta_k(X_t)] = \int_{\mathbb{R}^d} \left[\langle \nabla \eta_k(x), f(x, t) \rangle + \frac{g^2}{2} \Delta \eta_k(x) \right] p_t(x) \, dx. \quad (B)$$

Step 3: Transfer derivatives from η_k to p_t by integration by parts. Since $\eta_k = \mathcal{N}(x; x_0, k^{-2} I)$

decays super-polynomially in $|x - x_0|$, all boundary terms in integration by parts vanish.

Drift term:

$$\int \langle \nabla \eta_k, f p_t \rangle dx = - \int \eta_k(x) \nabla \cdot [f(x, t) p_t(x)] dx.$$

Diffusion term: Applying Green's second identity $\int u \Delta v dx = \int v \Delta u dx$ (boundary terms vanish by exponential decay of η_k):

$$\frac{\sigma^2}{2} \int \Delta \eta_k(x) p_t(x) dx = \frac{\sigma^2}{2} \int \eta_k(x) \Delta p_t(x) dx.$$

Substituting both into (B):

$$\frac{d}{dt} \mathbb{E}[\eta_k(X_t)] = \int_{\mathbb{R}^d} \eta_k(x) \left[-\nabla \cdot (f p_t) + \frac{\sigma^2}{2} \Delta p_t \right](x) dx. \quad (C)$$

Step 4: Take $k \rightarrow \infty$. Differentiating (*) in t (exchange of limit and derivative justified by dominated convergence for the Gaussian family) and applying (C):

$$\frac{d}{dt} p_t(x_0) = \lim_{k \rightarrow \infty} \int_{\mathbb{R}^d} \eta_k(x) \left[-\nabla \cdot (f p_t) + \frac{\sigma^2}{2} \Delta p_t \right](x) dx.$$

Since $\eta_k \rightarrow \delta_{x_0}$ and the bracket is continuous in x , the integral converges to the bracket evaluated at x_0 :

$$\frac{d}{dt} p_t(x_0) = -\nabla \cdot (f(x_0, t) p_t(x_0)) + \frac{\sigma^2}{2} \Delta p_t(x_0). \quad \square$$

A.3 Theorem 3.7: Langevin Stationary Distribution

Theorem 3.7 (Langevin dynamics has stationary distribution p). *The continuous-time Langevin SDE*

$$dX_t = \nabla_x \log p(X_t) dt + \sqrt{2} dW_t \quad (3.6)$$

has p as its unique stationary density (under standard regularity on p).

Before proving the theorem we first *derive* the specific form of the SDE, explaining in particular why the diffusion coefficient is $\sqrt{2}$ and not any other constant. After the derivation, the proof of the theorem proceeds in three parts: (A) existence via the infinitesimal generator, (B) detailed balance, which gives a second characterisation of stationarity and implies time-reversibility, and (C) a sketch of uniqueness.

Motivation: deriving the Langevin SDE and the role of $\sqrt{2}$

Setup. Consider a general Itô SDE with constant diffusion coefficient $\sigma > 0$:

$$dX_t = f(X_t) dt + \sigma dW_t. \quad (A.2)$$

The Fokker-Planck equation for the time-marginal density p_t is

$$\partial_t p_t = -\nabla \cdot (f p_t) + \frac{\sigma^2}{2} \Delta p_t.$$

We want p to be stationary ($\partial_t p_t|_{p_t=p} = 0$) and the process to satisfy *detailed balance*—that is, no net probability current flows at equilibrium.

Path 1: Zero probability current (Fokker–Planck). The probability current associated with Equation (A.2) is the vector field

$$J = f p - \frac{\sigma^2}{2} \nabla p.$$

Stationarity only requires $\nabla \cdot J = 0$ (divergence-free current), but the stronger condition of detailed balance requires $J = 0$ pointwise—no net local flow. Setting $J = 0$:

$$f p = \frac{\sigma^2}{2} \nabla p \implies f = \frac{\sigma^2}{2} \frac{\nabla p}{p} = \frac{\sigma^2}{2} \nabla \log p.$$

Thus the drift must equal $\frac{\sigma^2}{2} \nabla \log p$ for any $\sigma > 0$. The choice $\sigma^2 = 2$, i.e. $\sigma = \sqrt{2}$, normalises the coefficient to unity:

$$f = \nabla \log p, \quad \sigma = \sqrt{2}.$$

Path 2: Physical derivation—the fluctuation-dissipation theorem. The original Langevin equation in physics models a particle with mass m , friction coefficient γ , and potential $U(x)$:

$$m \ddot{x} = -\gamma \dot{x} - \nabla U(x) + \sqrt{2\gamma k_B T} \eta(t),$$

where $\eta(t)$ is white noise. The noise amplitude $\sqrt{2\gamma k_B T}$ is not a free parameter: the *fluctuation-dissipation theorem* fixes it so that the Boltzmann distribution $p(x) \propto e^{-U(x)/k_B T}$ is invariant. In the *overdamped* (high-friction) limit $m \rightarrow 0$ one drops the inertia term and divides by γ :

$$\dot{x} = -\frac{1}{\gamma} \nabla U + \sqrt{\frac{2k_B T}{\gamma}} \eta(t).$$

Setting $\gamma = 1$ (unit friction) and $k_B T = 1$ (unit temperature) and using $\nabla \log p = -\nabla U$ gives, in Itô form,

$$dX_t = \nabla \log p(X_t) dt + \sqrt{2} dW_t.$$

Hence $\sqrt{2}$ arises from the fluctuation-dissipation theorem at unit temperature and unit friction: it is the precise noise strength that balances the drift $\nabla \log p$ so that the system neither collapses nor disperses, but settles to p .

Path 3: Self-adjointness of the generator. For the SDE $dX_t = \frac{\sigma^2}{2} \nabla \log p dt + \sigma dW_t$ the generator is

$$\mathcal{L}f = \frac{\sigma^2}{2} [\langle \nabla \log p, \nabla f \rangle + \Delta f].$$

We verify that $\mathcal{L}$ is self-adjoint in $L^2(p)$, i.e. $\int f \mathcal{L}h dp = \int h \mathcal{L}f dp$. Applying integration by

parts to $\nabla \cdot (f \nabla h) = \langle \nabla f, \nabla h \rangle + f \Delta h$:

$$\begin{aligned}
 \int_{\mathbb{R}^d} f \mathcal{L}h p \, dx &= \frac{\sigma^2}{2} \int f \langle \nabla \log p, \nabla h \rangle p \, dx + \frac{\sigma^2}{2} \int f \Delta h p \, dx \\
 &= \frac{\sigma^2}{2} \int f \langle \nabla p, \nabla h \rangle \, dx + \frac{\sigma^2}{2} \int f p \Delta h \, dx \\
 &= -\frac{\sigma^2}{2} \int p \nabla \cdot (f \nabla h) \, dx + \frac{\sigma^2}{2} \int f p \Delta h \, dx \\
 &= -\frac{\sigma^2}{2} \int p \langle \nabla f, \nabla h \rangle \, dx - \frac{\sigma^2}{2} \int f p \Delta h \, dx + \frac{\sigma^2}{2} \int f p \Delta h \, dx \\
 &= -\frac{\sigma^2}{2} \underbrace{\int \langle \nabla f, \nabla h \rangle \, dp}_{=: \mathcal{E}(f, h)}.
 \end{aligned}$$

The *Dirichlet form* $\mathcal{E}(f, h) = \int \langle \nabla f, \nabla h \rangle \, dp$ is manifestly symmetric in f and h , so $\int f \mathcal{L}h \, dp = -\mathcal{E}(f, h) = \int h \mathcal{L}f \, dp$, confirming self-adjointness for *any* $\sigma > 0$. Choosing $\sigma = \sqrt{2}$ sets the overall scale of $\mathcal{E}$ to 1 and gives the cleanest form $\mathcal{L}f = \langle \nabla \log p, \nabla f \rangle + \Delta f$.

Summary. All three paths are equivalent characterisations of the same balance:

Path	Condition
Zero probability current	$J = fp - \frac{\sigma^2}{2} \nabla p = 0$
Fluctuation-dissipation	$\text{noise}^2 = 2\gamma k_B T, \gamma = k_B T = 1$
Dirichlet form	$\mathcal{E}(f, h) = \int \langle \nabla f, \nabla h \rangle \, dp$ (unit coefficient)

Each forces $\sigma^2 = 2$ when the drift is written without a prefactor as $\nabla \log p$.

Part A: Stationarity via the infinitesimal generator

Proof of existence. Goal. Show that if $X_0 \sim p$, then $X_t \sim p$ for all $t \geq 0$. Equivalently, for every smooth compactly supported test function $f \in C_c^\infty(\mathbb{R}^d)$, show that $\mu_t(f) := \mathbb{E}[f(X_t)]$ satisfies $\frac{d}{dt} \mu_t(f) = 0$ whenever $X_t \sim p$.

Step 1: Identify the generator. The Langevin SDE [Equation \(3.6\)](#) has drift $\nabla \log p$ and diffusion $\sqrt{2}$. By [Theorem 3.3](#) (Itô's formula applied to $f(X_t)$ with no explicit t -dependence):

$$df(X_t) = \underbrace{[\langle \nabla f, \nabla \log p \rangle + \Delta f]}_{\mathcal{L}f(X_t)}(X_t) \, dt + \langle \nabla f(X_t), \sqrt{2} \, dW_t \rangle.$$

The operator $\mathcal{L}f := \langle \nabla \log p, \nabla f \rangle + \Delta f$ is the *infinitesimal generator* of the Langevin SDE. Taking expectations and using that the stochastic integral is a martingale:

$$\frac{d}{dt} \mu_t(f) = \mathbb{E}[\mathcal{L}f(X_t)] = \int_{\mathbb{R}^d} \mathcal{L}f(x) p_t(x) \, dx. \tag{A.3}$$

Step 2: Evaluate $\int \mathcal{L}f p \, dx$ when $p_t = p$. Substituting $p_t = p$ and expanding $\mathcal{L}f$:

$$\int_{\mathbb{R}^d} \mathcal{L}f p \, dx = \int \langle \nabla \log p, \nabla f \rangle p \, dx + \int \Delta f p \, dx = \int \langle \nabla p, \nabla f \rangle \, dx + \int p \Delta f \, dx,$$

using $p \nabla \log p = \nabla p$ in the first term.

Integrate the first term by parts. Since $f \in C_c^\infty$, the boundary term $[f \nabla p]_{\partial\Omega}$ vanishes as $\Omega \rightarrow \mathbb{R}^d$:

$$\int \langle \nabla p, \nabla f \rangle \, dx = \int \nabla p \cdot \nabla f \, dx = - \int p \Delta f \, dx.$$

Combining:

$$\int \mathcal{L}f p \, dx = - \int p \Delta f \, dx + \int p \Delta f \, dx = 0.$$

Step 3: Conclude. From Equation (A.3), $\frac{d}{dt} \mu_t(f) = 0$ for all $f \in C_c^\infty(\mathbb{R}^d)$. Since test functions determine the distribution, $p_t \equiv p$ for all t , so p is stationary. $\square$

Part B: Detailed balance and reversibility

The Langevin SDE is *reversible* with respect to p : its generator $\mathcal{L}$ is self-adjoint in the weighted space $L^2(p) := L^2(\mathbb{R}^d, p \, dx)$.

Proof of detailed balance. Goal. Show that for all $f, h \in C_c^\infty(\mathbb{R}^d)$:

$$\int_{\mathbb{R}^d} f \mathcal{L}h \, dp = \int_{\mathbb{R}^d} h \mathcal{L}f \, dp. \quad (\text{A.4})$$

Step 1: Reduce to the Dirichlet form. Define the symmetric *Dirichlet form*

$$\mathcal{E}(f, h) := \int_{\mathbb{R}^d} \langle \nabla f, \nabla h \rangle p \, dx = \int_{\mathbb{R}^d} \langle \nabla f, \nabla h \rangle \, dp.$$

We claim $\int f \mathcal{L}h \, dp = -\mathcal{E}(f, h)$ for all $f, h \in C_c^\infty(\mathbb{R}^d)$. Since $\mathcal{E}$ is symmetric, this immediately gives Equation (A.4).

Step 2: Prove $\int f \mathcal{L}h \, dp = -\mathcal{E}(f, h)$. Expand $\mathcal{L}h = \langle \nabla \log p, \nabla h \rangle + \Delta h$ and use $p \nabla \log p = \nabla p$:

$$\int f \mathcal{L}h p \, dx = \int f \langle \nabla p, \nabla h \rangle \, dx + \int f p \Delta h \, dx. \quad (\text{A.5})$$

For the first term, integrate by parts: $\nabla \cdot (f \nabla h) = \langle \nabla f, \nabla h \rangle + f \Delta h$, so by the divergence theorem (boundary = 0 since f has compact support):

$$\int f \langle \nabla p, \nabla h \rangle \, dx = - \int p \nabla \cdot (f \nabla h) \, dx = - \int p \langle \nabla f, \nabla h \rangle \, dx - \int f p \Delta h \, dx.$$

Substituting into Equation (A.5):

$$\int f \mathcal{L}h p \, dx = - \int p \langle \nabla f, \nabla h \rangle \, dx - \int f p \Delta h \, dx + \int f p \Delta h \, dx = -\mathcal{E}(f, h). \quad \square$$

Remark A.1 (Physical meaning of detailed balance). Equation [Equation \(A.4\)](#) says that the probability current from any state A to any state B under p equals the reverse current from B to A . In physics this is called *detailed balance*; mathematically it is equivalent to p being a reversible (= symmetric) invariant measure of the process. The Langevin SDE is one of the few SDEs that automatically satisfies this property, which is the principal reason it is used in Bayesian sampling and generative modelling.

Part C: Uniqueness

Sketch of uniqueness. Uniqueness of the stationary distribution follows from two structural properties of the Langevin SDE (under mild regularity on p , e.g. locally Lipschitz $\nabla \log p$):

Strongly Feller. The transition semigroup $P_t f(x) = \mathbb{E}_x[f(X_t)]$ maps bounded measurable functions to continuous functions for every $t > 0$. This follows from ellipticity of $\mathcal{L}$ (the diffusion coefficient $\sqrt{2}$ is constant and positive, so the SDE is uniformly non-degenerate).

Irreducible. For any open set $U \subset \mathbb{R}^d$ and any starting point $x \in \mathbb{R}^d$, the process reaches U in finite time with positive probability. This follows from the support theorem for SDEs with non-degenerate noise (Stroock–Varadhan; the diffusion matrix is $2I_d$, full rank everywhere).

By the *Doob–Khasminskii theorem*, a strongly Feller and irreducible Markov process has at most one invariant probability measure. Combined with Part A (existence), p is the unique stationary distribution.

Quantitative rate. Under a convexity-at-infinity condition on $-\log p$ (e.g. $-\nabla^2 \log p \leq KI$ outside a compact set), Eberle et al. [\[13\]](#) construct an explicit synchronous-then-reflection coupling of two copies of the Langevin SDE and show that the Wasserstein-1 distance decays exponentially:

$$W_1(p_t, p) \leq C e^{-\lambda t} W_1(p_0, p),$$

for explicit constants $C, \lambda > 0$. This provides a *quantitative* version of uniqueness: convergence to p at an exponential rate from any starting distribution. $\square$

A.4 Theorem 3.10: DSM = ESM up to a constant

Theorem 3.10 (DSM = ESM up to a constant, Vincent [42](#)). *Define the ESM and DSM losses at noise level σ :*

$$J_{\text{ESM}}^\sigma(\theta) := \frac{1}{2} \mathbb{E}_{\tilde{x} \sim p_\sigma} [\|s_\theta(\tilde{x}) - \nabla_{\tilde{x}} \log p_\sigma(\tilde{x})\|^2], \quad (3.9)$$

$$J_{\text{DSM}}^\sigma(\theta) := \frac{1}{2} \mathbb{E}_{x_0 \sim p, \epsilon \sim \mathcal{N}(0, I)} [\|s_\theta(x_0 + \sigma\epsilon) + \epsilon/\sigma\|^2], \quad (3.10)$$

where $p_\sigma(\tilde{x}) = \int \mathcal{N}(\tilde{x}; x_0, \sigma^2 I) p(x_0) dx_0$ is the noise-convolved density and $\nabla_{\tilde{x}} \log p(\tilde{x}|x_0) = -\epsilon/\sigma$. Then

$$J_{\text{ESM}}^\sigma(\theta) = J_{\text{DSM}}^\sigma(\theta) + C, \quad (3.11)$$

with $C = \frac{1}{2} \mathbb{E} [\|\nabla \log p_\sigma\|^2] - \frac{d}{2\sigma^2}$ independent of θ .

Proof. We carry out the computation in five steps.

Step 1: Expand J_{ESM}^σ via $\|a - b\|^2 = \|a\|^2 - 2\langle a, b \rangle + \|b\|^2$.

$$J_{\text{ESM}}^\sigma(\theta) = \underbrace{\frac{1}{2}\mathbb{E}[\|s_\theta(\tilde{x})\|^2]}_{\text{(I)}} - \underbrace{\mathbb{E}[\langle s_\theta(\tilde{x}), \nabla_{\tilde{x}} \log p_\sigma(\tilde{x}) \rangle]}_{\text{(II) cross term}} + \underbrace{\frac{1}{2}\mathbb{E}[\|\nabla_{\tilde{x}} \log p_\sigma(\tilde{x})\|^2]}_{\text{(III) } \theta\text{-free}}. \quad (\text{A.6})$$

Step 2: Rewrite the cross term (II)—marginal score as a posterior expectation. By the law of total probability, $p_\sigma(\tilde{x}) = \int p(\tilde{x} | x_0) p(x_0) dx_0$. Differentiating under the integral and dividing by $p_\sigma(\tilde{x})$:

$$\nabla_{\tilde{x}} \log p_\sigma(\tilde{x}) = \frac{\nabla_{\tilde{x}} p_\sigma(\tilde{x})}{p_\sigma(\tilde{x})} = \int \frac{\nabla_{\tilde{x}} p(\tilde{x} | x_0)}{p(\tilde{x} | x_0)} \cdot \underbrace{\frac{p(\tilde{x} | x_0) p(x_0)}{p_\sigma(\tilde{x})}}_{= p(x_0 | \tilde{x}) \text{ (Bayes)}} dx_0 = \mathbb{E}_{x_0 | \tilde{x}}[\nabla_{\tilde{x}} \log p(\tilde{x} | x_0)]. \quad (\text{A.7})$$

The marginal score equals the posterior expectation of the conditional score. Substituting into (II) and applying the tower property $\mathbb{E}_{\tilde{x}}[\mathbb{E}_{x_0 | \tilde{x}}[\cdot]] = \mathbb{E}_{x_0, \tilde{x}}[\cdot]$:

$$\text{(II)} = -\mathbb{E}_{x_0, \tilde{x}}[\langle s_\theta(\tilde{x}), \nabla_{\tilde{x}} \log p(\tilde{x} | x_0) \rangle]. \quad (\text{A.8})$$

Step 3: Compute $\nabla_{\tilde{x}} \log p(\tilde{x} | x_0)$ for the Gaussian kernel. Under $\tilde{x} = x_0 + \sigma\epsilon$ with $\epsilon \sim \mathcal{N}(0, I_d)$,

$$\log p(\tilde{x} | x_0) = -\frac{\|\tilde{x} - x_0\|^2}{2\sigma^2} + \text{const}, \quad \nabla_{\tilde{x}} \log p(\tilde{x} | x_0) = -\frac{\tilde{x} - x_0}{\sigma^2} = -\frac{\epsilon}{\sigma}.$$

Substituting into Equation (A.8):

$$\text{(II)} = \mathbb{E}_{x_0, \epsilon}[\langle s_\theta(x_0 + \sigma\epsilon), \frac{\epsilon}{\sigma} \rangle]. \quad (\text{A.9})$$

Step 4: Expand J_{DSM}^σ via $\|a + b\|^2 = \|a\|^2 + 2\langle a, b \rangle + \|b\|^2$.

$$J_{\text{DSM}}^\sigma(\theta) = \underbrace{\frac{1}{2}\mathbb{E}[\|s_\theta(x_0 + \sigma\epsilon)\|^2]}_{\text{(I)}} + \underbrace{\mathbb{E}[\langle s_\theta(x_0 + \sigma\epsilon), \frac{\epsilon}{\sigma} \rangle]}_{\text{(II) same as Equation (A.9)}} + \underbrace{\frac{1}{2}\mathbb{E}[\|\frac{\epsilon}{\sigma}\|^2]}_{= d/(2\sigma^2), \theta\text{-free}}, \quad (\text{A.10})$$

using $\mathbb{E}[\|\epsilon\|^2] = d$ for $\epsilon \sim \mathcal{N}(0, I_d)$.

Step 5: Compare and conclude. The θ -dependent parts of Equation (A.6) and Equation (A.10) are identical:

$$\begin{aligned} J_{\text{ESM}}^\sigma(\theta) &= \underbrace{\frac{1}{2}\mathbb{E}[\|s_\theta\|^2] + \mathbb{E}[\langle s_\theta, \epsilon/\sigma \rangle]}_{\theta\text{-dependent, same in both}} + \frac{1}{2}\mathbb{E}[\|\nabla \log p_\sigma\|^2], \\ J_{\text{DSM}}^\sigma(\theta) &= \underbrace{\frac{1}{2}\mathbb{E}[\|s_\theta\|^2] + \mathbb{E}[\langle s_\theta, \epsilon/\sigma \rangle]}_{\theta\text{-dependent, same in both}} + \frac{d}{2\sigma^2}. \end{aligned}$$

Therefore

$$J_{\text{ESM}}^\sigma(\theta) = J_{\text{DSM}}^\sigma(\theta) + \underbrace{\frac{1}{2}\mathbb{E}[\|\nabla \log p_\sigma\|^2] - \frac{d}{2\sigma^2}}_{\text{C}},$$

where C is independent of θ . In particular, $\nabla_{\theta} J_{\text{ESM}}^{\sigma} = \nabla_{\theta} J_{\text{DSM}}^{\sigma}$, so both losses share the same minimiser. $\square$ $\square$

A.5 Theorem 3.17: Reverse-time SDE

Theorem 3.17 (Reverse-time SDE, Anderson 2). *Let $\{X_t\}_{t \in [0, T]}$ satisfy the forward SDE $dx = f(x, t) dt + g(t) dW_t$ with marginal densities $\{p_t\}$. Define the time-reversed process $\bar{X}_s := X_{T-s}$. Then $\bar{X}_s$ satisfies*

$$d\bar{X}_s = \left[-f(\bar{X}_s, T-s) + g(T-s)^2 \nabla_x \log p_{T-s}(\bar{X}_s) \right] ds + g(T-s) d\bar{W}_s, \quad (3.19)$$

where $\bar{W}_s$ is a Brownian motion running forward in s . The reversed process has the same marginals: $\bar{X}_s \sim p_{T-s}$.

The proof has two parts. Part A gives an intuitive, discrete-time derivation showing *why* the reverse drift must take the form $-f + g^2 \nabla \log p_t$ (following the Bayes-based approach of Song et al. [39] and Su 40). Part B verifies the result rigorously by checking that the proposed reverse SDE has the correct Fokker–Planck equation (following Anderson 2 and Holderrieth 17).

Part A: Discrete-time motivation via Bayes' theorem

Setup. Discretise the forward SDE $dx = f(x, t) dt + g(t) dW_t$ with step size $\Delta t > 0$. One Euler–Maruyama step gives

$$X_{t+\Delta t} = X_t + f(X_t, t) \Delta t + g(t) \sqrt{\Delta t} \xi, \quad \xi \sim \mathcal{N}(0, I_d), \quad (A.11)$$

so the forward transition density is

$$p(x_{t+\Delta t} | x_t) = \mathcal{N}(x_{t+\Delta t}; x_t + f(x_t, t) \Delta t, g(t)^2 \Delta t I).$$

Reverse transition via Bayes. The reverse step asks: given $X_{t+\Delta t} = x_{t+\Delta t}$, what is X_t ? By Bayes' theorem:

$$p(x_t | x_{t+\Delta t}) \propto p(x_{t+\Delta t} | x_t) p_t(x_t).$$

Taking logarithms (treating $x_{t+\Delta t}$ as fixed):

$$\log p(x_t | x_{t+\Delta t}) = \log p(x_{t+\Delta t} | x_t) + \log p_t(x_t) + \text{const.} \quad (A.12)$$

For the first term, the Gaussian log-density is

$$\log p(x_{t+\Delta t} | x_t) = -\frac{\|x_{t+\Delta t} - x_t - f(x_t, t) \Delta t\|^2}{2g(t)^2 \Delta t} + \text{const.}$$

For the second term, expand $\log p_t(x_t)$ around $x_{t+\Delta t}$. Since $x_t - x_{t+\Delta t} = O(\sqrt{\Delta t})$:

$$\log p_t(x_t) \approx \log p_t(x_{t+\Delta t}) + \nabla_x \log p_t(x_{t+\Delta t}) \cdot (x_t - x_{t+\Delta t}).$$

Substituting into Equation (A.12), letting $y := x_t - x_{t+\Delta t}$ and writing $s_t := \nabla_x \log p_t(x_{t+\Delta t})$:

$$\begin{aligned} \log p(x_t | x_{t+\Delta t}) &\approx -\frac{\|y + f\Delta t\|^2}{2g^2\Delta t} + s_t \cdot y + \text{const} \\ &= -\frac{\|y\|^2}{2g^2\Delta t} + y \cdot \left(-\frac{f\Delta t}{g^2\Delta t} + s_t\right) + \text{const}. \end{aligned} \quad (\text{A.13})$$

This is a Gaussian log-density in y with mean

$$\mu_y = g^2\Delta t \left(s_t - \frac{f}{g^2}\right) = g^2\Delta t s_t - f\Delta t = [g(t)^2 \nabla_x \log p_t(x_{t+\Delta t}) - f(x_{t+\Delta t}, t)]\Delta t$$

and variance $g(t)^2\Delta t I$. Thus

$$x_t | x_{t+\Delta t} \sim \mathcal{N}\left(x_{t+\Delta t} + [-f(x_{t+\Delta t}, t) + g(t)^2 \nabla_x \log p_t(x_{t+\Delta t})]\Delta t, g(t)^2\Delta t I\right). \quad (\text{A.14})$$

Reading off the reverse SDE. Equation (A.14) is exactly an Euler–Maruyama step of the SDE

$$dX_t = [-f(X_t, t) + g(t)^2 \nabla_x \log p_t(X_t)]dt + g(t) dW_t,$$

running *backwards* in time t (equivalently, forward in $s = T - t$). Substituting $\bar{X}_s = X_{T-s}$ gives exactly Equation (3.19).

Remark A.2 (Why the score appears). The term $g^2 \nabla_x \log p_t$ arises because the reverse transition must account for the fact that high-density regions at time t are over-represented among the states that transition to $x_{t+\Delta t}$; the score is the gradient of the log-marginal density that corrects for this bias.

Part B: Rigorous verification via Fokker–Planck

Proof of Theorem 3.17. Write $\bar{p}_s := p_{T-s}$ for the target density at reverse time s . We verify that $\bar{p}_s$ satisfies the Fokker–Planck equation of the proposed reverse SDE Equation (3.19), whose drift and diffusion are $\bar{f}(x, s) := -f(x, T-s) + g(T-s)^2 \nabla_x \log p_{T-s}(x)$ and $\bar{g}(s) := g(T-s)$.

Step 1: FP equation of the reverse SDE. The Fokker–Planck equation for a density q_s evolving under the reverse SDE is

$$\partial_s q_s = -\nabla \cdot (\bar{f} q_s) + \frac{\bar{g}^2}{2} \Delta q_s. \quad (\text{A.15})$$

Step 2: Time-derivative of $\bar{p}_s$. Since $\bar{p}_s = p_{T-s}$ and the forward density satisfies $\partial_t p_t = -\nabla \cdot (f p_t) + \frac{g^2}{2} \Delta p_t$, the chain rule gives

$$\partial_s \bar{p}_s = -\partial_t p_t|_{t=T-s} = \nabla \cdot (f p_{T-s}) - \frac{g^2}{2} \Delta p_{T-s}. \quad (\text{A.16})$$

Step 3: Plugging $\bar{p}_s$ into Equation (A.15). Expand $\bar{f} = -f + g^2 \nabla \log p_{T-s}$ and use $p_{T-s} \nabla \log p_{T-s} = \nabla p_{T-s}$:

$$\begin{aligned} -\nabla \cdot (\bar{f} p_{T-s}) + \frac{\bar{g}^2}{2} \Delta p_{T-s} &= \nabla \cdot (f p_{T-s}) - g^2 \nabla \cdot (\nabla p_{T-s}) + \frac{g^2}{2} \Delta p_{T-s} \\ &= \nabla \cdot (f p_{T-s}) - g^2 \Delta p_{T-s} + \frac{g^2}{2} \Delta p_{T-s} \\ &= \nabla \cdot (f p_{T-s}) - \frac{g^2}{2} \Delta p_{T-s}. \end{aligned}$$

This equals $\partial_s \bar{p}_s$ from Equation (A.16).

Step 4: Conclude. The density $\bar{p}_s = p_{T-s}$ satisfies the Fokker–Planck equation of the reverse SDE Equation (3.19) with initial condition $\bar{p}_0 = p_T$. Under mild regularity (locally Lipschitz f , non-degenerate g), Fokker–Planck solutions are unique given the initial condition, so $\bar{p}_s = p_{T-s}$ is the unique marginal density of Equation (3.19). $\square$

A.6 Mass Conservation (Theorem 4.2)

Theorem 4.2 (Mass Conservation / Continuity Equation). *Let p_t be a probability path and u_t a locally Lipschitz, integrable velocity field. The following are equivalent:*

- (i) $\partial_t p_t + \nabla \cdot (p_t u_t) = 0$ (continuity equation) for $t \in [0, 1]$.
- (ii) u_t generates p_t : the flow model $X_t = \psi_t(X_0)$ satisfies $X_t \sim p_t$ for all t .

Proof. We prove each direction.

(ii) $\Rightarrow$ (i). Suppose $X_t = \psi_t(X_0) \sim p_t$ for all t , where $\dot{\psi}_t(x) = u_t(\psi_t(x))$. Represent the density via the Dirac delta:

$$p_t(x) = \mathbb{E}_{X_0 \sim p_0} [\delta(x - \psi_t(X_0))].$$

Differentiate under the expectation (justified by regularity of u_t):

$$\partial_t p_t(x) = \mathbb{E}_{X_0} [-\dot{\psi}_t(X_0) \cdot \nabla_x \delta(x - \psi_t(X_0))] = \mathbb{E}_{X_0} [-u_t(\psi_t(X_0)) \cdot \nabla_x \delta(x - \psi_t(X_0))].$$

Using $f(y) \nabla_x \delta(x - y) = \nabla_x [f(x) \delta(x - y)]$ and integrating (the same calculation as the Fokker–Planck derivation of Chapter 3 with $g_t = 0$):

$$\partial_t p_t(x) = -\nabla_x \cdot [\mathbb{E}_{X_0} [u_t(\psi_t(X_0)) \delta(x - \psi_t(X_0))]] = -\nabla_x \cdot [u_t(x) p_t(x)],$$

where in the last step we used that under the Dirac delta $\psi_t(X_0) = x$.

(i) $\Rightarrow$ (ii). Suppose p_t satisfies the continuity equation. Let $\tilde{p}_t$ denote the density of $\psi_t(X_0)$. By the direction (ii) $\Rightarrow$ (i) just proved, $\tilde{p}_t$ also satisfies the continuity equation with the same u_t and initial condition $\tilde{p}_0 = p_0$. Under the stated regularity assumptions, the continuity equation has a unique solution (see 25, Appendix A), hence $\tilde{p}_t = p_t$ for all t . $\square$

A.7 Marginalization Trick (Theorem 4.12)

Theorem 4.12 (Marginalization trick). *Suppose for each x_1 , the velocity $u_t(\cdot|x_1)$ generates $p_{t|1}(\cdot|x_1)$. Then the marginal velocity field*

$$u_t(x) := \mathbb{E} [u_t(X_t|X_1) \mid X_t = x] = \int u_t(x|x_1) p_{1|t}(x_1|x) dx_1, \quad (4.11)$$

where $p_{1|t}(x_1|x) = p_{t|1}(x|x_1) q(x_1)/p_t(x)$ is the posterior, generates the marginal path p_t .

Proof. It suffices to verify the continuity equation $\partial_t p_t(x) + \nabla_x \cdot (u_t(x) p_t(x)) = 0$.

By assumption, for each x_1 , the pair $(u_t(\cdot|x_1), p_{t|1}(\cdot|x_1))$ satisfies the conditional continuity equation:

$$\partial_t p_{t|1}(x|x_1) + \nabla_x \cdot (u_t(x|x_1) p_{t|1}(x|x_1)) = 0. \quad (\text{A.17})$$

Integrate [Equation \(A.17\)](#) against $q(x_1)$ and exchange differentiation with integration (dominated convergence under the integrability assumption):

$$\begin{aligned} \partial_t p_t(x) &= \int \partial_t p_{t|1}(x|x_1) q(x_1) dx_1 \\ &= - \int \nabla_x \cdot (u_t(x|x_1) p_{t|1}(x|x_1)) q(x_1) dx_1 \\ &= - \nabla_x \cdot \int u_t(x|x_1) p_{t|1}(x|x_1) q(x_1) dx_1. \end{aligned}$$

Apply Bayes' theorem: $p_{t|1}(x|x_1) q(x_1) = p_{1|t}(x_1|x) p_t(x)$. Therefore

$$\int u_t(x|x_1) p_{t|1}(x|x_1) q(x_1) dx_1 = p_t(x) \int u_t(x|x_1) p_{1|t}(x_1|x) dx_1 = p_t(x) u_t(x),$$

where the last equality is the definition [Equation \(4.11\)](#) of the marginal velocity. Substituting back:

$$\partial_t p_t(x) = - \nabla_x \cdot (u_t(x) p_t(x)).$$

By [Theorem 4.2](#), u_t generates p_t . □

A.8 Proposition 5.1: DDPM $\rightarrow$ VP-SDE weak convergence

We give a rigorous formulation of the discrete-to-continuous limit stated in [Proposition 5.1](#).

Setup. Fix a smooth noise schedule $\beta: [0, 1] \rightarrow (0, \infty)$ and set $\beta_k^{(T)} := \beta(k/T)/T$ for $k = 1, \dots, T$. Define the rescaled discrete chain $X_s^{(T)} := x_{\lfloor sT \rfloor}^{(T)}$ for $s \in [0, 1]$, where $x_0^{(T)} = x_0$ is a fixed initial condition. The VP-SDE is

$$dX_s = -\frac{\beta(s)}{2} X_s ds + \sqrt{\beta(s)} dW_s, \quad X_0 = x_0.$$

Euler–Maruyama identification. At step k with $s = k/T$, the DDPM transition gives

$$\begin{aligned} X_{s+1/T}^{(T)} &= \sqrt{1 - \beta(s)/T} X_s^{(T)} + \sqrt{\beta(s)/T} \xi_k, \quad \xi_k \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I) \\ &= X_s^{(T)} - \frac{\beta(s)}{2T} X_s^{(T)} + \sqrt{\frac{\beta(s)}{T}} \xi_k + O(T^{-3/2}), \end{aligned}$$

where the Taylor expansion $\sqrt{1-u} = 1 - u/2 + O(u^2)$ was used. This is the Euler–Maruyama discretization of the VP-SDE with step size $h = 1/T$:

$$X_{s+h}^{\text{EM}} = X_s^{\text{EM}} - \frac{\beta(s)}{2} X_s^{\text{EM}} h + \sqrt{\beta(s)h} \xi_k.$$

Weak convergence. Since the VP-SDE has smooth Lipschitz coefficients $f(x, s) = -\beta(s)x/2$ and $g(s) = \sqrt{\beta(s)}$ (bounded and Lipschitz in s by regularity of β), the standard Euler–Maruyama

weak convergence theorem applies [23, Theorem 9.7]: for any $\varphi \in C_b^4(\mathbb{R}^d)$ and any $s \in [0, 1]$,

$$|\mathbb{E}[\varphi(X_s^{(T)})] - \mathbb{E}[\varphi(X_s)]| = O(T^{-1}).$$

In particular, $X_s^{(T)} \rightharpoonup X_s$ weakly as $T \rightarrow \infty$, and the marginal distributions converge: $p_s^{(T)} \rightharpoonup p_s$ in the sense of weak convergence of probability measures.

Remark on the reverse chain. The same argument applies to the reverse chain: the DDPM reverse step (Chapter 2) with learned $\mu_\theta(x_t, t)$ is the Euler–Maruyama step of the reverse-time VP-SDE (Equation (3.19)). As $T \rightarrow \infty$, the quality of DDPM samples is therefore bounded by (i) the accuracy of the score network $s_\theta \approx \nabla \log p_t$ and (ii) the $O(T^{-1})$ discretization error.

A. Bibliography

- [1] Michael S. Albergo, Nicholas M. Boffi, and Eric Vanden-Eijnden. Stochastic interpolants: A unifying framework for flows and diffusions. URL <http://arxiv.org/abs/2303.08797>.
- [2] Brian D. O. Anderson. Reverse-time diffusion equation models. 12(3):313–326. ISSN 0304-4149. doi: 10.1016/0304-4149(82)90051-5.
- [3] Jacob Austin, Daniel D. Johnson, Jonathan Ho, Daniel Tarlow, and Rianne van den Berg. Structured denoising diffusion models in discrete state-spaces. In *Advances in Neural Information Processing Systems*, volume 34, pages 17981–17993. Curran Associates, Inc. URL <https://proceedings.neurips.cc/paper/2021/hash/958c530554f78bcd8e97125b70e6973d-Abstract.html>.
- [4] Valentin De Bortoli, James Thornton, Jeremy Heng, and Arnaud Doucet. Diffusion schrödinger bridge with applications to score-based generative modeling. URL <http://arxiv.org/abs/2106.01357>.
- [5] Gabriel Cardoso, Yazid Janati, Sylvain Le Corff, and Eric Moulines. Monte carlo guided denoising diffusion models for bayesian linear inverse problems.
- [6] Stanley H. Chan. Tutorial on diffusion models for imaging and vision. URL <http://arxiv.org/abs/2403.18103>.
- [7] Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David K. Duvenaud. Neural ordinary differential equations. In *Advances in Neural Information Processing Systems*, volume 31. Curran Associates, Inc., . URL <https://proceedings.neurips.cc/paper/2018/hash/69386f6bb1dfed68692a24c8686939b9-Abstract.html>.
- [8] Sitan Chen, Sinho Chewi, Jerry Li, Yuanzhi Li, Adil Salim, and Anru R. Zhang. Sampling is as easy as learning the score: Theory for diffusion models with minimal data assumptions, . URL <http://arxiv.org/abs/2209.11215>.
- [9] Hyungjin Chung, Jeongsol Kim, Michael T. Mccann, Marc Louis Klasky, and Jong Chul Ye. Diffusion posterior sampling for general noisy inverse problems. In *The Eleventh International Conference on Learning Representations*. URL <https://openreview.net/forum?id=OnD9zGAGT0k>.
- [10] Giannis Daras, Hyungjin Chung, Chieh-Hsin Lai, Yuki Mitsufuji, Jong Chul Ye, Peyman Milanfar, Alexandros G. Dimakis, and Mauricio Delbracio. A survey on diffusion models for inverse problems. URL <http://arxiv.org/abs/2410.00083>.
- [11] Prafulla Dhariwal and Alexander Nichol. Diffusion models beat GANs on image synthesis. In *Advances in Neural Information Processing Systems*, volume 34, pages 8780–8794. Curran Associates, Inc. URL <https://proceedings.neurips.cc/paper/2021/hash/49ad23d1ec9fa4bd8d77d02681df5cfa-Abstract.html>.

- [12] Zehao Dou and Yang Song. Diffusion posterior sampling for linear inverse problem solving — a filtering perspective.
- [13] Andreas Eberle, Arnaud Guillin, and Raphael Zimmer. Couplings and quantitative contraction rates for Langevin dynamics. 47(4):1982–2010. ISSN 0091-1798. doi: 10.1214/18-AOP1299.
- [14] Berthy T. Feng, Jamie Smith, Michael Rubinstein, Huiwen Chang, Katherine L. Bouman, and William T. Freeman. Score-based diffusion models as principled priors for inverse imaging. URL <http://arxiv.org/abs/2304.11751>.
- [15] Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. In *NeurIPS 2021 Workshop on Deep Generative Models and Downstream Applications*. URL <https://openreview.net/forum?id=qw8AKxfYbI>.
- [16] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. URL <http://arxiv.org/abs/2006.11239>.
- [17] Peter Holderrieth. The Fokker–Planck equation and diffusion models. Blog post, 2023. URL <https://www.peterholderrieth.com/blog/2023/The-Fokker-Planck-Equation-and-Diffusion-Models/>.
- [18] M. F. Hutchinson. A stochastic estimator of the trace of the influence matrix for Laplacian smoothing splines. 18(3):1059–1076. ISSN 0361-0918. doi: 10.1080/03610918908812806.
- [19] Aapo Hyvärinen. Estimation of non-normalized statistical models by score matching. 6: 695–709. ISSN 1532-4435. URL <https://jmlr.org/papers/v6/hyvarinen05a.html>.
- [20] Ulugbek S. Kamilov, Charles A. Bouman, Gregory T. Buzzard, and Brendt Wohlberg. Plug-and-play methods for integrating physical and learned models in computational imaging. 40(1):85–97. ISSN 1053-5888. doi: 10.1109/MSP.2022.3199595. URL <https://ieeexplore.ieee.org/document/10004791/>.
- [21] Bahjat Kawar, Michael Elad, Stefano Ermon, and Jiaming Song. Denoising diffusion restoration models. URL <http://arxiv.org/abs/2201.11793>.
- [22] Diederik P. Kingma and Max Welling. Auto-encoding variational Bayes. URL <http://arxiv.org/abs/1312.6114>.
- [23] Peter E. Kloeden and Eckhard Platen. *Numerical Solution of Stochastic Differential Equations*, volume 23 of *Applications of Mathematics*. Springer, Berlin, Heidelberg, 1992. ISBN 978-3-540-54062-5. doi: 10.1007/978-3-662-12616-5.
- [24] Chieh-Hsin Lai, Yuhta Takida, Toshimitsu Uesaka, Naoki Murata, Yuki Mitsufuji, and Stefano Ermon. On the equivalence of consistency-type models. URL <http://arxiv.org/abs/2306.00367>.
- [25] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. URL <https://papers.cool/arxiv/2210.02747>.
- [26] Xingchao Liu, Chengyue Gong, and Qiang Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. URL <https://papers.cool/arxiv/2209.03003>.

- [27] Calvin Luo. Understanding diffusion models: A unified perspective. URL <http://arxiv.org/abs/2208.11970>.
- [28] Morteza Mardani, Jiaming Song, Jan Kautz, and Arash Vahdat. A variational perspective on solving inverse problems with diffusion models. URL <http://arxiv.org/abs/2305.04391>.
- [29] Alexander Quinn Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139, pages 8162–8171. PMLR. URL <https://proceedings.mlr.press/v139/nichol21a.html>.
- [30] Chicago Y. Park, Yuyang Hu, Michael T. McCann, Cristina Garcia-Cardona, Brendt Wohlberg, and Ulugbek S. Kamilov. Plug-and-play priors as a score-based method. In *2025 IEEE International Conference on Image Processing (ICIP)*, pages 49–54. doi: 10.1109/ICIP55913.2025.11084503. URL <http://arxiv.org/abs/2412.11108>.
- [31] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10684–10695. IEEE. doi: 10.1109/CVPR52688.2022.01042. URL https://openaccess.thecvf.com/content/CVPR2022/html/Rombach_High-Resolution_Image_Synthesis_With_Latent_Diffusion_Models_CVPR_2022_paper.html.
- [32] Tim Salimans and Jonathan Ho. Progressive distillation for fast sampling of diffusion models. In *The Tenth International Conference on Learning Representations*. URL <https://openreview.net/forum?id=TIIdIXIpzhoI>.
- [33] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models, . URL <http://arxiv.org/abs/2010.02502>.
- [34] Jiaming Song, Arash Vahdat, Morteza Mardani, and Jan Kautz. Pseudoinverse-guided diffusion models for inverse problems. .
- [35] Jiaming Song, Qinheng Zhang, Hongxu Yin, Morteza Mardani, Ming-Yu Liu, Jan Kautz, Yongxin Chen, and Arash Vahdat. Loss-guided diffusion models for plug-and-play controllable generation. In *Proceedings of the 40th International Conference on Machine Learning*, pages 32483–32498. PMLR, . URL <https://proceedings.mlr.press/v202/song23k.html>.
- [36] Yang Song and Stefano Ermon. Generative modeling by estimating gradients of the data distribution. In *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc., . URL <https://proceedings.neurips.cc/paper/2019/hash/3001ef257407d5a371a96dcd947c7d93-Abstract.html>.
- [37] Yang Song and Stefano Ermon. Improved techniques for training score-based generative models, . URL <http://arxiv.org/abs/2006.09011>.
- [38] Yang Song, Prafulla Dhariwal, Mark Chen, and Ilya Sutskever. Consistency models, . URL <http://arxiv.org/abs/2303.01469>.

- [39] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations, . URL <http://arxiv.org/abs/2011.13456>.
- [40] Jianlin Su. Generative diffusion models (V): The general SDE framework. Blog post, kexue.fm, 2022. URL <https://kexue.fm/archives/9209>. In Chinese.
- [41] Singanallur V. Venkatakrishnan, Charles A. Bouman, and Brendt Wohlberg. Plug-and-play priors for model based reconstruction. In *2013 IEEE Global Conference on Signal and Information Processing*, pages 945–948. IEEE. doi: 10.1109/GlobalSIP.2013.6737048. URL <http://ieeexplore.ieee.org/document/6737048/>.
- [42] Pascal Vincent. A connection between score matching and denoising autoencoders. 23 (7):1661–1674. ISSN 0899-7667. doi: 10.1162/NECO_a_00142. URL <https://direct.mit.edu/neco/article/23/7/1661-1674/7677>.
- [43] Yinhuai Wang, Jiwen Yu, and Jian Zhang. Zero-shot image restoration using denoising diffusion null-space model. URL <http://arxiv.org/abs/2212.00490>.
- [44] Bingliang Zhang, Wenda Chu, Julius Berner, Chenlin Meng, Anima Anandkumar, and Yang Song. Improving diffusion inverse problem solving with decoupled noise annealing. In *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 20895–20905. doi: 10.1109/CVPR52734.2025.01946. URL <http://arxiv.org/abs/2407.01521>.
- [45] Yuanzhi Zhu, Kai Zhang, Jingyun Liang, Jiezhong Cao, Bihan Wen, Radu Timofte, and Luc Van Gool. Denoising diffusion models for plug-and-play image restoration. URL <http://arxiv.org/abs/2305.08995>.